

Retail Trading and Return Predictability in China

中国零售交易与收益可预测性研究

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Abstract 摘要

Using comprehensive account-level data, we separate Chinese retail investors into five groups and document strong heterogeneity in trading dynamics and performances. Retail investors with smaller account sizes cannot predict future returns correctly, display daily momentum patterns, fail to process public news, and show overconfidence and gambling preferences; while retail investors with larger account balances predict future returns correctly, display contrarian patterns, and incorporate public news in trading. With Barber et al. (2009) performance measures, smaller retail investors suffer from poor stock selection abilities and trading costs, while large retail investors' stock selection abilities are offset by trading costs.

利用全面的账户级数据, 我们将中国散户投资者分为五组, 并记录了交易动态和表现的强烈异质性。账户规模较小的散户投资者不能正确预测未来回报, 不能显示每日的动量模式, 不能处理公共新闻, 不能显示过度自信和赌博偏好, 而账户余额较大的散户投资者则能正确预测未来回报, 显示反转模式, 并在交易中纳入公共新闻。根据 Barber 等人(2009)的业绩指标, 较小的散户投资者的选股能力和交易成本较低, 而大型散户投资者的选股能力被交易成本抵消。

Keywords: retail investors, Chinese stock market, return predictability, information content. JEL

关键词: 散户投资者, 中国股市, 收益可预测性, 信息内容

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Retail investors are important participants in financial markets, and many studies using data from the United States (U.S.) and Europe are devoted to understanding their trading motives, performance, and role in information transmission and price discovery.¹ China's equity market, which is the second largest in the world, provides an important setting for studying retail investors. According to the annual report of the Shanghai Stock Exchange in 2016, retail investors contribute about 85% of the daily trading volume on the exchange, whereas institutional investors contribute only around 15%. Behind the high trading volumes are tens of millions of retail investors in China, which accounts for the largest population of retail investors in the world. The dominance and prevalence of retail trading in China brings retail investors to the central stage of the capital market, making it crucial for researchers, regulators, and practitioners to understand their investment choices and how those choices affect information transmission and price discovery.

散户投资者是金融市场的重要参与者，许多利用美国和欧洲数据进行的研究致力于了解散户投资者的交易动机、表现以及在信息传递和价格发现中的作用。¹ 中国股票市场是世界第二大股票市场，为研究散户投资者提供了重要的环境。根据上海证券交易所 2016 年的年度报告，散户投资者贡献了交易所日交易量的 85%，而机构投资者只贡献了约 15%。高交易量的背后是中国数以千万计的散户投资者，他们是世界上散户投资者人数最多的国家。中国零售交易的主导地位和流行程度，将散户投资者带到了资本市场的中心舞台，因此，研究人员、监管机构和从业人员必须了解他们的投资选择，以及这些选择如何影响信息传递和价格发现。

An easy reference for answering these questions might appear to be the earlier studies that use data from the U.S. or Europe. However, the development of the Chinese stock market has been substantially different from that of the U.S. or Europe. For instance, the U.S. market displays typical features of developed capital markets, such as the dominance of institutional investors in terms of trading and holding, mature information environment, and established legislature system. Meanwhile,

Chinese stock market displays typical features of emerging markets, with rising institutional investors accounting for only a small part of the daily trading volume, high valuations (or lower earnings yields), high volatilities, and high turnovers. Given these significant differences, it is not clear whether previous studies conclusions regarding retail investors in other developed

回答这些问题的一个简单参考可能是早期使用美国或欧洲数据的研究。然而，中国股票市场的发展已经与美国或欧洲大不相同。例如，美国市场具有发达资本市场的典型特征，如机构投资者在交易和持有方面的主导地位、成熟的信息环境以及成熟的立法体系。与此同时，中国股市呈现出新兴市场的典型特征：机构投资者的增加仅占日交易量的一小部分，估值较高(或收益率较低)，波动性较大，成交量较高。鉴于这些显著的差异，目前尚不清楚先前的研究是否得出了关于其他发达国家散户投资者的结论

¹ Related studies include Barber and Odean (2000, 2001, 2008), Barber, Odean and Zhu (2008), Kaniel, Saar, and Titman (2008), Kelley and Tetlock (2013), Welch (2022), Barber, Huang, Odean and Schwartz (2022), Boehmer, Jones, Zhang, and Zhang (2021), Eaton, Green, Roseman and Wu (2022), and many others. 相关研究包括 Barber 和 Odean (2000,2001,2008) , Barber, Odean 和 Zhu (2008) , Kaniel, Saar 和 Titman (2008) , Kelley 和 Tetlock (2013) , Welch (2022) , Barber, Huang, Odean 和 Schwartz (2022) , Boehmer, Jones, Zhang 和 Zhang (2021) , Eaton, Green, Roseman 和 Wu (2022)等等。

countries would readily apply to China, and a focused study of Chinese retail investors become essential.

一些国家将很容易向中国提出申请，对中国散户投资者进行重点研究就变得至关重要。

Our study focuses on two core questions to understand retail investors in China: First, how do retail investors contribute to the price discovery process? In other words, can they predict or fail to predict future price movements? Second, if retail investors succeed or fail in predicting future price movements, what are the driving forces of their trading behaviors? Liquidity, information, or behavioral biases? Clearly, millions of Chinese retail investors in our sample are not homogenous individuals, and the answers to these two questions likely differ drastically for different groups of retail investors. Therefore, we rely on the rich cross section of our data to examine the heterogeneity of retail investors and how their trading interacts with stock returns, information flows, and behavioral features.

我们的研究集中在两个核心问题，以了解散户投资者在中国：第一，散户投资者如何贡献的价格发现过程？换句话说，他们能否预测未来的价格走势？其次，如果散户投资者在预测未来价格走势方面成功或失败，他们交易行为的驱动力是什么？流动性，信息，还是行为偏差？显然，在我们的样本中，数以百万计的中国散户投资者并不是同质的个人，对于不同的散户投资者群体，这两个问题的答案可能会有很大的不同。因此，我们依靠丰富的横截面数据来检验散户投资者的异质性，以及他们的交易如何与股票收益、信息流和行为特征相互作用。

We are grateful that a major stock exchange in China provides us the access to proprietary account-level trading and holdings data from 2016 to 2019, accounting for over 53 million retail accounts. To comply with regulatory requirements, the exchange categorizes all Chinese retail accounts into five groups based on their account balances: less than 100,000 CNY (RT1), between 100,000 and 500,000 CNY (RT2), between 500,000 and 3,000,000 CNY (RT3), between

3,000,000 and 10,000,000 CNY (RT4), and greater than 10,000,000 CNY (RT5). The five groups account for 58.7%, 28.6%, 10.9%, 1.4%, and 0.4% of the total number of accounts, respectively. In other words, the majority of Chinese retail investors have account sizes below 500,000 CNY.²

我们感谢中国的一家大型证券交易所为我们提供了 2016 年至 2019 年的自营账户交易和持股数据，其中超过 5300 万个零售账户。为了符合监管要求，交易所将所有中国零售账户按账户余额分为五类：小于 10 万元人民币(RT1)、10 万至 50 万元人民币(RT2)、50 万至 300 万元人民币(RT3)、300 万至 1000 万元人民币(RT4)和 1000 万元人民币以上(RT5)。这五个组别分别占帐户总数的 58.7% 、 28.6% 、 10.9% 、 1.4% 及 0.4% 。换句话说，大多数中国散户投资者的账户规模低于 50 万元人民币。²

To answer the first research question on how retail investors contribute to price discovery, we directly examine whether retail investors' buy and sell activities predict future stock returns. If the market is perfectly efficient, stock prices would follow random walks, and retail trading would

为了回答第一个关于散户投资者如何参与价格发现的研究问题，我们直接考察了散户投资者的买卖活动是否能够预测未来的股票收益。如果市场是完全有效的，股票价格将遵循随机游走，散户交易将

² Additional gender and age information shows that most Chinese retail investors are young or middle-aged males. 更多的性别和年龄信息显示，大多数中国散户投资者是年轻或中年男性。

not predict future returns. If the market is not perfectly efficient and some investors have value-relevant information for future stock prices, their order flows would positively predict future returns. On the other hand, if some investors have an information disadvantage, or fail to incorporate timely information into their trades, their order flows might negatively predict future returns. Using daily retail order imbalances from each retail group, we examine their predictive power for future stock returns at different horizons. The smaller retail investors, RT1-RT4, predict next-day returns with negative and significant coefficients. That is, the prices of stocks they purchase experiences negative returns the following day, whereas the stocks they sell experiences positive returns. By contrast, the largest retail investors, RT5, positively and significantly predict next-day returns, indicating that they purchase and sell stocks in directions consistent with future price movements. When we examine longer horizons, the above-mentioned predictive patterns persist for about 8 to 9 weeks. These patterns remain quite robust when we form long-short strategies on order flow information, and for subsets of stocks with differences in size, value, liquidity, and share price levels.

不能预测未来的回报。如果市场不是完全有效的，并且一些投资者拥有未来股票价格的价值相关信息，那么他们的订单流将会正向预测未来的回报。另一方面，如果一些投资者在信息方面处于劣势，或者未能及时将信息纳入交易，他们的指令流可能会对未来收益产生负面预测。利用每个零售集团的每日订单不平衡，我们检验他们在不同视野下对未来股票收益的预测能力。较小的散户投资者，RT1-RT4，用负系数和显著系数预测次日收益。也就是说，他们购买的股票的价格在第二天经历负回报，而他们卖出的股票经历正回报。相比之下，最大的散户投资者，RT5，积极和显著地预测第二天的回报，表明他们买卖股票的方向与未来的价格走势一致。当我们研究更长的视野时，上述预测模式会持续大约 8 到

9 周。当我们根据订单流信息形成多空策略时，以及对于规模、价值、流动性和股价水平不同的股票子集，这些模式仍然相当稳健。

The data allow us to examine the return predictive patterns among the characteristics that are not easily available in most other datasets. For instance, we directly observe the counterparties of each trade and find that the order flows from small retail investors always predict returns negatively and do not depend on counterparties. Interestingly, for large retail and institutional investors, their order flows typically predict future returns positively, but they become negative when they trade on the same side as small retail investors, suggesting that their prediction becomes erroneous when on the same side as smaller retail investors. We also examine how holding horizons are related to the predictive patterns of retail order flow. Small retail investors' negative return predictive power is stronger over the short horizon and slowly diminishes when holding

这些数据使我们能够检查在大多数其他数据集中不容易获得的特征之间的返回预测模式。例如，我们直接观察每笔交易的交易对手，发现来自小散户投资者的订单流总是预测收益为负，而不依赖于交易对手。有趣的是，对于大型散户和机构投资者而言，他们的指令流通常会正面预测未来的回报，但当他们与小型散户投资者处于同一边时，他们的预测就会变为负面，这表明当他们与小型散户投资者处于同一边时，他们的预测就会出错。我们还研究了持有期限与零售订单流的预测模式之间的关系。小型散户投资者的负回报预测能力在短期内更强，在持有时慢慢减弱

horizons become longer, indicating that a longer holding horizon might lead to better returns. In contrast, large retail investors' positive return predictive power is stronger over the short horizon, which suggests that they might trade on time-sensitive information, and the predictive power becomes weaker for longer horizons.

视野变长，表明持有视野变长可能会带来更好的回报。相比之下，大型散户投资者的正回报预测能力在短期内较强，这意味着他们可能会利用对时间敏感的信息进行交易，而在长期内，预测能力会变弱。

For the second research question regarding trading motives of retail investors, previous literature provides multiple explanations, such as order flow persistence, liquidity provision, behavioral biases, and information (dis)advantages. These explanations also naturally connect with the predictive pattern of retail order flows for future returns. We adopt the two-stage decomposition procedure in Boehmer et al. (2021) to examine whether these hypotheses explain the trading activities of different retail investor groups and how these trading activities contribute to the predictive patterns for future returns. Our results show that small retail investors display significant order flow persistence, and that their trades have momentum patterns at daily horizons that demand immediate liquidity. Smaller retail investors also display strong behavioral biases, such as overconfidence and gambling preferences, and fail to predict and process informational earnings news. In explaining order flow's predictive power for future returns, order persistence, daily momentum trading patterns, behavioral biases, and information disadvantages all contribute to the negative predictive power of smaller retail investors. In contrast, the largest retail investors display contrarian trading patterns, trade against the behavioral biases of other retail groups, and are capable of predicting and processing earnings news. All of these features contribute to the positive predictive power of the largest retail investors.

对于第二个关于散户投资者交易动机的研究问题，以往的文献提供了多种解释，如订单流持续性、流动性供给、行为偏差和信息优势。这些解释也自然地与零售订单流对未来回报的预测模式联系在一起。我们采用 Boehmer 等人(2021)的两阶段分解程序来检验这些假设是否解释了不同散户投资者群体的交易活动，以及这些交易活动如何对未来收益的预测模式产生影响。我们的结果表明，小型散户投资者显示出显著的订单流持续性，并且他们的交易在每天的视野中具有动量模式，需要立即的流动性。较小的散户投资者也表现出强烈的行为偏差，例如过度自信和赌博偏好，并且不能预测和处理信息性的盈利消息。在解释订单流对未来收益的预测能力时，订单持续性、每日动量交易模式、行为偏差和信息劣势都会导致较小散户投资者的负预测能力。相比之下，最大的散户投资者表现出反向交易模式，与其他散户集团的行为偏差进行交易，并能够预测和处理收益新闻。所有这些特征都有助于最大的散户投资者的积极预测能力。

To further understand the performance of retail trading flows, we follow Barber et al. (2009) to construct net buy and sell portfolios, track the net positions of each group of retail investors, and compute the performance of the net buy and sell portfolios. Consistent with previous findings

为了进一步了解零售交易流的表现，我们跟随 Barber 等人(2009)构建净买卖投资组合，跟踪每组零售投资者的净头寸，并计算净买卖投资组合的表现。与之前的研究结果一致

that smaller retail investors negatively predict future returns, while large investors positively predict future returns, we find that the net buy and sell portfolios tracking smaller retail investors' order flows have an annual return of -5.61%, while the net buy and sell portfolios tracking large retail investors have an annual return of -0.29%. We also decompose the performance into three components: stock selection, market timing, and trading cost. We find that stock selection and trading costs are the two major drivers of total returns in the tracking portfolios.

我们发现，追踪较小散户投资者订单流量的净买卖投资组合的年回报率为 -5.61%，而追踪较大散户投资者的净买卖投资组合的年回报率为 -0.29%。我们还将绩效分解为三个组成部分：股票选择、市场时机和交易成本。我们发现，股票选择和交易成本是追踪投资组合总收益的两个主要驱动因素。

This study is closely related to the retail investor literature. As mentioned earlier, previous studies on retail investors primarily use data from the U.S. and other markets, and tend to treat retail investors as one group.³ For instance, using data from a discount broker in the U.S., Barber and Odean (2000, 2001, 2008) document many behavioral biases. Kaniel et al. (2008), Barber et al. (2009), Kelley and Tetlock (2013), and Boehmer et al. (2021) use different datasets from the U.S. and find that retail trading can positively predict the cross-section of future returns. Outside the U.S., Grinblatt and Keloharju (2000), Linnainmaa (2010), and Grinblatt, Keloharju, and Linnainmaa (2012) focus on Finland data; Bach, Calvet, and Sodini (2020) focus on Swedish data; Dorn, Huberman, and Sengmueller (2008) study data from Germany. All of these studies provide important results regarding retail investors' trading activities. Interestingly though, the results are not always consistent, and evidence from China will provide more insights into this literature.

这项研究与散户投资者文献密切相关。如前所述，以往关于散户投资者的研究主要使用来自美国和其他市场的数据，倾向于将散户投资者作为一个群体来对待。例如，使用来自美国折扣经纪人的数据，Barber 和 Odean (2000,2001,2008)记录了许多行为偏见。Kaniel 等人

(2008), Barber 等人(2009), Kelley 和 Tetlock (2013), 以及 Boehmer 等人(2021)使用来自美国的不同数据集, 发现零售交易可以正向预测未来收益的横截面。在美国之外, Grinblatt 和 Keloharju (2000), Linnainmaa (2010), Grinblatt, Keloharju 和 Linnainmaa (2012)关注芬兰的数据; Bach, Calvet 和 Sodini (2020)关注瑞典的数据; Dorn, Huberman 和 Sengmueller (2008)关注德国的研究数据。所有这些研究都提供了关于散户投资者交易活动的重要结果。有趣的是, 这些结果并不总是一致的, 来自中国的证据将为这些文献提供更多的见解。

This study is also related to the recent studies of the rapidly growing Chinese stock market. Liu, Stambaugh and Yuan (2019), and Liu, Zhou, and Zhu (2021) establish asset pricing factors for stock returns. For Chinese retail investors, An, Lou, and Shi (2022) study the wealth redistribution role of financial bubbles and crashes over July 2014 and December 2015,

这项研究也与最近对快速增长的中国股票市场的研究有关。Liu, Stambaugh and Yuan (2019), and Liu, Zhou, and Zhu (2021)为股票回报建立了资产定价因素。对于中国散户投资者来说, An, Lou 和 Shi (2022)研究了 2014 年 7 月和 2015 年 12 月期间金融泡沫和崩溃的财富再分配作用,

³To facilitate reading, Appendix Table I provides a summary table of previous papers on retail investors using data from different markets, and Appendix Table II provides comparisons between the findings of this study with the results from previous literature.

为方便阅读, 附录表 i 提供了以往利用不同市场数据编制的一般投资者文件摘要表, 附录表 II 则提供了本研究结果与以往文献结果的比较。

documenting a net transfer of 250 billion CNY from the poor to ultra-wealthy retail investors over this period. Liu, Peng, Xiong, and Xiong (2021) and Liao, Peng, and Zhu (2021) both focus on the behavioral features of Chinese retail investors and document overconfidence, gambling preferences, and extrapolative expectations in these investors. Li, Geng, Subrahmanyam, and Yu (2017), Chen, Gao, He, Jiang, and Xiong (2019), Jiang, Liu, Peng, and Wang (2022), Hu, Liu, and Xu (2021), and Titman, Wei, and Zhao (2022) all focus on an earlier Chinese sample period and examine behavioral biases and reactions to corporate events. These Chinese studies primarily rely on low-frequency data, or data from a small number of brokerages covering a small part of the market, or investigate issues other than return predictability. As a result, there remains no direct study on the heterogeneity of retail investors' trading behavior, their return predictive power, and how they process information using high-frequency trading data in one major stock market dominated by retail investors.

记录显示，在此期间，有 2,500 亿元人民币从穷人手中净转移到超级富有的散户投资者手中。Liu, Peng, Xiong, and Xiong (2021)和 Liao, Peng, and Zhu (2021)都关注中国散户投资者的行为特征，并记录了这些投资者的过度自信、赌博偏好和外推预期。Li, Geng, Subrahmanyam, and Yu (2017) , Chen, Gao, He, Jiang, and Xiong (2019) , Jiang, Liu, Peng, and Wang (2022) , Hu, Liu, and Xu (2021) , and Titman, Wei, and Zhao (2022)都关注早期的中国样本期，研究行为偏差和对公司事件的反应。这些中国研究主要依赖于低频率数据，或少数几家经纪公司覆盖市场一小部分的数据，或者研究回报可预测性以外的问题。因此，对散户投资者交易行为的异质性、他们的收益预测能力以及他们如何利用高频交易数据在一个以散户投资者为主的主要股票市场中处理信息，目前还没有直接的研究。

Compared with previous studies, this study makes three important contributions. First, our study, with its large market coverage of the market for a recent sample period, is one of the

most thorough and comprehensive studies of Chinese retail investors, and provides many important implications for regulators, practitioners, and academic researchers. Second, we separate retail investors into groups based on account size and provide unique and direct evidence of investor heterogeneity in terms of return predictability, counterparties, holding horizons, and performance. Third, we examine different hypotheses regarding return prediction patterns for different retail investor groups and provide clear evidence on the sources of the negative or positive predictive power of different retail investors.

与之前的研究相比，这项研究有三个重要贡献。首先，我们的研究是对中国散户投资者最全面和最深入的研究之一，在最近的一个样本期间，我们的研究覆盖了大量的市场，为监管机构、从业人员和学术研究者提供了许多重要的启示。其次，我们根据账户规模将散户投资者分成不同的群体，并提供独特而直接的证据，证明投资者在回报可预测性、交易对手、持有期限和业绩方面的异质性。第三，我们考察了不同散户投资者群体收益预测模式的不同假设，并提供了关于不同散户投资者负向或正向预测能力来源的明确证据。

I. Data 资料

A. Data on Retail Investors

A. 零售投资者数据

Our data comes from one major stock exchange in China, which contains the trading and holding history of all stocks listed on the exchange from all investors, between January 2016 and June 2019. This proprietary dataset contains roughly 53 million accounts, and based on investor identities, they are grouped into three major categories: retail (RT), institutional (INST), and corporations (CORP). Retail investors are further stratified into five groups based on their account sizes at the beginning of each year, which is the average portfolio value (including equity holdings in all A-share listed firms, plus cash balance) over the previous twelve months.⁴ As mentioned in the introduction, there are five subgroups: below 100,000 CNY (RT1), 100,000-500,000 CNY (RT2), 500,000 - 3 million CNY (RT3), 3 million - 10 million CNY (RT4), and above 10 million CNY (RT5). Since the focus of this study is how retail trades are related to stock prices in the cross-section, we sum up individual investors' trading information at seven investor group levels (RT1-RT5, INST, and CORP) for each stock each day.

我们的数据来自中国的一家主要证券交易所，其中包含了 2016 年 1 月至 2019 年 6 月期间所有投资者在该交易所上市的所有股票的交易和持有历史。这个专有数据集包含大约 5300 万个账户，根据投资者身份，它们被分为三大类：零售(RT)、机构(INST)和公司(CORP)。一般投资者会根据每年年初的户口规模进一步分为五组，即过去 12 个月的平均投资组合价值(包括所有 a 股上市公司的持股及现金结余)。4 正如引言中提到的，有五个分组：低于 10 万元人民币(RT1)、10 万至 50 万元人民币(RT2)、50 万至 300 万元人民币(RT3)、300 万至 1000 万元人民币(RT4)及 1000 万元人民币以上(RT5)。由于本研究的重点是散户交易与股票价格在横截面上的关系，我们每天在七个投资者群体水平(RT1-RT5, INST, 和 CORP)对每只股票总结个人投资者的交易信息。

Panel A of Table I presents the summary statistics for investor accounts. During the sample period, the total number of active accounts for retail investors, institutions, and corporations are 53.4 million, 40,000, and 47,000, respectively. Within the retail investor category, there are 31.4 million, 15.3 million, 5.8 million, 0.7 million, and 0.2 million accounts for RT1 to RT5, respectively. Clearly, most of the retail investors have accounts less than 500,000 CNY. The overall trading volume on this exchange averages 201 billion CNY per day, with retail investors,

表一的 a 小组显示了投资者账户的概要统计数据。在样本期间，一般投资者、机构及公司的活跃帐户总数分别为 5340 万、40000 及 47000 个。在散户投资者类别中，rt1 至 rt5 分别有 3140 万个、1530 万个、580 万个、70 万个和 20 万个账户。显然，大多数散户投资者的账户少于 50 万元人民币。这个交易所的总交易量平均每天 2010 亿元人民币，

⁴The annual calculation of account balances is designed by the exchange. There is a concern that investors might migrate to different groups after the initial grouping at the beginning of the year. We acknowledge this possibility. However, the fact that the grouping is redone each year eases the concern to some extent. The cutoff numbers to separate retail investors into different groups are also chosen by the exchange to comply with exchange regulations. For instance, derivatives and leverage trading are only allowed for investors with account size higher than 0.5 million CNY in order to protect the investors with lower account balances.

账户余额的年度计算是由交易所设计的。有人担心，投资者可能会在年初的初始分组之后转移到不同的集团。我们承认这种可能性。然而，每年重新分组的事实在某种程度上缓解了人们的担忧。为了遵守交易所的规定，交易所还选择了将散户投资者分成不同群体的截止数字。例如，为了保护账户余额较低的投资者，只允许账户规模大于 50 万元人民币的投资者进行衍生品和杠杆交易。

institutions, and corporations accounting for 81%, 17%, and 2% of the total trading volume, respectively. Within the retail investor sector, the trading volumes for RT1 to RT5 are 5%, 17%, 27%, 13%, and 19% of the total trading volume, respectively, indicating that RT3 is the most active trading subgroup. Regarding stock holdings, retail investors' holdings account for 22%, institutions are 17%, and corporations are 62%. Within the retail investor sector, the account values for RT1 to RT5 are 1%, 4%, 6%, 3%, and 7% of the total tradable market cap, respectively. With low holdings and high trading volumes, Chinese retail investors trade quite actively; in some cases, they might even trade excessively.

机构和公司分别占总交易量的 81% , 17% 和 2% 。在一般投资者方面, rt1 至 rt5 的交易量分别占总交易量的 5% 、 17% 、 27% 、 13% 及 19% , 显示 rt3 是最活跃的交易分组。在股票持有量方面, 散户投资者占 22% , 机构投资者占 17% , 公司投资者占 62% 。在零售投资者部门, rt1 至 rt5 的账户价值分别为可交易市值总额的 1% 、 4% 、 6% 、 3% 和 7% 。由于持有量低, 交易量大, 中国散户投资者交易相当活跃; 在某些情况下, 他们甚至可能过度交易。

[Insert Table I about here]
[在这里插入表 1]

To understand the relative importance of different investment groups' trading over time, we plot the time series of cross-sectional means of various investors' trading activities in Figure I. Panel A presents each group's trading volume as a percentage of the total trading volume. Again, the RT3 group has the highest trading volume, accounting for about 30% of total trading. Interestingly, institutional trading gradually increases over time, from 10% in 2016 to over 20% in 2019. The corporations barely trade and account for a negligible amount of trading volume. Panel B displays the percentage of shares held by each group, and it can be seen that the time-series patterns in holdings are stable. Corporations hold a large chunk of shares, accounting for over 60% of total

shares; institutions hold about 17% shares; large and medium retail investors hold about 3% to 6% shares, and the smallest retail investors RT1 hold around 1% of the shares. In Panel C, we compute relative trading activeness with respect to holdings for each group, by scaling each group's trading share volumes by each group's holdings at stock level. The retail investors all follow similar time-series patterns, trading around 10% of their holdings each day on average, which is quite high. We observe a slow decrease in trading after 2018 to about 5%, but there is also

为了理解不同投资集团的交易随时间变化的相对重要性，我们在图一中绘制了不同投资者交易活动的横截面方法的时间序列。同样，rt3 集团拥有最高的交易量，约占总交易量的 30%。有趣的是，随着时间的推移，机构交易逐渐增加，从 2016 年的 10% 到 2019 年的超过 20%。这些公司几乎没有交易，在交易量中所占的份额微不足道。面板 b 显示了每个集团所持股份的百分比，可以看出，持股的时间序列模式是稳定的。公司持有大量股份，占总股份的 60% 以上；机构持有约 17% 的股份；大中型散户投资者持有约 3% 至 6% 的股份，最小的散户投资者持有约 1% 的股份。在面板 c 中，我们计算每个集团相对于其持股的交易活跃度，方法是按照每个集团在股票水平上的持股量来衡量每个集团的交易份额。散户投资者都遵循类似的时间序列模式，平均每天交易大约 10% 的股票，这是相当高的。我们观察到 2018 年之后交易量缓慢下降到 5% 左右，但是也有

a large but short-term surge up to 15% at the beginning of 2019. Considering that the market return is -22.2% during 2018, and it reverts back to 21.2% during the first half of 2019, it is likely that the large negative returns reduce retail trading enthusiasms, while large positive returns reignite the enthusiasms. By the end of the 2019, retail trading returns to the average level of 10% of holdings. We notice that among the five groups of retail investors, RT5 trade slightly less than the rest of the retail investors. We also observe a clear increasing pattern in the activeness of institutional trading, which increases from 7% in 2016 to 16% in 2019. Finally, the corporates in general rarely trade, and their trading accounts for about 2% of their holdings.

在 2019 年初，一个大的但是短期的增长高达 15%。考虑到市场回报率在 2018 年为 -22.2%，在 2019 年上半年又回到 21.2%，大量的负回报可能会降低零售交易的热情，而大量的正回报则会重新点燃这种热情。到 2019 年底，零售交易回报率平均为 10%。我们注意到，在五组散户投资者中，rt5 的交易量略低于其他散户投资者。我们亦发现机构交易活跃程度明显上升，由二零一六年的 7% 上升至二零一九年的 16%。最后，一般来说，企业很少交易，他们的交易占他们持股的 2% 左右。

Overall, in the Chinese stock market, retail investors dominate in trading, while corporations dominate in holdings. The retail investors' dominance in trading in the Chinese stock market is likely the joint result of market development, regulations, and investor preferences. This pattern is not particularly rare for emerging markets, but is quite different from that of developed markets.⁵

总的来说，在中国股票市场，散户投资者在交易中占主导地位，而公司在持股中占主导地位。散户投资者在中国股市交易中的主导地位可能是市场发展、监管和投资者偏好共同作用的结果。这种模式在新兴市场并不罕见，但与发达市场有很大不同

[Insert Figure I about here]
[这里插入图 i]

B. Data on Stock Returns and Firm Characteristics

乙、股票回报及公司特征数据

We obtain data on stock returns, volumes, and accounting information from Wind Information Inc. (WIND), the largest financial data provider in China. For consistency with the retail data, the sample period runs from January 2016 to June 2019. We adopt the filters in Liu et al. (2019) and exclude stocks with less than 15 days of trading records during the most recent

我们从中国最大的金融数据提供商万得资讯公司(WIND Information inc.)获得股票回报、成交量和会计信息的数据。为了与零售数据保持一致，样本期从 2016 年 1 月至 2019 年 6 月。我们采用 Liu 等人(2019)的过滤器，排除最近一次交易记录少于 15 天的股票

⁵We present additional summary statistics on retail trading volumes and holdings in Appendix Table III. The results show that small retail investors prefer to trade and hold small, low earning/price ratio, and high turnover firms, while the largest retail investors' trading and holdings are tilted towards larger, high earnings-to-price ratio firms. In terms of sectors, the small retail investors prefer to trade and hold the alternative energy sector, and prefer not to trade and hold banks and life insurance firms, while the institutions and corporates behave in opposite ways. Finally, small retail investors tend to use small order sizes, while large retail investors tend to use large orders sizes, while institutions use all order sizes.

我们在附录表三中提供了有关零售交易量和持有量的额外汇总统计数据。结果表明，小型散户投资者倾向于交易和持有小型、低收益/价格比和高成交量的公司，而大型散户投资者的交易和持股倾向于较大型、高收益/价格比的公司。就部门而言，小型散户投资者更愿意交易和持有替代能源部门，而不愿意交易和持有银行和寿险公司，而机构和公司则表现相反。最后，小型散户投资者倾向于使用小额订单，而大型散户投资者倾向于使用大额订单，而机构投资者则使用所有订单。

month. Liu et al. (2019) also eliminate stocks that become public within the past six months, stocks with fewer than 120 days of trading records during the past 12 months, and the smallest 30% of all firms listed in the Chinese A-share market. This study does not exclude these stocks from the main results because retail investors trade actively in small stocks and during a stock's IPO period. We present the results with all the filters from Liu et al. (2019) in our robustness checks, and the findings are similar with the additional filters. Starting from March 31, 2010, margin buy and short-sell are allowed on Chinese stock exchanges for subsets of stocks. We include these leveraged trades in our main results and provide an additional analysis that excludes leveraged trading in our robustness checks. We merge the exchange and WIND data by stock ticker and present account summary statistics. Our final sample covers over 1.1 million stock-day observations, and on each day, we have an average of around 1,200 firms.

月。Liu 等人(2019)还排除了在过去 6 个月内上市的股票, 在过去 12 个月内交易记录少于 120 天的股票, 以及在中国 a 股市场上市的所有公司中最小的 30% 的股票。这项研究并没有将这些股票排除在主要结果之外, 因为散户投资者在小型股和 IPO 期间交易活跃。在我们的稳健性检查中, 我们用 Liu 等人(2019)的所有过滤器呈现了结果, 结果与附加过滤器相似。从 2010 年 3 月 31 日起, 中国股票交易所允许对股票子集进行融资融券交易。我们将这些杠杆交易纳入我们的主要结果, 并提供了一个额外的分析, 排除了在我们的稳健性检查杠杆交易。我们通过股票代码和当前账户汇总统计数据合并交易所和 WIND 数据。我们的最终样本涵盖了超过 110 万个股票日的观察数据, 平均每天我们有大约 1200 家公司。

Table I Panel B presents several key variables for the Chinese capital market. Market capitalization is the product of the previous month's closing price and total A shares outstanding. The average Chinese firm capitalization is 20.1 billion CNY, or 3 billion USD, about half of the cross-sectional average in the U.S stock market during the same period, which is 6.9 billion USD. The

earnings-to-price ratio (EP ratio) is the ratio of the most recently reported quarterly net profit, excluding non-recurrent gains/losses over last month-end's market capitalization. According to Liu et al. (2019), the EP ratio captures the value effect. The average EP ratio is 0.0075 in China, while the average EP ratio is 0.0272 in U.S stock market, which indicates high valuation ratios in China. The average daily stock return is -0.01% for Chinese stocks, while the average daily stock return is 0.04% in the U.S. stock market over the same sample period.⁶ Finally, we compute daily

表一 b 显示了中国资本市场的几个关键变量。市值是上个月收盘价和 a 股总流通量的乘积。中国企业的平均市值为 201 亿元人民币，约合 30 亿美元，约为同期美国股市横截面平均值(69 亿美元)的一半。市盈率是指最近公布的季度净利润，不包括上月末市值的非经常性收益/亏损。根据 Liu et al. (2019)，市盈率反映了价值效应。中国股票市场的平均收益率为 0.0075，而美国股票市场的平均收益率为 0.0272，表明中国股票市场的估值比较高。中国股市的平均日收益率为 -0.01%，而美国股市同期的平均日收益率为 0.04%。最后，我们计算每天

⁶ Previous literature using the U.S. data shows that microstructure frictions can generate noise in daily return measures. For instance, Blume and Stambaugh (1983) show that daily returns computed from the end-of-day closing prices can have an upward bias due to bid-ask bounce. We compute this bias measure using the closing bid and ask prices for all A-share stocks listed on this major stock exchange. The average bias measure is generally below 0.0002% across all
以往的文献使用美国的数据表明，微观结构摩擦可以产生噪声的日回报措施。例如，Blume 和 Stambaugh (1983) 指出，由于买卖反弹，从当日收盘价计算出来的每日回报率可能有上升的倾向。我们使用收盘报价来计算这种偏差，并且要求在这个主要股票交易所上市的所有 a 股股票的价格。平均偏差指标一般低于 0.0002%

turnover as daily share trading volume divided by tradable shares outstanding. The average daily turnover in China is 2.45%, indicating that it takes $1 \div 2.45\% = 41$ days to turn around all shares, which is much larger than the daily turnover of 1.11% in the U.S. during the same period. These summary statistics show that an average Chinese firm has smaller capitalization, higher valuation ratio, lower returns, and higher turnover than an average U.S. firm.

每日成交量除以流通股。中国股市的平均日成交量为 2.45%，这意味着所有股票需要 $12.45\% = 41$ 天的周转时间，远远高于美国股市同期的 1.11%。这些统计数据表明，一个普通的中国公司比一个普通的美国公司拥有更小的市值，更高的估值比率，更低的回报率 and 更高的营业额。

C. The Order Imbalance Measure 订单不平衡度量

The order flows from different groups of investors are measured using order imbalances, as in Chordia and Subrahmanyam (2004). For stock i , day d , and investor group G , we compute

如 Chordia 和 Subrahmanyam (2004)所示，来自不同投资者群体的订单流量是通过订单失衡来衡量的。对于股票 i ， d ，和投资者组 g ，我们计算

$$OibRT1 = \frac{\sum_{j \in G} V_{i,j}^B - \sum_{j \in G} V_{i,j}^S}{\sum_{j \in G} V_{i,j}^B + \sum_{j \in G} V_{i,j}^S}, \quad (1)$$

where the numerator is the *difference* between buy and sell share volumes⁷ for stock i on day d , summed over all individual j 's within group G , and the denominator is the *sum* of buy and sell share volumes for stock i on day d , summed over all individual j 's within group G . The data allows us to directly observe each trade's direction. When a set of investors buy more than they sell, the order imbalance is positive and vice versa. We compute the order imbalance measure for each investor group as $OibRT1$ to $OibRT5$, $OibINST$, and $OibCORP$.

其中分子是 d 天股票 i 的买入和卖出股票成交量之差，相加于 g 组内所有个体 j 之和，分母是 d 天股票 i 的买入和卖出股票成交量之和，相加于 g 组内所有个体 j 之和。数据允许我们直接观察每个交易的方向。当一组投资者买入多于卖出时，订单失衡是正的，反之亦然。我们计算每个投资者群体的订单不平衡度量值为 $oibrt1$ 到 $OibRT5$ ， $OibINST$ 和 $OibCORP$ 。

Table I Panel C reports the summary statistics for the order imbalance measures. The average order imbalances for RT1 to RT5, institutions, and corporations range between -0.021 and 0.019, are all close to zero. The small magnitude of these average order imbalance measures indicates that most buys and sells within each investor group cancel each other out. The standard deviations of order imbalances are larger for large retail investors (0.352) and institutions (0.455),

表一面板 c 报告了订单不平衡度量的汇总统计数据。Rt1 到 RT5，机构和公司的平均订单不平衡在 -0.021 到 0.019 之间，都接近于零。这些平均订单不平衡指标的小幅度表明，每个投资者群体内的大多数买入和卖出相互抵消。对于大型散户投资者(0.352)和机构投资者(0.455)，订单失衡的标准差更大，

stocks, which is negligible compared to the bias computed in Blume and Stambaugh (1983). Therefore, we compute daily returns using daily close prices without the Blume and Stambaugh (1983) adjustments.

与 Blume 和 Stambaugh (1983)计算的偏差相比，这种偏差可以忽略不计。因此，我们使用每日收盘价来计算每日收益，而没有 Blume 和 Stambaugh (1983)的调整。

⁷ In this study, “share volume” refers to number of shares traded on that day, and “cash volume” refers to the product of share volume and the execution price.

在本研究中，“股票交易量”是指当日交易的股票数量，“现金交易量”是指股票交易量与执行价格的乘积。

as compared to small and medium retail investors (between 0.166 and 0.250), indicate that there is more cross-stock variation in large retail investor and institutional trading activity. The one-day autocorrelation coefficient, AR1, for these *Oib* measures are 0.243, 0.259, 0.216, 0.059, and 0.102 for RT1 to RT5, respectively, suggesting that small and medium retail order imbalances are generally more persistent than large retail imbalances.

与中小型散户投资者(介乎 0.166 至 0.250 之间)比较, 显示大型散户投资者及机构投资者的交易活动存在较大的跨股票差异。这些 *Oib* 指标的单日自相关系数 *ar1* 分别为 0.243、0.259、0.216、0.059, *rt1* 至 *rt5* 的单日自相关系数为 0.102, 表明中小型零售订单失衡通常比大型零售订单失衡更持久。

In terms of order flow correlations across the seven groups, the order flows from smaller retail investors, *OibRT1*, *OibRT2*, and *OibRT3*, are highly correlated, with correlation coefficients primarily higher than 0.60. *OibRT4* is still positively correlated with *OibRT1*-*OibRT3*, but with lower correlation coefficients of around 0.20. The largest retail investors' order imbalance, *OibRT5*, is negatively correlated with all four other groups, with correlations around -0.15, indicating that this group of retail investors might have different trading patterns from the others. Institutional order imbalances are negatively correlated with all five retail groups, with correlations between - 0.380 and -0.188, again implying different trading patterns from retail investors, even the largest retail investors. As we find earlier, corporations barely trade and their correlations with the rest of the investor categories are all lower than 10%.⁸

根据七个集团之间的订单流相关性, 来自较小散户投资者 *OibRT1*、*oibrt2* 和 *oibrt3* 的订单流高度相关, 相关系数主要高于 0.60。*Oibrt4* 仍与 *oibrt1*-*oibrt3* 呈正相关, 但相关系数较低, 约为 0.20。最大的散户投资者订单失衡 *oibrt5* 与其他四个群体呈负相关, 相关系数在 -0.15 左右, 表明这一群体的散户投资者可能有着与其他群体不同的交易模式。机构订单失衡与所有 5 家零售集团都呈负相关, 相关系数在 -0.380 和 -0.188 之间, 这再次表

明，与散户投资者(甚至是最大的散户投资者)的交易模式不同。正如我们之前发现的那样，企业几乎不进行交易，它们与其他投资者类别的相关性都低于 10%

In this data section, we include *OibINST* and *OibCORP* to ensure the completeness of the summary statistics. Given that corporations are long-term investors and rarely trade, and that our study focuses on trading behavior, we exclude corporations from the remaining empirical results. In terms of institutional investors, given that retail investors are commonly assumed less sophisticated in comparison to institutional investors, we retain institutions in our main results for comparison purposes.

在这个数据部分，我们包括 *OibINST* 和 *OibCORP*，以确保总结统计的完整性。鉴于公司是长期投资者，很少进行交易，我们的研究侧重于交易行为，我们将公司排除在剩余的实证结果之外。在机构投资者方面，由于一般认为一般投资者较机构投资者成熟，我们在主要结果中保留机构投资者的资料，以作比较之用。

II. Can Retail Order Flows Predict Future Stock Returns?

零售订单流能否预测未来股票收益？

⁸ Appendix Figure I presents the time-series plot of the order imbalance measures for each investor group, and we find no evidence of time trends or breaks over our sample period.
附录图 i 显示了每个投资者群体的订单不平衡度量的时间序列图，我们没有发现在我们的样本周期内有时间趋势或中断的证据。

A. Predicting Next Day Stock Returns Using Retail Order Flows

To investigate the roles different retail investors play in the price discovery process, we first examine retail order flow’s predictive power for the next day’s return. We adopt the two-stage Fama and MacBeth (1973) regression to examine the predictive patterns of order flows for next-day returns. At the first stage, we estimate the following cross-sectional regression for each day d within each investor group G :

为了研究不同散户投资者在价格发现过程中所扮演的角色，我们首先考察了散户订单流对第二天收益的预测能力。我们采用两阶段 Fama 和 MacBeth (1973) 回归来检验订单流对次日收益的预测模式。在第一阶段，我们估计每个投资者组 g 内每天 d 的以下横截面回归：

$$, = 0, + 1, \quad , -1 + 2, ' \quad , -1 + 1, , \quad (2)$$

$$, = 0, +1, -1 + 2, -1 + 1, \quad (2)$$

where the dependent variable r_{it} is the stock return for firm i on day d , and the independent variable z_{it} is a vector of variables that explain the return. The error term ϵ_{it} is the error term. The regression equation is:

variables include order imbalance measures from investor group G from the previous day, $_{t-1}$, and control variables, $_{t-1}$. We follow previous literature for the choice of 变量包括投资者 g 组前一天的订单不平衡度量, -1 和控制变量, -1 。我们根据以前的文献选择

control variables. To control for potential momentum/reversal from past returns, we include returns from the previous day, $Ret(-1)$, returns from the previous week, $Ret(-6,-2)$, and returns from the previous month, $Ret(-27, -7)$. For size, value, and liquidity effects, we include log market size ($Size$), earnings-to-price ratio (EP), and turnover as controls, all computed from the previous month-end.

控制变量。为了控制过去收益的潜在动量/逆转，我们包括前一天的收益， $Ret(-1)$ ，前一周的收益， $Ret(-6, -2)$ ，以及前一个月的收益， $Ret(-27, -7)$ 。对于规模、价值和流动性效应，我们将对数市场规模(Size)、盈利价格比(EP)和成交量作为控制因素，所有这些都是从上个月月底计算出来的。

From the first stage estimation, we obtain the daily time series of coefficients for each investor group $G, \{0, 1, 2'\}$. For the second stage estimation, we conduct statistical

从第一阶段估计出发，我们得到了每个投资者群体 $g, \{0, 1, 2'\}$ 系数的日时间序列。对于第二阶段的估计，我们进行统计

inference based on the means, $\{0, 1, 2'\}$, and standard errors of the first stage coefficients. To be specific, we compute standard errors following the method in Newey and West (1987), with the optimal bandwidth chosen by using Newey and West (1994) approach.⁹ If the order flow variable from a specific investor group G predicts future returns in the correct direction, defined
基于平均值 $\{0, 1, 2'\}$ 和第一阶段系数的标准误差的推断。具体地说，我们按照 Newey 和 West (1987) 的方法计算标准误差，并用 Newey 和 West (1994) 的方法选择最佳带宽

⁹We also consider robustness check using BIC to select optimal lag number. Results are similar and available on request.

我们还考虑使用 BIC 进行鲁棒性检查，以选择最佳滞后数。结果是相似的，并可根据要求。

as an occasion when more past purchases are associated with higher future returns and more past sales are associated with lower future returns, we expect the coefficient β_1 to be significantly positive, and vice versa.¹⁰

当过去的购买量越多，未来的回报越高，而过去的销售量越多，未来的回报越低时，我们预计系数 β_1 将显著为正，反之亦然

The estimation results for equation (2) are reported in Table II, which displays distinctive predictive patterns across different groups of retail investors. For the smallest retail investor group, RT1, the coefficient on the retail order flow variable is -0.0093, with a significant t -statistic of -17.36. The negative coefficient indicates that if retail investors RT1 buy more than they sell on a given day, the next-day return on that stock is significantly negative. To understand the economic magnitude of the coefficients, we report the interquartile range for $OibRT1$ at the bottom of the table. Multiplying the interquartile range, 0.2222, by the regression coefficient of -0.0093, generates an interquartile daily return difference of -21 basis points (more than 50% annualized!). The predictive patterns are qualitatively similar for retail investors in groups RT2-RT4. All coefficients are negative and statistically significant, and the daily interquartile return differences are -17, -11, and -2 basis points for RT2, RT3, and RT4, respectively. That is to say, the first four groups of investors trade in the incorrect direction versus future price movements: the stocks they buy have lower returns, and the stocks they sell have higher returns. It is possible that these retail investors are short of finance literacy, or their trading displays behavioral biases, which lead to the wrong return predictions. Interestingly, when we move from smaller account sizes to larger ones, the negative coefficients become smaller, indicating that larger retail investors trade more correctly than smaller ones. Indeed, for the largest retail investors, RT5, the coefficient on the past day order imbalance is 0.0012, which is positive and significant with a t -statistic of 10.48. The interquartile

方程式(2)的估计结果见表二，表二显示了不同散户投资者群体的独特预测模式。对于最小的零售投资者群体 RT1，零售订单流量变量的系数为 -0.0093，显著 t 统计量为 -17.36。负系数表明，如果散户投资者 $rt1$ 在某一天买入多于卖出，那么该股票第二天的回报率就是显著的负值。为了理解这些系数的经济规模，我们在表格底部报告了 $oibrt1$ 的四分差范围。将

四分位间距 0.2222 乘以 -0.0093 的回归系数，就得到了 -21 个基点的四分位间日回报差(年化回报率超过 50% !).Rt2-rt4 组的散户投资者的预测模式在质量上是相似的。所有系数均为负值，具有统计学意义，RT2、rt3 和 rt4 的每日四分位间回报差异分别为 -17、-11 和 -2 个基点。也就是说，前四类投资者的交易方向相对于未来的价格走势而言是错误的：他们购买的股票回报率较低，而他们卖出的股票回报率较高。这些散户投资者可能缺乏金融知识，或者他们的交易表现出行为偏差，从而导致错误的回报预测。有趣的是，当我们从较小的账户转向较大的账户时，负系数会变小，表明较大的散户投资者的交易比较正确。事实上，对于最大的一般投资者(RT5)而言，过去一天订单失衡系数为 0.0012，这是正数，而且有显著意义，t 统计值为 10.48。四分位数

¹⁰As an alternative to the Fama-Macbeth regression, we also adopt the portfolio sorting approach, using order imbalance measures from different groups as sorting variables. Results are similar to the results of the Fama-Macbeth regression, and are discussed in robustness check in Section IV.D.

作为 Fama-Macbeth 回归的替代方法，我们还采用了投资组合排序方法，使用来自不同组的顺序不平衡度量作为排序变量。结果与 Fama-Macbeth 回归的结果类似，并在第 IV.D 节的稳健性检查中进行了讨论。

daily return difference is five basis points per day (over 12% per year). It appears that the trading of the largest retail investors predicts the cross-section of future stock price movements in the correct direction.

每日回报差额为每日 5 个基点(每年超过 12%)。似乎最大的散户投资者的交易预测了未来股票价格走势的正确方向。

[Insert Table II about here]
[在这里插入表 2]

For comparison, the coefficient on the previous day order imbalance is 0.0016 for institutions, with a *t*-statistic of 18.06. That is to say, institutional order flows predict future stock price movements in the correct direction, and the interquartile return difference is 10 basis points per day (over 25% per year), which is approximately twice the magnitude of the RT5 estimate. This finding is consistent with many previous studies showing that institutional investors are generally more informed than retail investors.¹¹

作为比较，机构的前一天订单失衡系数为 0.0016，*t* 统计为 18.06。也就是说，机构订单流量预测未来股票价格走势的方向是正确的，四分位间回报率差异为每天 10 个基点(每年超过 25%)，大约是 rt5 估计值的两倍。这一发现与之前许多研究结果一致，表明机构投资者通常比散户投资者更有见识

For the control variables, the coefficients on the previous day return have mixed signs, while the coefficients on the previous week and previous month returns are all negative and significant, indicating strong reversals over the weekly and monthly horizons. Size is mostly insignificant, whereas the earnings-to-price variables are always positive and significant, indicating a strong value effect. The coefficients on turnover are always negative and significant, suggesting that higher turnover leads to lower returns in the future. These findings are consistent with those of previous studies on the Chinese stock market, such as Liu et al. (2019). These

results also confirm that the predictive power of the various order flow variables for future stock returns is not a manifestation of the size, value, liquidity, or momentum/reversal effect.

对于控制变量，前一天回报的系数有混合符号，而前一周和前一个月回报的系数都是负的和显著的，表明在周和月期间有强烈的反转。大小通常是微不足道的，而盈利-价格变量总是正的和显著的，表明一个强大的价值效应。营业额的系数总是显著的负值，这表明更高的营业额将导致未来更低的回报。这些发现与之前对中国股市的研究结果一致，如 Liu 等(2019)。这些结果也证实了各种订单流量变量对未来股票收益的预测能力不是大小、价值、流动性或动量/反转效应的表现。

B. Predictive Patterns with Different Counter-parties **不同交易对手的预测模式**

¹¹Readers might wonder whether the RT5 are corporate insiders, which allow them to have inside information and positive prediction for future returns. China has strict rules on the windows during which insiders can trade related stocks. For instance, they cannot trade before significant corporate announcements, such as earnings announcements. Later results show that RT5 trade actively around earnings announcements, which support the notion that they are unlikely to be corporate insiders.

读者可能想知道 rt5 是否是企业内部人士，这使他们有内部信息和对未来回报的积极预测。中国对内部人士交易相关股票的窗口有严格的规定。例如，他们不能在重要的公司公告(如收益公告)之前进行交易。后来的研究结果表明，rt5 在收益公告周围交易活跃，这支持了他们不太可能是公司内部人士的观点。

Trading dynamics among counterparties is an interesting and important concept that helps to understand how different groups of investors interact with each other. However, most existing studies cannot offer much insight into this because it is difficult to pin down each trade's counterparties, due to data limitation.¹² We are fortunate to have the whole trading history from the exchange, which makes it possible for us to identify the counterparties' group for each trade. Therefore, in this subsection, we closely examine the trading dynamics among different groups of investors, and whether different trading pairs are associated with different return predictive patterns.

交易对手之间的交易动态是一个有趣而重要的概念，它有助于理解不同的投资者群体之间是如何相互作用的。然而，由于数据的限制，很难确定每笔交易的交易对手，因此现有的大多数研究无法对此提供更多的见解。¹² 我们很幸运地掌握了该交易所的整个交易历史，这使我们能够确定每笔交易的交易对手群体。因此，在本小节中，我们仔细研究了不同投资者群体之间的交易动态，以及不同的交易对是否与不同的收益预测模式相关联。

We group trades by counterparties from the buy and sell sides of each trade in three steps. First, because corporations rarely trade and account for less than 2% of daily trading volume, we exclude trades with corporations. Second, we regroup the remainder of the six groups into three larger groups to reduce overall dimensions: 1) RT1 to RT4 are bundled together as one group because their order imbalances are positively correlated, and they share similar predictive patterns; and 2) RT5 and INST are considered as two separate groups because their order flows are negatively correlated with those from smaller retail investors, and the order flows from RT5 and INST are also negatively correlated. For the final step, with three groups of investors and two sides of trades (buy or sell), we divide all the remaining trades into six bins: BBS, BSB, BSS, SBB, SBS, and SSB. The first letter indicates the trade direction of RT1-RT4, the second indicates the trade direction of RT5,

and the third indicates INST. For instance, “BBS” means RT1-RT4 “buy”, RT5 “buy” and INST “sell”. Table III Panel A presents the proportion of each type of trade. The highest

我们将交易对手从每笔交易的买卖双方分为三个步骤进行分组。首先，由于公司很少交易，占日交易量的 2% 以下，我们排除了与公司的交易。其次，我们将六个组的其余部分重新分组为三个较大的组，以减少总体维度：1) rt1 到 rt4 被捆绑在一起作为一个组，因为它们的订单不平衡是正相关的，它们共享相似的预测模式；2) rt5 和 INST 被视为两个独立的组，因为它们的订单流量与来自小型散户投资者的订单流量是负相关的，并且 rt5 和 INST 的订单流量也是负相关的。最后一步，由三组投资者和交易双方(买入或卖出)组成，我们将所有剩余的交易划分为六个箱子：BBS、BSB、BSS、SBB、SBS 和 SSB。第一个字母表示 rt1-rt4 的交易方向，第二个字母表示 rt5 的交易方向，第三个字母表示 INST。例如，“BBS”表示 RT1-RT4“买入”，RT5“买入”和 INST“卖出”。表三面板 a 显示了每种交易类型的比例。最高

¹²Exceptions include Boehmer, Sang, and Zhang (2021), which examines the trading patterns of retail investors following insider trading using retail trading trades identified from TAQ and Thomson Reuters Insiders data; and Van Kervel and Menkveld (2019), which examines the high-frequency trading around large institutional orders. Unlike the rest of the studies, these two papers can identify the counterparties of the trades in their sample, but their samples only cover limited subsets of the market.

例外包括 Boehmer, Sang 和 Zhang (2021)，他们利用 TAQ 和汤森路透(Thomson Reuters)知情人数据中确定的零售交易来研究内幕交易后散户投资者的交易模式；Van Kervel 和 Menkveld (2019)，他们研究围绕大型机构指令的高频交易。与其他研究不同的是，这两篇论文能够在样本中识别交易对手，但是他们的样本仅仅涵盖了市场中有限的子集。

proportions are “BSS” and “SBB”, both above 20%, where RT5 is on the same side as institutions. The lowest proportions are for “BSB” and “SBS”, both lower than 13%, indicating that RT1-RT4 are less likely to be on the same side as institutions.

比例为“ BSS”和“ SBB”，均超过 20%，其中 rt5 与机构处于同一边。最低的比例是“ BSB”和“ SBS”，均低于 13%，表明 rt1-rt4 不太可能与机构处于同一边。

To address whether the return predictive patterns change when the counterparties are different, we modify equation (2) to create equation (3), as follows:

为了解决当交易对手不同时，收益预测模式是否会发生变化，我们修改方程(2)以创建方程(3)，如下所示:

$$r_{g,t} = \alpha_0 + \sum_{k=1}^6 \alpha_k I_{k,t} + \epsilon_{g,t}$$

$$r_{g,t} = \alpha_0 + \sum_{k=1}^6 \alpha_k I_{k,t} + \epsilon_{g,t}$$

$$I_{k,t} = \begin{cases} 1 & \text{if } r_{g,t} \in \text{bin } k \\ 0 & \text{otherwise} \end{cases}$$

$$I_{k,t} = \begin{cases} 1 & \text{if } r_{g,t} \in \text{bin } k \\ 0 & \text{otherwise} \end{cases}$$

$$I_{k,t} = \begin{cases} 1 & \text{if } r_{g,t} \in \text{bin } k \\ 0 & \text{otherwise} \end{cases}$$

$$I_{k,t} = \begin{cases} 1 & \text{if } r_{g,t} \in \text{bin } k \\ 0 & \text{otherwise} \end{cases}$$

According to the six bins, we define the corresponding indicator variable $I_{k,t}$, which takes the value of one when the trade belongs to the k -th bin, and zero otherwise. We estimate equation (3) for each of the three investor groups, RT1-RT4, RT5, and INST, using the Fama-MacBeth regression. The first stage coefficient α_0 , captures the predictive power for the k -th combination on day d for investor group G , and the second stage coefficient α_k , (the mean of α_k), captures the average predictive power for the k -th combination on day d for investor group G .

当交易属于 k -th bin 时，值为 1，否则为 0。我们使用 Fama-MacBeth 回归估计三个投资者群体，RT1-RT4，rt5 和 INST 的方程式(3)。第一阶段系数 α_0 表示投资者群体 g 在 d 天的 k -th 组合的预测能力，第二阶段系数 α_k 表示投资者群体 g 在 d 天的 k -th 组合的平均预测能力

the first stage time-series of coefficient) measures the average predictive power for the k -th combination for group G .

第一阶段时间序列的系数)衡量的平均预测功率的 k 组合为 g 组。

[Insert Table III about here]
[在这里插入表 III]

Panel B of Table III reports the estimation results. Our prior is that given that the smaller investors, RT1-RT4, predict returns negatively, they might be at a disadvantage while trading against the large retail investors and the institutional investors. We focus on the order flows from RT1-RT4 in the first specification. The coefficients on the order flows from RT1-RT4 are consistently negative, implying that small retail investors’ negative predictive power is prevalent and does not depend on

counterparties. We notice that the coefficient magnitudes are relatively large when RT1-RT4 are on the opposite side of both RT5 (BSS) and INST (SBB), which supports our prior that RT1-RT4 might be at a disadvantage when trading against both RT5 and INST. In specification II, we focus on the predictive pattern of RT5 with different counterparties. The

表三 b 小组报告了估计结果。我们的先验是，鉴于较小的投资者，RT1-RT4，预测收益为负，他们可能处于不利地位，在交易对大型散户投资者和机构投资者。在第一个规范中，我们关注 rt1-rt4 的订单流。Rt1-rt4 订单流的系数一直为负值，说明小散户投资者的负预测能力普遍存在，且不依赖于交易对手。我们注意到 rt1-rt4 在 RT5(BSS)和 INST (SBB)的对立面时，系数幅度相对较大，这支持了 rt1-rt4 在 rt5 和 INST 交易时可能处于不利地位的观点。在规范 II 中，我们关注 rt5 与不同对手的预测模式。他说

coefficients are mostly positive, consistent with RT5's positive predictive power for returns in general. Interestingly, the coefficients turn to negative for the cases of BBS and SSB, when RT5 are on the same side of RT1-RT4, but on the opposite side of INST. This suggests that when RT5 sides with smaller retail investors, and trades against INST, their prediction becomes erroneous. Similar patterns also apply to INST in regression III, which has positive coefficients when INST trades on the same side of RT5, but the coefficients become negative when INST trades on the opposite side of RT5. That is, when RT5 and INST agree with each other, and trade against the smaller retail investors, their prediction for future return is correct, but when RT5/INST agree with the smaller retail investors, but disagree with each other, they have the incorrect signs for predicting future returns.¹³

系数大多是正的，与 rt5 对回报的正预测能力一致。有趣的是，当 rt5 位于 rt1-rt4 的同一侧，而在 INST 的相反侧时，BBS 和 SSB 的系数均为负值。这表明，当 rt5 站在较小的散户投资者一边，与 INST 交易时，他们的预测就会出错。类似的模式也适用于回归 III 中的 INST，当 INST 在 rt5 的同一边交易时，系数为正，但当 INST 在 rt5 的另一边交易时，系数为负。也就是说，当 rt5 和 INST 相互认同，并且与较小的散户投资者进行交易时，他们对未来回报的预测是正确的，但当 RT5/INST 与较小的散户投资者认同，但彼此不同意时，他们对未来回报的预测就出现了错误的迹象

C. Predictive Patterns with Different Holding Horizons **不同持股视野下的预测模式**

Another key variable for investment is the holding horizon. The statistics in Table I show that Chinese retail investors have high turnover ratios, which lead to short holding horizons. Does the predictive power of order flows change when holding horizons vary? Conventional wisdom suggests that shorter holding horizons are likely to be associated with excessive trading, and possibly poorer returns. Due to data limitation, it is difficult for most existing studies to

directly verify how holding horizons are associated with predictive power for a large sample. Taking advantage of our data, we compute holding horizons for each investor group, and examine how holding horizons are related to predictive patterns of retail order flow.

投资的另一个关键变量是持有期限。表 1 中的统计数据显示，中国散户投资者拥有较高的周转率，这导致了短期持股。当持有期限变化时，订单流的预测能力会发生变化吗？传统观点认为，持仓时间较短可能与过度交易有关，也可能与回报率较低有关。由于数据的限制，现有的大多数研究很难直接验证持有期限与大样本预测能力之间的关系。利用我们的数据，我们计算每个投资者群体的持有期限，并研究持有期限与零售订单流的预测模式之间的关系。

The first step is to find a measure to capture holding horizons. Direct and accurate holding horizon calculation is quite challenging, as there are partial orders, cross trading, and we need to make assumptions regarding the in and out orders. To overcome this, we borrow the simple and

第一步是找到一种方法来捕捉持有期限。直接和准确的持有期限计算是相当具有挑战性的，因为有部分订单，交叉交易，我们需要对进出订单作出假设。为了克服这个问题，我们借用了简单的

¹³We also form portfolios based on order imbalance and trading counterparties, and present the results in Appendix Table IV Panel A. The results are qualitatively similar to these in Table III. 我们还根据订单不平衡和交易对手形成投资组合，并在附录表 IV 面板 a 中展示结果。结果在质量上与表三中的结果相似。

intuitive days-to-cover-ratio (DTCR) idea from the short selling literature, which measures the number of days it takes for all short shares to be covered, as in Diether, Lee, and Werner (2008) and Boehmer and Wu (2013). That is, we define the holding period of average type G investors for

从卖空文献中得到的直观的天数与覆盖率(DTCR)概念，它衡量所有卖空股票被覆盖的天数，如 Diether, Lee, 和 Werner (2008 年)和 Boehmer and Wu (2013 年)。也就是说，我们定义了 g 型投资者的平均持股时间

stock i on day d as

D) 日的股票 i = h_i 或投资者集团持有的股票总数

$$DTCR_i = \frac{h_i}{V_i}$$

, or the total shares held by investor group

G divided by the daily shares traded on stock i for type G investors. For example, if 20% of the total shares are held by RT5, and the RT5 daily trading volume for this stock is 1%, then it takes 20 days for the entire holding stock by group G to be covered, and the days-to-cover-ratio for RT5 would be 20 days. Panel A of Table IV reports the cross-sectional distribution of DTCRs for each investor group. Overall, the DTCRs are similar among RT1-RT4, with means of approximately 50 days, whereas RT5 and INST have longer horizons of approximately 200 days.

G 除以 g 型投资者在股票 i 上交易的每日股票。例如，如果 $rt5$ 持有总股份的 20%，而该股票的 $rt5$ 日交易量为 1%，那么按 g 组分列的整个持股需要 20 天才能被覆盖，而 $rt5$ 的日覆盖率为 20 天。表四的面板 a 报告了每个投资者群体的 dcr 的横截面分布。总体而言， $rt1$ - $rt4$ 中的 dcr 相似，平均约为 50 天，而 $rt5$ 和 $INST$ 的视野更长，约为 200 天。

To examine how DTCR, a rough measure for holding horizons, is associated with the return predictive pattern, we simplify our analysis by selecting three benchmark horizons: 10, 20, and 60 days, which roughly correspond to the 25th, 50th and 75th DTCRs, across different groups. Next, we separate all stock-day-investor group observations, depending on the DTCR calculated from the previous day, into four bins: (0,10], (10,20], (20,60], and above 60 days. We modify equation (2) to create equation (4), which incorporates the differences in DTCR as follows:

$$\begin{aligned} \gamma_{\alpha} &= 0, +(\sum_{\alpha=1}^n 1, \gamma_{\alpha}, \gamma_{\alpha}) \quad \gamma_{\alpha} = -1, +2', \gamma_{\alpha} = -1 + 3, \gamma_{\alpha} \\ \gamma_{\alpha} &= 0, +(\sum_{\alpha=1}^n 1, \gamma_{\alpha}), -1, +2', -1 + 3, \gamma_{\alpha} \end{aligned} \quad \begin{aligned} (4) \\ (4) \end{aligned}$$

zero otherwise. If conventional wisdom is correct, shorter holding horizons may be associated with poorer predictive power. Since the DTCR measure is typically correlated with trading volumes, which can have predictive power for returns by itself, as in Gervais, Kaniel, and Mingelgrin (2001), we include a volume variable as an additional control variable. In particular, we follow Gervais et al. (2001) and include a dummy variable equal to one if the stock has been included in the Dow Jones Industrial Average index. We also include a dummy variable equal to one if the stock is in the S&P 500 index. Finally, we include a dummy variable equal to one if the stock is in the Russell 2000 index. All dummy variables are defined at the time of the first trade.

除此之外一无所获。如果传统观点是正确的，那么较短的持有视野可能与较差的预测能力有关。由于 DTCR 度量通常与交易量相关，而交易量本身就具有预测回报的能力，如在 Gervais、Kaniel 和 Mingelgrin (2001)中，因此我们将一个交易量变量作为额外的控制变量。特别是，我们遵循 Gervais 等

al. (2001) and construct the relative trading volume (*Rvol*) measure, which is the trading volume on the stock-day divided by the average daily stock trading volume from previous 49 days.¹⁴

Other control variables are the same as those in equation (2).

构造了相对交易量度量，即股票当日的交易量除以前 49 天的平均日交易量。¹⁴ 其他控制变量与方程(2)中的控制变量相同。

[Insert Table IV about here]

[在这里插入表 IV]

The estimates of β_1 are presented in Panel B of Table IV. Take RT1 as an example. For
估计数为 β_1 ，见表四 b 小组。以 rt1 为例。因为

the short horizon of less than 10 days, the coefficient is -0.0171 and highly significant. When the holding horizon become longer, the coefficient slightly decreases in magnitude to -0.0086 for 10 to 20 days, -0.0064 for 20 to 60 days, and -0.0074 for longer than 60 days. This decay shows that mistakes in trading direction of the smallest retail investors slowly diminish when holding horizons become longer. The changes are more dramatic for RT2 to RT4, which begin with negative and significant coefficients for shorter horizons, and eventually become mostly insignificant for horizons longer than 60 days. These findings are consistent with the conventional wisdom that shorter holding horizons are likely associated with poorer returns, while longer horizons improve performances. Interestingly, for RT5, the patterns are different. For horizons shorter than 10 days, the $c1$ coefficient is 0.0056 and highly significant, but the $c1$ coefficient quickly reduces to 0.0015 (still highly significant) for holding horizons of 10 to 20 days, and eventually becomes insignificant for holding horizons longer than 60 days. One possibility for the decreasing coefficients and statistical significance is that RT5 might trade on time-sensitive information, which is gradually incorporated into price, leading to the pattern of strong predictive power of shorter holding horizons than longer holding horizons.

短于 10 天的时间范围内，系数为 -0.0171，且非常显著。当保持时间变长时，系数在 10-20 天内略有减小，为 -0.0086,20-60 天为 -0.0064,60 天以上为 -0.0074。这种衰减表明，当持有期限变长时，最小散户投资者在交易方向上的错误会慢慢减少。Rt2 到 rt4 的变化更

为显著，在较短的视野中以负的和显著的系数开始，在超过 60 天的视野中最终变得微不足道。这些发现与传统观点一致，即持股时间越短，回报越低，而持股时间越长，表现越好。有趣的是，对于 RT5，模式是不同的。对于短于 10 天的视野， $c1$ 系数为 0.0056，具有高度显著性，但对于持续 10 至 20 天的视野， $c1$ 系数很快降至 0.0015(仍具有高度显著性)，而对于持续 60 天以上的视野， $c1$ 系数最终变得微不足道。系数和统计显著性下降的一种可能性是， $rt5$ 可能交易对时间敏感的信息，这些信息逐渐被纳入价格，导致持有期限短于持有期限长的模式具有强大的预测能力。

The pattern for INST is somewhat similar to that for RT5, whereas the rate of reduction for the coefficients is much slower. In fact, for horizons shorter than 10 days, the $c1$ coefficient is

INST 的模式与 $rt5$ 的模式有些相似，而系数的降低速率要慢得多。事实上，对于短于 10 天的视野， $c1$ 系数是

¹⁴We thank our referee for this suggestion.
我们感谢我们的裁判提出这个建议。

0.0017, and it slowly decreases to 0.0008 for holding horizons longer than 60 days. In this case, it is possible that the INST's order flow contains some time-sensitive information, which explains the reduction; it is also possible that the INST order flow contains information related to firm fundamentals, which takes longer for the information to be revealed and thus, *slow* decay. Of course, most of these coefficients are positive and significant, indicating that, even for longer holding horizons, the order flows from RT5 and INST still contain price-relevant information in the right direction.¹⁵

0.0017，如果持续时间超过 60 天，它会慢慢减少到 0.0008。在这种情况下，INST 的订单流可能包含一些对时间敏感的信息，这就解释了减少的原因；也可能 INST 的订单流包含与公司基本面有关的信息，这就需要更长的时间来显示信息，因此，缓慢衰减。当然，这些系数中的大多数都是正值和显著的，这表明，即使持有时间较长，来自 rt5 和 INST 的订单流仍然包含正确方向的价格相关信息

D. Predicting Long Term Stock Returns Using Retail Order Flows **利用零售订单流预测长期股票收益**

Previous exercises focus on next-day return prediction, and it is natural to ask whether the predictive patterns continue for longer terms. If the predictive pattern quickly vanishes or reverses, the return predictability may be driven by short-term noise. If the predictive pattern persists over longer horizons, the return predictability is more likely to be linked to firm fundamentals or persistent biases. Therefore, we extend the Fama-MacBeth specification in equation (2) to create equation (5), where longer horizons are up to 12 weeks:

以前的练习集中在第二天的回报预测上，很自然的，我们会问预测模式是否会持续更长的时间。如果预测模式迅速消失或逆转，回报的可预测性可能是由短期噪音驱动的。如果预测模式在更长的时间范围内持续存在，那么回报的可预测性更有可能与坚实的基本

面或持续的偏差有关。因此，我们将公式(2)中的 Fama-MacBeth 规范扩展为公式(5)，其中较长的视野长达 12 周：

$$\begin{aligned} &= 0 + 1 + 2' + 4, \\ &= 0, + 1 + 2' + 4, \end{aligned} \quad (5)$$

That is, we use the previous day order imbalance from group G ,

be more specific,
end of week w , where $w=1, \dots, 12$. For instance, when $w=1$,
 $w=12$.

,-1, to predict weekly returns over the next w -th week. To
, is calculated as a 5-day return from the beginning to the
, is the cumulative return over days d to $d+4$, and when
, is the cumulative return over day

也就是说，我们使用来自 g 组的前一天订单不平衡， -1 ，来预测下一周的周收益。更具体地说，计算为从周初到周末的 5 天回报，其中 $w = 1, \dots, 12$ 。例如，当 $w = 1$ 时，是 d 到 $d + 4$ 的累积收益，当 $w = 12$ 时，是一天的累积收益

$d+55$ to $d+59$. If order imbalances have only short-lived predictive power for future returns, we should observe the coefficient $d1$ quickly reverses. Alternatively, if the specified retail order

$D + 55$ 到 $d + 59$ 。如果订单失衡对未来收益只有短暂的预测能力，我们应该观察到系数

$d1$ 迅速反转。或者，如果指定的零售订单

¹⁵As an alternative to the Fama-MacBeth regression, we construct double sort portfolios based on order imbalance and holding horizons. The results are reported in Appendix Table IV Panel B, and are qualitatively similar to those in Table IV. 作为 Fama-MacBeth 回归的替代方法，我们构造了基于订单不平衡和持有视野的双排序投资组合。结果在附录表 IV b 中报告，并且在质量上与表 IV 中的结果相似。

imbalance has a longer predictive power, the coefficient $d1$ may slowly decrease, rather than quickly reverse.

不平衡具有更长的预测能力，系数 $d1$ 可能会慢慢下降，而不是迅速逆转。

[Insert Table V about here]
[在这里插入表 v]

We present the estimates of coefficient $d1$ in equation (5) in Table V. For the smallest retail investor RT1, the coefficients on $OibRT1$ are negative and monotonically increase from -0.0226 at week one to -0.0005 at week 12, while the coefficients are statistically significant until week 9. Reverse patterns are not observed, which implies that the predictive power is long lasting. Similar patterns are observed for $OibRT2$ and $OibRT3$. For $OibRT4$, the coefficient is -0.0019 and highly significant at week 1, but quickly reverses and becomes insignificant, indicating that its predictive power for future returns might be temporary. The positive predictive power of $OibRT5$ and $OibINST$ also significantly persist for approximately 8 to 9 weeks, and there are no significant reversals. The general persistence of the positive predictive pattern for RT5 and INST indicates that the predictive power is likely rooted in information related to fundamentals, or against persistent noise/behavioral biases.¹⁶

对于最小的零售投资者 RT1, $oibrt1$ 上的系数是负的，从第 1 周的 -0.0226 单调增加到第 12 周的 -0.0005，而这些系数在第 9 周之前是有统计学意义的。没有观察到反向模式，这意味着预测能力是持久的。 $Oibrt2$ 和 $oibrt3$ 观察到类似的模式。对 $oibrt4$ 而言，该系数为 -0.0019，在第 1 周时非常显著，但很快就会反转，变得微不足道，这表明它对未来回报的预测能力可能是暂时的。 $Oibrt5$ 和 $OibINST$ 的正向预测能力也显著持续了大约 8 到 9 周，并且没有显著的逆转。 $Rt5$ 和 $INST$ 的正预测模式的普遍持续性表明，预测能力可能植根于与基本面有关的信息，或针对持续的噪音/行为偏差

III. What Drives the Order Imbalance Predictive Power for Future Returns? 是什么驱动了订单不平衡对未来回报的预测能力？

A. Alternative Hypotheses for Explaining Retail Order Flow and Its Return Predictive Power **解释零售订单流及其回报预测能力的另类假设**

Given the large differences in the predictive power of future returns for different investor groups' order flows, it is important to understand the driving forces behind these order flows. 考虑到不同投资者对未来收益预测能力的巨大差异

Previous literature provides several hypotheses for explaining investor order flows, which may help to explain the heterogeneous predictive patterns of different investor groups for future returns. Below we outline four main hypotheses, and these hypotheses are not mutually exclusive.

群体的秩序流动，了解这些秩序流动背后的驱动力很重要。以前的文献提供了几个解释投资者指令流的假设，这可能有助于解释不同投资者群体对未来回报的异质预测模式。下面我们概述四个主要的假设，这些假设并不是相互排斥的。

¹⁶ We also consider an alternative specification for the long-term predictability. To match the weekly returns on the left-hand side, we use the previous week order imbalance, -1 , to predict future weekly returns. The results are reported in Appendix Table V are similar to those in Table V. 我们还考虑了长期可预测性的替代规范。为了匹配左侧的周收益，我们使用前一周的订单不平衡， -1 ，来预测未来的周收益。附录表 v 中报告的结果与表 v 中的结果类似。

First, Chordia and Subrahmanyam (2004) state that order flows tend to be persistent, and that persistent buying/selling pressure could directly lead to the predictability of future returns. Here we adopt the previous day order imbalance measure, OIB_{t-1} , as the proxy for order

首先, Chordia 和 Subrahmanyam (2004)指出, 订单流动趋于持续, 持续的买卖压力可能直接导致未来回报的可预测性。在这里, 我们采用前一天的订单不平衡度量, OIB_{t-1} , 作为订单的代理

persistence.

坚持不懈。

Second, Kaniel et al. (2008) argue that retail traders in the U.S. are mostly contrarian, providing liquidity to the market, and afterwards receiving positive premium for liquidity provision. Following this logic, if the retail trades are momentum trades that demand liquidity, it is then possible that momentum trades will have a negative relation with future returns; and if the retail trades are contrarian and provide liquidity, these contrarian trades will have a positive relation with future returns. For this hypothesis, we choose returns from the previous day, week, and month as proxies for momentum or contrarian trading.

第二, Kaniel 等人(2008)认为, 美国的零售交易者大多是反向投资者, 他们为市场提供流动性, 然后通过提供流动性获得正溢价。按照这个逻辑, 如果零售交易是需要流动性的动量交易, 那么动量交易可能与未来回报呈负相关; 如果零售交易是反向交易并提供流动性, 那么这些反向交易将与未来回报呈正相关。对于这个假设, 我们选择前一天、前一周和前一个月的收益作为动量或反向交易的替代指标。

Third, Liu et al. (2021) connect retail trading motives to behavioral biases and find that overconfidence and gambling preferences are the two dominant behavioral biases that affect trade of Chinese retail investors. For the overconfidence measure, we follow Barber et al. (2009) and Liu et al. (2021), and proxy it with the average of the daily investor group's stock turnover from previous 20 days, ST_{t-1} .¹⁷ For gambling preferences at the stock level, GL_{t-1} ,

第三, 刘等人(2021)将零售交易动机与行为偏好联系起来, 发现过度自信和赌博偏好是影响中国零售投资者交易的两个主要行为偏好。对于过度自信的度量, 我们跟随 Barber 等人(2009)和 Liu 等人(2021), 并用投资者群体过去 20 天的日均股票交易量来代替它, ST_{t-1} .¹⁷ 对于股票水平的赌博偏好, GL_{t-1} ,

we follow Bali et al. (2011) and compute the maximum daily returns from the previous 20 days.¹⁸
我们跟随 Bali 等人(2011), 计算前 20 天的最大日回报率

¹⁷For the overconfidence proxy, Barber and Odean (2000) use each household's portfolio's turnover. Our data has a different structure and cannot be used to construct household level proxies. Considering our study focuses on investor groups rather than individual households, we carry the spirit from the previous literature and use the stock-level turnover from each investor group.

对于过度自信的代理，Barber 和 Odean (2000)使用每个家庭的投资组合的周转率。我们的数据具有不同的结构，不能用于构建家庭级别的代理。考虑到我们的研究集中在投资者群体而不是个人家庭，我们借鉴了以往文献的精神，并使用了每个投资者群体的股票水平成交量。

¹⁸An alternative measure for gambling preference proxy is introduced in Liu et al. (2021), which relies on events when the stock return hits a 10% price limit. However, the 10% price limit only accounts for 0.07% of the total sample, and is not suitable for our purposes using on daily*stock returns. Thus, we choose the maximum daily return as our main gambling measure. We also consider other alternative proxies for gambling preferences, such as idiosyncratic volatility and skewness. These proxies deliver similar results to those using maximum daily returns, and are available upon request.

Liu 等人(2021)提出了一种赌博偏好代理的替代方法，该方法依赖于当股票收益达到 10% 的价格上限时发生的事件。然而，10% 的限价只占样本总数的 0.07%，并不适用于我们每日 * 股票回报的用途。因此，我们选择最大日回报率作为我们主要的赌博指标。我们还考虑了其他代表赌博偏好的替代指标，比如特殊波动性和偏斜性。这些代理提供了类似的结果，那些使用最大日收益，并可根据要求提供。

Finally, Kelley and Tetlock (2013) find that retail investors, especially the aggressive ones, may have valuable information about fundamental firm news, and thus, their trading could correctly predict the direction of future returns. We measure firm-level fundamental news by the cumulative abnormal returns (*CAR*) over the earnings announcement period. Unlike the proxies for order persistence, liquidity provision, and behavioral biases, which can be computed for each stock on each day, the news proxies are only available on earnings news days, which account for 1.58% of stock days, rendering our two-stage estimation (to be introduced in Section III.B.) imprecise. To cope with this missing data issue for the news hypothesis, we first consider the order persistence, liquidity provision, and behavioral bias hypotheses in Section III.B, and focus on the news hypothesis using an event-day approach in Section III.C.

最后，Kelley 和 Tetlock (2013)发现，散户投资者，尤其是激进的投资者，可能掌握有关基本面公司新闻的有价值信息，因此，他们的交易可以正确地预测未来回报的方向。我们衡量企业层面的基本面消息的累积超常收益(*CAR*)在收益公布期间。新闻代理不同于订单持续性、流动性供应和行为偏差的代理，后者可以在每天为每只股票计算，新闻代理只能在盈利新闻日使用，这占股票日的 1.58%，因此我们的估计分为两个阶段(将在第三节 b 部分介绍)不精确。为了解决新闻假说中缺失的数据问题，我们首先考虑了第三节 b 中的订单持续性、流动性提供和行为偏差假说，并在第三节 c 中采用事件日方法重点研究了新闻假说。

B. A Two-Stage Decomposition for Order Flows' Return Predictive Power **订单流返回预测能力的两阶段分解**

To find out whether the above hypotheses explain the trading behavior of different retail investor groups and their predictive power for future stock returns, we adopt a two-stage decomposition method as in Boehmer et al. (2021).¹⁹ For the first stage, we use the above hypotheses to explain the retail flow measures to find out which ones are important drivers for the

order flows. This step also helps to decompose the retail order flows into hypothesis-implied components for each hypothesis. For the second stage, we investigate which of the hypothesis-implied components contributes to the predictive pattern of different investor order flow measures.

为了验证上述假设是否能够解释不同散户投资者群体的交易行为及其对未来股票收益的预测能力，我们采用了 Boehmer 等人(2021)提出的两阶段分解方法。在第一阶段，我们使用上述假设来解释零售流量度量，以找出哪些是订单流量的重要驱动因素。这一步也有助于将零售订单流分解为每个假设的假设隐含组件。在第二阶段，我们调查哪个假设隐含的成分有助于不同的投资者订单流量度量的预测模式。

Table VI Panel A reports the first-stage estimation results. In the first row, the coefficients of the lagged order flow variables are always positive and significant, indicating that order persistence is an important driver of order flow. For the next three rows, order flows are connected with returns from the previous day, week, and month. The order imbalances of RT1, RT2, and RT3

表六 a 小组报告第一阶段的估计结果。在第一行中，滞后订单流变量的系数总是正的且有意义，表明订单持续性是订单流的一个重要驱动因素。在接下来的三行中，订单流与前一天、前一周和前一个月的返回相关联。RT1、rt2 和 rt3 的顺序不平衡

¹⁹A step-by-step description of the two-stage decomposition is provided in Appendix A.
附录 a 提供了两阶段分解的逐步描述。

load positively and significantly on the previous day's return, indicating that these investors buy more if the previous day's return is positive, and sell more if the previous day's return is negative. This corresponds to a daily momentum trading strategy, which demands immediate liquidity. For larger retail investors in RT4 and RT5, order imbalances load negatively and significantly on returns from the previous day, indicating that they are contrarian investors who buy low, sell high, and possibly provide immediate liquidity. If we extend the horizon to the previous one week or one month, the coefficients on all returns are negative and significant, indicating that all retail investors become contrarian and buy losers, and sell winners over the longer term.²⁰

在前一天的回报率上显著增加正值，表明如果前一天的回报率为正值，这些投资者会买入更多，如果前一天的回报率为负值，则会卖出更多。这相当于每日动量交易策略，需要立即的流动性。对于 rt4 和 rt5 的大型散户投资者，指令失衡对前一天的回报造成负面和显著的影响，表明他们是反向投资者，低买高卖，并可能提供即时的流动性。如果我们把期限延长到前一个星期或一个月，那么所有回报的系数都是负的，而且是显著的，这表明所有散户投资者都变成了反向投资者，买进输家，卖出赢家。²⁰

[Insert Table VI about here]
[在这里插入表六]

The next two rows present the results on how behavioral biases are related to order flows. The coefficients on the overconfidence proxy are all positive and significant for RT1-RT4, indicating that overconfidence might be a strong driver of trading by these retail investors. For the largest retail group, RT5, the coefficient becomes -0.0881, with a significant *t*-stat of -8.80. In other words, the largest retail investors' trades are in the opposite direction of the overconfidence proxy. In terms of gambling preference, for RT2-RT4, the coefficients are always positive and significant, indicating that these retail investors prefer to buy stocks with lottery features.²¹ When

接下来的两行显示了行为偏差如何与订单流相关的结果。Rt1-rt4 的过度自信代理系数均为正数，显著正数，显示过度自信可能是这些一般投资者交易的强劲推动因素。对于最大的零售集团，RT5，系数变成 -0.0881，一个重要的 t 统计 -8.80。换句话说，最大的散户投资者的交易与过度自信代理的方向相反。在博彩偏好方面，对 RT2-RT4，系数总是正的和显著的，表明这些散户投资者更倾向于购买彩票特征的股票。²¹ 何时

²⁰Our finding that large retail investors are contrarian and smaller ones are momentum traders over daily horizon differs from some previous studies. For instance, contrarian patterns are documented in Kaniel et al. (2008) using monthly horizons in the U.S., and Barrot et al. (2016) using daily and weekly horizons in France. Using U.S. data, Kelley and Tetlock (2013) and Boehmer et al. (2021) both find that retail trades follow momentum over daily horizons, but are contrarian at weekly horizons. In our setting, we find the trading patterns from investors with smaller account sizes are similar to those in Kelley and Tetlock (2013) and Boehmer et al. (2021), while the investors with the largest account sizes behave similarly to the patterns documented in Kaniel et al. (2008) and Barrot et al. (2016). We also examine whether the momentum/contrarian patterns are potentially related to average holding periods of stock. Interacting the return of the previous day with holding horizons, we find small retail investors' momentum are stronger for short holding periods, while large retail investors' contrarian is similar for both short and long holding periods. The results are available on request.

我们发现，大型散户投资者是反向投资者，小型散户投资者是动量交易者，这与以前的一些研究不同。例如，Kaniel 等人(2008)利用美国的月视野记录了反向投资模式，Barrot 等人(2016)利用法国的日视野和周视野记录了反向投资模式。利用美国的数据，Kelley 和 Tetlock (2013)以及 Boehmer 等人(2021)都发现，零售交易在日线上遵循势头，但在周线上是反向的。在我们的设置中，我们发现账户规模较小的投资者的交易模式类似于 Kelley 和 Tetlock (2013)和 Boehmer 等人(2021)的交易模式，而账户规模最大的投资者的行为类似于 Kaniel 等人(2008)和 Barrot 等人(2016)的交易模式。我们还研究了动量/反转模式是否可能与股票的平均持有期有关。结果发现，小散户投资者在短期持有时的反转势头较强，而大散户投资者在短期和长期持有时的反转势头相似。结果可以根据要求提供。

²¹The increasing coefficients from RT1 to RT4 should not be interpreted as RT4's gambling preference being stronger than RT1, because these variables are not standardized and cannot be compared directly. If we standardize them and make them comparable across investor groups, RT3 has the strongest gambling preference. The results are available on request.

从 rt1 到 rt4 的增长系数不应该被解释为 rt4 的赌博偏好强于 RT1，因为这些变量没有标准化，不能直接进行比较。如果我们将它们标准化，使它们在投资者群体之间具有可比性，rt3 就具有最强的赌博偏好。结果可以根据要求提供。

we move on to RT5, the coefficient is -0.0671 with a significant t -stat of -2.53, which indicates that the largest retail investors' trades are in the opposite direction of the gambling motive.

我们再来看 RT5，系数是 -0.0671，显著的 t -stat 是 -2.53，这表明最大的散户投资者的交易与赌博动机相反。

We report the second stage decomposition results in Panel B of Table VI. We consider the first retail group, RT1, as an example. The coefficient estimate on *Oib(Persistence)* is -0.0301, with a t -statistic of -16.73, which implies that order persistence significantly and negatively contributes to the predictive power of the RT1 trading flow. The coefficient estimate on *Oib(Liquidity)* is -0.0135, with a t -statistic of -4.48, which suggests that daily momentum trading probably significantly and negatively contributes to the predictive power of the RT1 trading flow. The coefficient of *Oib(Overconf)* is -0.1006, with a t -statistic of -4.00, and the coefficient of *Oib(Gamble)* is -0.2877, with a t -statistic of -4.89. These two significant coefficients imply that the two behavioral biases significantly and negatively contribute to the predictive power of RT1 trading flow. For the *Oib(Other)* component, the coefficient is -0.0086, with a significant t -statistic of -15.54, indicating that there is residual information other than those incorporated in the three hypotheses, which significantly contributes to RT1's negative predictive pattern for future returns. In terms of the economic magnitude, we compute the interquartile range of all five components of the order imbalance measure. For the smallest retail group RT1, if we move from the 25th percentile to the 75th percentile in the distribution, the interquartile differences in future one-day stock returns are -0.1177%, -0.0290%, -0.0337%, -0.0314%, and -0.1778% for *Oib(Persistence)*, *Oib(Liquidity)*, *Oib(Overconf)*, *Oib(Gamble)*, and *Oib(Other)*, respectively. In other words, order persistence, liquidity demand, overconfidence, and gambling preferences all contribute to the negative predictive power of RT1 for next-day returns, with the first term having the largest magnitude. Similar patterns are observed for other smaller retail investor groups RT2-RT4.

我们在表 VI 的面板 b 中报告了第二阶段的分解结果。我们以第一个零售集团 rt1 为例。持续性的系数估计为 -0.0301, t 统计量为 -16.73, 这意味着订单持续性对 rt1 交易流的预测能力有显著的负向影响。流动性系数估计值为 -0.0135, t 统计值为 -4.48, 表明日动量交易可能对 rt1 交易流的预测能力有显著的负向影响。Oib (Overconf)系数为 -0.1006, t 统计量为 -4.00, Oib (Gamble)系数为 -0.2877, t 统计量为 -4.89。这两个重要的系数意味着这两个行为偏差对 rt1 交易流的预测能力有显著的负面影响。对于 Oib (Other)部分, 系数为 -0.0086, 显著 t 统计量为 -15.54, 表明除了这三个假设所包含的信息之外, 还有其他残余信息, 这显著地促成了 rt1 对未来收益的负面预测模式。在经济规模方面, 我们计算了顺序不平衡度量的所有五个组成部分的四分差范围。对于最小的零售集团 RT1, 如果我们从分布的第 25 个百分位移到第 75 个百分位, 未来一天股票收益的四分位差分别是 -0.1177%, -0.0290%, -0.0337%, -0.0314% 和 -0.1778%, 分别为持久性的 Oib, 流动性的 Oib, 赌博的 Oib 和其他的 Oib。换句话说, 订单持续性、流动性需求、过度自信和赌博偏好都有助于 rt1 对次日收益的负预测能力, 其中第一项的预测能力最大。其他较小的散户投资者群体也观察到类似的模式 RT2-RT4。

Regarding the largest retail investors, RT5, the patterns are quite different. In terms of the coefficient estimates, we find that order persistence, overconfidence, gambling preferences, and others are all positive and significant. In terms of economic magnitude, if we move from the 25th percentile to the 75th percentile in the distribution, the interquartile differences in future one-day stock returns are 0.0281%, 0.0097%, 0.0257%, 0.0425%, and 0.0490% for *Oib(Persistence)*, *Oib(Liquidity)*, *Oib(Overconf)*, *Oib(Gamble)*, and *Oib(Other)*, respectively. This indicates that order persistence, liquidity provision, and trading against behavioral biases all contribute to RT5's positive predictive pattern for future returns.

对于最大的散户投资者，RT5，模式是完全不同的。就系数估计而言，我们发现订单持续性、过度自信、赌博偏好等等都是正值和显著的。就经济规模而言，如果我们从分布中的第 25 个百分位数移动到第 75 个百分位数，未来一日股票收益的四分位差分别为 0.0281% 、0.0097% 、0.0257% 、0.0425% 和 0.0490% ，其中 *Oib (Persistence)*、*Oib (Liquidity)*、*Oib (Overconf)*、*Oib (Gamble)*和 *Oib (Other)*分别为 0.0281% 、0.0257% 、0.0425% 和 0.0490% 。这表明，订单持续性，流动性供应，和交易对行为偏差都有助于 rt5 的未来回报的积极预测模式。

Overall, our decomposition exercise shows that a substantial part of the negative predictive power of retail investors with smaller account sizes comes from order persistence, liquidity demand, and behavioral biases, while the positive predictive power of retail investors with larger account balances mostly comes from order persistence and trading against overconfidence and gambling preferences.²² Across all investor groups, the significance and large magnitude of the “other” component indicate that existing hypotheses cannot fully explain trading behaviors and their predictive power for returns. Then, what does “other” stand for? One possibility is information, which we take a close look at in the next subsection.

总的来说，我们的分解研究表明，账户规模较小的散户投资者的负面预测能力很大一部分来自订单持续性、流动性需求和行为偏差，而账户余额较大的散户投资者的正面预测能力大多来自订单持续性和交易对过度自信和赌博偏好的反对。那么，“其他”代表什么呢？一种可能性是信息，我们将在下一小节详细讨论。

C. A Close Look at the Information Channel **密切关注信息渠道**

It is important to understand how various retail investors participate in the information discovery process. As mentioned earlier, the most influential information at the firm level is earnings news, hence we follow Kelley and Tetlock (2013) and measure firm-level information by the cumulative abnormal returns (CAR) over the earnings announcement period. Notice that

了解不同的散户投资者如何参与信息发现过程是很重要的。正如前面提到的，企业层面上最有影响力的信息是盈利新闻，因此我们跟随 Kelley 和 Tetlock (2013)，通过在盈利公告期间的累积异常收益(CAR)来衡量企业层面的信息。注意

²²In Appendix Table VI, we examine the robustness of two-stage decomposition by adding additional two weeks lag of order flow in the first stage, construct the new “persistence” component, and re-estimate the second stage decomposition. The results are robust.

在附录表 VI 中，我们通过在第一阶段增加额外的两周时滞来检验两阶段分解的鲁棒性，构造新的“持久性”组件，并重新估计第二阶段分解。结果是稳健的。

earnings news only happens quarterly rather than daily, so the daily Fama and MacBeth (1973) estimation adopted for the two-stage estimation might not be proper for understanding how Chinese retail investors process information. As an alternative, in this section, we focus on event-days to study this issue. We capture each retail investor group's participation in the information discovery process in three steps.

盈利新闻只发生在季度，而不是每天，因此，两阶段估计采用的每日 Fama 和 MacBeth (1973)估计，可能不适合理解中国散户投资者如何处理信息。作为一种替代方法，在本节中，我们关注事件日来研究这个问题。我们通过三个步骤记录了每个散户投资者群体在信息发现过程中的参与情况。

First, we examine whether different retail investors can predict earnings news the next day. A positive answer indicates that these investors anticipate the information before it becomes public, either because they have access to private information or because they have better skills in prediction. We estimate the following cross-sectional specification for each quarter q :

首先，我们检验不同的散户投资者是否能够预测第二天的收益消息。一个积极的答案表明，这些投资者在信息公开之前就预测到了这些信息，要么是因为他们能够获得私人信息，要么是因为他们在预测方面有更好的技能。我们估计每个季度的截面规格如下：

$$\begin{aligned} & \text{AR}_{i,t} = \alpha_0 + \alpha_1 \text{OI}_{i,t-1} + \alpha_2 \text{OI}_{i,t-2} + \alpha_3 \text{OI}_{i,t-3} + \alpha_4 \text{OI}_{i,t-4} + \alpha_5 \text{OI}_{i,t-5} + \epsilon_{i,t} \\ & \text{AR}_{i,t} = \alpha_0 + \alpha_1 \text{OI}_{i,t-1} + \alpha_2 \text{OI}_{i,t-2} + \alpha_3 \text{OI}_{i,t-3} + \alpha_4 \text{OI}_{i,t-4} + \alpha_5 \text{OI}_{i,t-5} + \epsilon_{i,t} \end{aligned} \quad (6)$$

Assuming that the earnings announcement day is day d , we compute the cumulative returns over day $d-1$ and day d and subtract the market returns over the same period to obtain cumulative abnormal returns for each stock, $\text{AR}_{i,t}$.²³ The main predictive variable on the right-hand side is the order imbalance measure from day $d-2$. Notice that each firm has only one earnings day per quarter, and equation (6) is estimated for each quarter in the cross-section to ensure that we cover all firms in each quarter. Statistical inferences are based on the quarterly time-series of the estimated coefficients, and standard errors are computed using Newey-West

of the equation is the order imbalance measure from day $d-2$. Notice that each firm has only one earnings day per quarter, and equation (6) is estimated for each quarter in the cross-section to ensure that we cover all firms in each quarter. Statistical inferences are based on the quarterly time-series of the estimated coefficients, and standard errors are computed using Newey-West

(1987) method with the optimal bandwidth chosen by using Newey and West (1994) approach. If retail order flows can predict earnings surprises in the right direction, coefficient e_1 should be significantly positive, and vice versa. Panel A of Table VII presents the estimation results. For retail investors RT1-RT3, the coefficients e_1 are -0.0251, -0.0234, and -0.0166, respectively, with highly significant t -statistics. These negative and significant coefficients indicate that these 方程的关键是从第 $d-2$ 天开始的顺序不平衡度量。注意，每个公司每个季度只有一个盈利日，并且在横截面上估计每个季度的方程式(6)，以确保我们涵盖每个季度的所有公司。基于估计系数的季度时间序列进行统计推断，利用 Newey-West (1987)方法计算标准误差，并利用 Newey 和 West (1994)方法选择最佳带宽。如果零售订单流可以正确地预测盈利意外，那么系数 e_1 应该是正的，反之亦然。表 7 的 a 小组显示了估计结果。对于散户投资者 RT1-RT3，系数 e_1 分别为 -0.0251、-0.0234 和 -0.0166，具有非常显著的 t 统计量。这些负的和显着的系数表明，这些

²³ We also examine wider window such as CAR(-1,1) and CAR(-3,3), the results are similar and available on request. 我们还研究了更广泛的窗口，如 CAR (- 1,1)和 CAR (- 3,3)，结果是相似的，可根据要求提供。

investors incorrectly predict earnings surprises. Coefficient $e1$ for RT4 is close to zero and insignificant. In contrast, the coefficient $e1$ for RT5 is 0.0023, positive and statistically significant, implying that these investors can correctly predict future earnings surprises.

投资者错误地预测了意外收益。Rt4 的系数 $e1$ 接近于零，微不足道。相比之下，rt5 的 $e1$ 系数为 0.0023，正值且具有统计学意义，这意味着这些投资者可以正确预测未来的收益意外。

[Insert Table VII about here]
[在这里插入表 7]

connect to Second, we examine whether different retail groups can process contemporaneous public news. Here, the dependent variable is retail order flow, o_{it} , which we
第二，我们研究不同的零售集团是否能够处理同时期的公共新闻。在这里，因变量是零售订单流，我们连接到
contemporaneous earnings news, e_{it} ,
同期收益新闻,-1,

$$\begin{aligned} &= 0 + 1 + 2' + 6. \\ &= 0 + 1 + 2' + 6. \end{aligned} \tag{7}$$

$$\begin{aligned} &, -1, -1, -1, \end{aligned} \tag{7}$$

If a particular type of retail order imbalance can process contemporaneous and public earnings news in the right direction, the associated coefficient $f1$ is significantly positive and vice versa. Panel B of Table VII reports these results. For retail investors RT1-RT4, the coefficients $f1$ are -1.9225, -1.8291, -1.4349, and -0.8781, respectively, with highly significant t -statistics. These negative and significant coefficients indicate that these retail investor groups process contemporaneous public earnings news in the wrong direction. By contrast, the coefficient $f1$ for RT5 is 0.1583, with a t -statistic of 2.12, implying that RT5 might be able to correctly process contemporaneous public earnings news.

如果某种特定类型的零售订单失衡能够正确处理同期和公共收益新闻，则相关系数 $f1$ 为显著正值，反之亦然。表七的 b 小组报告了这些结果。对于散户投资者，rt1-rt4 的系数 $f1$ 分别为 -1.9225、-1.8291、-1.4349 和 -0.8781，具有非常显著的 t 统计量。这些负的和重要的系数表明，这些零售投资者群体处理当时的公共收益新闻的方向是错误的。相比之下，

rt5 的 f1 系数为 0.1583, t 统计量为 2.12, 这意味着 rt5 可能能够正确处理当时的公共收益新闻。

Third, we examine whether the predictive power of retail order flows for future returns improves or deteriorates on event days to understand how much the information hypothesis helps explain the return predictive patterns observed in Section II. Here, we add the event-day dummy and interaction term:

第三, 我们检验零售订单流对未来收益的预测能力是否在事件日提高或恶化, 以了解信息假说在多大程度上有助于解释第二节中观察到的收益预测模式。在这里, 我们添加了事件日假设和交互项:

$$r_{it} = \alpha_0 + (\alpha_1 + \alpha_2 D_{it}) r_{it} + \alpha_3 D_{it} + \alpha_4 D_{it} r_{it} \quad (8)$$

$$r_{it} = \alpha_0 + (\alpha_1 + \alpha_2 D_{it}) r_{it} + \alpha_3 D_{it} + \alpha_4 D_{it} r_{it} \quad (8)$$

The event dummy $\text{event}_{i,t}$, -1 , is equal to one if firm i has earnings news on day $d-1$ and zero otherwise. 如果公司 i 在第 $d-1$ 天和第 0 天都有收益消息, 那么事件假人 -1 等于 1

otherwise. For non-news days, the predictive power of retail trades is measured by the coefficient g_1 ; for news days, the predictive power is measured by (g_1+g_2) . If coefficient g_2 is significantly different from zero, the group of retail investors anticipates future stock returns differently on these news days. In the U.S., firm earnings announcements are chosen by firms and are scattered throughout the year. In China, all firms are required to report their financial statements to regulators before the four preset deadline dates each year. Consequently, firms mostly announce their earnings within a short period before these deadline dates, and there would be zero announcements outside these short periods. To ensure that we have enough observations to estimate the Fama and Macbeth (1973) coefficients in equation (8), we only include days with at least 5% of the total number of firms with earnings announcements, which gives us 68 days, or 8% of the total days in our sample.

否则。对于非新闻日, 零售交易的预测能力用系数 g_1 来衡量; 对于新闻日, 预测能力用 $(g_1 + g_2)$ 来衡量。如果系数 g_2 与 0 有显著差异, 那么散户投资者在这些新闻日对未来股票收益的预期就会有所不同。在美国, 公司的收益公告是由公司选择的, 并且分散在全年。在中国, 所有的公司都被要求在每年四个预先设定的截止日期之前向监管机构报告财务报表。因此, 公司大多在这些截止日期之前的短期内公布他们的收益, 而在这些短期之外没有任何公告。为了确保我们有足够的观察数据来估计方程式(8)中的 Fama 和 Macbeth (1973)系数, 我们只包括至少占公司总数 5% 的发布收益公告的天数, 这给了我们 68 天, 或我们样本中总天数的 8% 。

Table VII and Panel C present the results. We consider the smallest retail investor, RT1, as an example. The coefficient on order imbalance, g_1 , is -0.0079 and statistically significant, indicating that, on average, the trades from RT1 negatively predict future returns. When there is earnings announcement news, the coefficient on the interaction of the event dummy and the order imbalance is -0.0080 and is statistically significant, implying that the negative prediction of RT1 for future stock returns doubles on earnings news days. This is consistent with our earlier finding that smaller retail

investors fail to predict and process earnings news, which leads to more negative predictions of returns on event days. Similar patterns are observed in RT2, RT3, and RT4. For the largest retail investor, RT5, the coefficients g_1 and g_2 are 0.0005 and 0.0014, respectively, both of which are statistically significant. In other words, large retail investors' predictive power for future

表七和 c 小组介绍了结果。我们以最小的散户投资者 rt1 为例。订单失衡系数 g_1 为 -0.0079, 具有统计学意义, 表明 rt1 的交易平均对未来回报具有负面预测作用。当有盈余发布消息时, 事件假人与订单失衡的相互作用系数为 -0.0080, 具有统计学意义, 这意味着 rt1 对未来股票收益的负预测在盈余发布日翻了一番。这与我们早些时候的发现是一致的, 即较小的散户投资者不能预测和处理收益消息, 这导致对事件日的收益预测更为负面。在 RT2, rt3 和 rt4 中也观察到类似的模式。对于最大的散户投资者 RT5, 系数 g_1 和 g_2 分别为 0.0005 和 0.0014, 两者都具有统计学意义。换句话说, 大型散户投资者对未来的预测能力

returns quadruples on earnings news days, possibly because these retail investors can correctly predict and process earnings news, which enhances their ability to predict future stock returns.

这可能是因为这些散户投资者能够正确预测和处理收益信息，从而提高了他们预测未来股票收益的能力。

Overall, our results reveal interesting heterogeneous patterns in how retail investors predict and process public information. Smaller retail investors are unable to predict future news and lack the skills to correctly process public news, while the largest retail investors and institutions can correctly anticipate future earnings news and incorporate contemporaneous news into their trading. The differences in the information-processing abilities of different retail investors clearly contribute to the differences in their predictive power for future returns.²⁴

总的来说，我们的研究结果揭示了散户投资者如何预测和处理公共信息的有趣的不同模式。规模较小的散户投资者无法预测未来的新闻，也缺乏正确处理公共新闻的技能，而规模最大的散户投资者和机构能够正确预测未来的盈利新闻，并将当时的新闻纳入自己的交易。不同散户投资者在信息处理能力上的差异显然导致了他们对未来回报的预测能力的差异

IV. Trading Performances Using Tracking Portfolios of Retail Trading **利用零售交易跟踪组合的交易业绩**

A. The BLLO Method **基础研究方法**

With the short- and long-term predictive patterns documented in the previous sections, it is natural to ask how good the trading performance would be if we track the order flows from different groups of retail investors. To answer this question, we follow Barber, Lee, Liu, and Odean (2009, BLLO hereafter), which designs a simple and intuitive method to compute the trading performances of different types of investors in Taiwan Stock Exchange over 1995 to

1999. A step-by-step description of BLLO's method is provided in Appendix B. Here we provide a brief description of the BLLO's method, which has four steps.

根据上文所述的短期和长期预测模式，我们很自然地会问，如果我们跟踪来自不同群体的散户投资者的订单流，交易表现会有多好。为了回答这个问题，我们跟随 Barber, Lee, Liu 和 Odean (2009, BLLO 以后)，设计了一个简单而直观的方法来计算 1995 年到 1999 年台湾证券交易所不同类型的投资者的交易表现。对 BLLO 方法的逐步描述见附录 b。在这里，我们提供了 BLLO 方法的简要描述，其中有四个步骤。

First, BLLO separate all investors into “individuals”, “corporations”, “dealers”, “foreigners”, and “mutual funds”, using identities provided by the exchange. Second, within each investor group, BLLO compute the aggregate daily net buy and net sell positions and create daily matching trading portfolios, the “net buy” and “net sell” portfolios. Third, BLLO track these “net

首先，BLLO 使用交易所提供的身份将所有投资者分为“个人”、“公司”、“交易商”、“外国人”和“共同基金”。其次，在每个投资者群体中，BLLO 计算每日净买入和净卖出头寸的总和，并创建每日匹配的交易组合，即“净买入”和“净卖出”组合。第三，BLLO 跟踪这些“净值”

²⁴As an alternative, we use the Financial News Database of Chinese Listed Companies (CFND) to investigate whether the results from earnings news can be extended to other news. The results confirm our findings in Section III, and are available on request. 作为替代方案，我们使用中国上市公司财务新闻数据库(CFND)来研究盈余新闻的结果是否可以推广到其他新闻。这些结果证实了我们在第三部分的发现，并且可以根据要求提供。

buy” and “net sell” portfolios over a holding horizon of n days, and compute cumulative cash flows and returns from this tracking strategy, and treat them as proxies for trading performances of different investor groups. After computing the total performance, BLLO decomposes the total into three intuitive components, stock selection, market timing, and trading cost.

在 n 天的持有期限内买入和净卖出投资组合，计算这种跟踪策略的累积现金流和收益，并将它们作为不同投资者群体交易业绩的代理。在计算总体表现之后，BLLO 将总体表现分解为三个直观的组成部分，股票选择，市场时机和交易成本。

We adopt the BLLO method in our study for three reasons. First, the data structure of our sample is quite similar to those of BLLO's, therefore it is relatively straightforward to apply to our data. Second, the BLLO method is simple and easy to interpret, and it allows us to compare our results with their results. Third, the decomposition of total performance into stock selection, market timing, and trading cost helps us to connect our earlier results on the predictive power of order flows in the cross section. Notice that the performance estimates from the BLLO method are not realized gains and losses from investors. Rather, they are estimates of a trading strategy that follows signals from retail trading, namely, the net positions of each investor group, which reflects the investment skills of different groups of investors.

我们在研究中采用 BLLO 方法有三个原因。首先，我们样本的数据结构与 BLLO 的数据结构非常相似，因此应用于我们的数据相对简单。其次，BLLO 方法简单易懂，允许我们将结果与他们的结果进行比较。第三，将整体绩效分解为股票选择、市场时机和交易成本，有助于我们将先前的结果与横截面上订单流的预测能力联系起来。注意，基于 BLLO 方法的绩效评估并不是从投资者那里实现的收益和损失。相反，它们是根据散户交易信号(即反映不同投资者群体投资技能的每个投资者群体的净头寸)估算出的交易策略。

We make three parameter choices within the BLLO framework. First, BLLO assumes that the net buy and sell portfolios are held for n days and sets n to different horizons. As the main

discussion of BLLO focuses on the holding horizon of 140 days ($n=140$), we choose the same horizon for ease of comparison. Second, BLLO assumes that the net buy (sell) portfolios are independent for each day d , and annualizes the performance by aggregating the daily net buy and sell portfolios over each year. Here we relax the independent assumption, and compute the standard error estimates using the Newey and West (1987) method, with the optimal bandwidth chosen by using Newey and West (1994) procedure.²⁵ Third, to have a perspective of total performance as a percentage of the total investment of these tracking portfolios, which is similar to the idea of return

我们在 BLLO 框架内进行三个参数选择。首先, BLLO 假设净买卖投资组合持有 n 天, 并将 n 设置为不同的视野。由于 BLLO 的主要讨论集中在 140 天的持有期限($n = 140$), 我们选择相同的期限以便于比较。其次, BLLO 假设净买入(卖出)投资组合对于每天 d 是独立的, 并且通过对每年的净买入和卖出投资组合进行聚合, 使业绩年化。在这里, 我们放松了独立假设, 使用 Newey 和 West (1987)方法计算标准误差估计, 并使用 Newey 和 West (1994)方法选择最佳带宽。第三, 从总业绩占这些跟踪投资组合总投资的百分比的角度看问题, 这与回报的概念相似

²⁵We conduct robustness check using optimal lag numbers from Bayesian Information Criterion (BIC). The results are similar and available on request.

我们使用贝斯信息量准则(BIC)的最优滞后数进行稳健性检查。结果是相似的, 并可根据要求提供。

on investments, BLLO measures total investment as the aggregate holding value for each group of investors. For the ease of comparison, we choose the aggregate holding values in Panel A of Table I and present the return on investment as the total performance over aggregate holding.

在投资方面，《基本法》以投资总额作为每组投资者的总持有价值。为了便于比较，我们在表 i 的面板 a 中选择了总持有量值，并将投资回报率表示为总持有量之上的总表现。

B. Performances of Chinese Investors' Trading Using the BLLO Method **基于 BLLO 方法的中国投资者交易绩效研究**

Panel A of Table VIII reports the performance of the tracking portfolios. The annualized return to investment ranges between -5.61% and -2.80% for a 140-day holding horizon for RT1 to RT4, and RT5 have a close to zero, -0.29%, return to investment, while the institutions have a return to investment of 1.15%. It might not be too surprising to find that BLLO's net buy and sell tracking portfolio approach provides negative returns on investments for RT1-RT4, given that our earlier results show that RT1-RT4 predict future returns negatively and significantly. Similarly, positive returns on investments for INST are also expected because earlier evidence shows that institutional order flows predict future returns positively and significantly. It is intriguing to find that the net buy and sell portfolios tracking the order flows of RT5 deliver negative returns on investments, while earlier results show that RT5 predicts future returns positively and significantly.

表 8 的 a 小组报告了追踪投资组合的表现。投资回报率介乎 -5.61% 至 -2.80% 之间，rt1 至 rt4 为期 140 天，rt5 的投资回报率接近零，-0.29%，而机构的投资回报率为 1.15%。鉴于我们早期的研究结果显示 rt1-rt4 对未来回报的预测是负面且显著的，因此，发现 BLLO 的净买卖跟踪投资组合方法对 rt1-rt4 的投资回报为负可能并不太令人惊讶。同样，INST 的投资回报也是预期的，因为早期的证据表明，机构订单流量预测未来的回报是积极和显著的。有趣的是，跟踪 rt5 指令流的净买卖投资组合带来负的投资回报，而较早的结果显示 rt5 预测未来的回报是正的且显著的。

The answer is in the next three columns, which provide the three components of total performance. For RT5's annualized return to investment of -0.29%, the stock selection component contributes a significant and positive coefficient of 1.05%, which is consistent with the earlier positive cross-sectional return predictive power; the market timing component contributes an insignificant coefficient of -0.30%, indicating that RT5 might not be able to time the market; and the trading cost component contributes a significant coefficient of -1.03%. That is, even though RT5 is capable of stock selection, their low market timing ability and significant transaction cost cause the tracking portfolio to have a close-to-zero total return on investment.

答案在接下来的三列中，它们提供了总体性能的三个组成部分。Rt5 的年投资回报率为 -0.29%，其中股票选择部分贡献了 1.05% 的显著正系数，这与早先的正截面回报预测能力一致；市场时机部分贡献了 -0.30% 的微不足道系数，这表明 rt5 可能无法确定市场时机；交易成本部分贡献了 -1.03% 的显著系数。也就是说，即使 rt5 能够进行股票选择，但其较低的市场时机选择能力和巨大的交易成本导致跟踪投资组合的总投资回报率接近于零。

The decomposition of RT1-RT4 is different from that of RT5 in the sense that their groups of investors do not possess either stock selection or market timing ability, and so all three components of total returns on investment are all negative, mostly attributable to poor stock selection and significant trading cost. For instance, the trading cost component ranges between -1.38% (RT1) and -1.80% (RT2), which suggests that RT2 pays the most for trading relative to their holding. In the case of institutional investors, the results are quite similar to RT5, except that they have superior stock selection ability (StkSelect=1.38%), market timing ability (MktTiming=0.19%), and lower trading costs (TrdCost=-0.41%), and the total return on investment is positive and significant.²⁶

Rt1-rt4 与 rt5 的分解不同之处在于, rt1-rt4 的投资者群体既不具备选股能力, 也不具备选择市场时机的能力, 因此投资总回报的三个组成部分均为负值, 主要原因是选股能力差和交易成本高。例如, 交易成本部分介于 -1.38% (RT1)和 -1.80% (RT2)之间, 这表明相对于他们持有的股票, rt2 支付的交易费用最高。对于机构投资者而言, 除了具有较高的选股能力(StkSelect = 1.38%)、市场择时能力(MktTiming = 0.19%)和较低的交易成本(TrdCost =-0.41%)外, 其结果与 rt5 非常相似, 且总投资回报率为正值且显著。²⁶

[Insert Table VIII about here]
[在这里插入表 8]

C. Performances for Momentum and Contrarian Trading Using the BLLO Method **基于BLLO 方法的动量和反转交易的表现**

The previous section shows that different retail investors have different momentum and contrarian patterns, which contribute to their different predictive patterns for future returns. For instance, smaller retail investors display daily momentum patterns, which contribute to their negative return predictive pattern, while the largest retail investors display daily contrarian trading patterns, which contribute to their positive predictive pattern for future returns. Does the

momentum or contrarian trading pattern also play a significant role in the performance of net buy-and sell-tracking portfolios?

前一部分显示，不同的散户投资者有不同的动量和反转模式，这有助于他们对未来回报的不同预测模式。例如，较小的散户投资者显示每日动量模式，这有助于他们的负回报预测模式，而最大的散户投资者显示每日反向交易模式，这有助于他们对未来回报的正预测模式。动量或反向交易模式是否也在净买入和卖出跟踪投资组合的表现中扮演重要角色？

To answer this question, we separate the total performances in equation (12) into two scenarios, “daily momentum” and “daily contrarian”. If the investor group G ’s order imbalance times previous day return, $(\sum_{i \in G} \Delta O_i) * (\Delta R_{i-1})$, is positive, we classify the stock i on day t as “daily momentum”. If it is negative, we classify it as “daily contrarian”.
为了回答这个问题，我们将方程(12)中的总体表现分为两种情景，“日动量”和“日反转”。如果投资者群体 g 的订单不平衡乘以前一天的回报， $(\sum_{i \in g} \Delta O_i) * (\Delta R_{i-1})$ ，是正的，我们在当天分类股票 i 为“日动量”。如果是负的，我们将其分类为“日反转”。

²⁶We compare Table VIII Panel A with BLLO’s results for the Taiwan stock market and find that the general patterns are similar. In Appendix Figure II, we also conduct a similar exercise for shorter holding periods. The results display similar patterns as in Panel A of Table VI.

我们比较表 VIII 面板 a 和 BLLO 的台湾股票市场的结果，发现一般的模式是相似的。在附录图二中，我们还对较短的持有期进行了类似的练习。结果显示与表 VI 中的图 a 相似的模式。

d for investor group G to be “daily momentum” because they buy when price rises and they sell when price drops; if $(\text{buy}_{it} - \text{sell}_{it}) * (\text{price}_{it} - 1)$ is negative, then we classify the stock i on day d for investor group G to be “daily contrarian”. We then calculate the performance for each scenario. 如果 $(\text{buy}_{it} - \text{sell}_{it}) * (\text{price}_{it} - 1)$ 为负值, 那么我们将投资者组 g 在 d 日的 i 分类为“每日反转”。然后我们计算每种情况下的表现。

Panel B of Table VIII reports the results. Taking RT1 for example, following the daily momentum pattern, the annualized return on investment for RT1 is -4.57% with t -statistic of -8.47, which is significantly worse than the annualized return on investment of RT1 at -1.04% with t -statistic of -2.57 when they follow contrarian strategy. The results show that momentum trading results in worse performance than that of contrarian trading. Similar patterns exist for RT2-RT4. For RT5, the annualized return on investment is -1.61% when their trades display daily momentum patterns and 1.32% when they display daily contrarian patterns. For INST, the annualized performances are positive in both momentum (0.51%) and contrarian (0.64%) conditions, while the contrarian conditions have slightly higher returns on investments. Regarding the three components of the total performances, the stock selection performances are in general better under the “daily contrarian” column than “daily momentum” column; the market timing component are mostly insignificant under “daily momentum”, but significantly negative under “daily contrarian”, the trading costs are similar for momentum and contrarian columns. In other words, investors generally perform better when they pursue daily contrarian rather than daily momentum trading strategies.

表八 b 小组报告了结果。以 rt1 为例, 遵循日动量模式, rt1 的年化投资收益率为 -4.57%, t 统计量为 -8.47, 显著低于 rt1 的年化投资收益率为 -1.04%, t 统计量为 -2.57。结果表明, 动量交易比反向交易的表现更差。Rt2-rt4 也存在类似的模式。对于 RT5, 当他们的交易显示每日动量模式时, 年投资回报率为 -1.61%, 当他们显示每日反转模式时, 年投资回报率为 1.32%。INST 的年化业绩在动量条件(0.51%)和反向条件(0.64%)下均为正值, 而反向条件下的投资回报率略高。就总体表现的三个组成部分而言, “每日反转”栏

下的股票选择表现一般优于”每日动量”栏下的股票选择表现;”每日动量”栏下的市场时机组成部分大多微不足道，但”每日反转”栏下的市场时机组成部分明显为负值，动量栏和反转栏下的交易成本相似。换句话说，当投资者追求每日反转策略而不是每日动量交易策略时，他们的表现通常更好。

D. Performances around Earnings Announcements Using the BLLO Method **利用BLLO 方法在盈利公告周围的表现**

Given that earnings announcements are the most informative firm-level events, we are also curious to understand how they contribute to the overall performance of the tracking portfolios. Our prior finding is that if investors have an information advantage around earnings announcements, they would have better performance around earnings announcements, while if

鉴于收益公告是公司层面上信息量最大的事件，我们也很想知道它们如何对跟踪投资组合的整体表现做出贡献。我们先前的发现是，如果投资者在收益公告周围有信息优势，他们在收益公告周围会有更好的表现

investors have disadvantages around these event days, they would have worse losses on these days. There are 15,702 quarterly earnings announcement events during our 3.5-year sample. As in Section III.C, we consider both the day before earnings announcements and the announcement day as event days, accounting for 2.98% ($=15,702*2/1,053,148$) of the total stock-day observations. For this analysis, we compute the total performance on these event days and compare it with the overall performance on all days.

投资者在这些事件发生的日子处于劣势，他们在这些日子会遭受更大的损失。在我们 3.5 年的样本中，有 15702 个季度收益公告事件。正如第 iii. c 节所述，我们把公布业绩前一天和公布当天都视为事件日，占总股票日观察数的 2.98% ($= 15,702 * 2/1,053,148$)。对于这个分析，我们计算这些事件日的总体表现，并与所有日子的总体表现进行比较。

We present the results in Table VIII Panel C. For ease of comparison, we present the estimates over the earnings event days under “EA”, and its proportion to all days under “EA/all days”. The annualized total performance over earnings days for RT1 is -0.39%, which accounts for 6.89% of the total performance of all days. The stock selection, market timing, and trading cost components are -0.36%, 0.02%, and -0.04%, accounting for 8.96%, -8.63%, and 3.06% of all days, respectively. The coefficients on total performance, stock selection, and trading costs are all statistically significant. Considering that earnings days only account for 2.98% of all days, the total performance and stock selection component of the tracking portfolio following RT1 order flows is particularly bad on earnings days, which echoes our findings in Section III. The trading cost on EA days accounts for approximately 3% of all days, which is close to the overall number of EA days. Similar patterns are observed for RT2-RT4. For RT5, the annualized EA total performance of RT5 is insignificant. For the stock selection component, RT5’s annualized return on investment is 0.14%, accounting for 13.81% of RT5’s total stock selection, with a *t*-statistic of 2.56. This finding is consistent with our earlier

result that RT5 might be better informed around EA days. For INST, the total performance over EA days is 0.11%, accounting for 9.80% of the total performance,

为了便于比较，我们在“EA”项下列出了收益事件日的估计数，以及它与“EA/all days”项下所有日的比例。Rt1 收益日的年度总表现为 -0.39%， 占所有日总表现的 6.89%。股票选择、上市时机和交易成本构成分别为 -0.36%、0.02% 和 -0.04%， 分别占总日数的 8.96%、-8.63% 和 3.06%。总体表现、股票选择和交易成本的系数都具有统计学意义。考虑到收益日仅占总日数的 2.98%， rt1 指令流之后的跟踪投资组合的总体业绩和股票选择部分在收益日特别糟糕，这与我们在第三节中的发现相呼应。EA 日的交易成本约占所有交易日的 3%， 接近 EA 日的总数。Rt2-rt4 也观察到类似的模式。对于 RT5， rt5 的年度 EA 总体性能是微不足道的。在选股部分， rt5 的年投资回报率为 0.14%， 占 rt5 总选股的 13.81%， t 统计量为 2.56。这一发现与我们早期的结果一致， 即 rt5 可能在 EA 日期前后更好地获得信息。对于 INST 来说， EA 日的总体表现是 0.11%， 占总体表现的 9.80%，

with a significant t -statistic. The stock selection component shows a similar pattern. These findings indicate that institutional investors perform better on EA days than on all days.²⁷

有显着的 t 统计。股票选择部分显示了类似的模式。这些发现表明，机构投资者在 EA 日的表现要好于所有日子

V. Robustness

健壮

A. Subperiods: Predictive Patterns for Different Years

A. 子周期: 不同年份的预测模式

Do the predictive patterns of retail order flows differ when market conditions change? Our 3.5-year sample period provides a good setting with different market conditions. The market return is -8.4% in 2016, 12.1% in 2017, -22.2% in 2018, and 21.2% in 2019. With two positive and negative returns, our sample period provides settings for both market-ups and-downs. To examine the predictive patterns of order flows for different years, we modify the Fama and MacBeth (1973) specification in equation (2) and interact the order flow variables with year dummies. The coefficients on the interaction terms provide information on whether the predictive pattern changes on each year.

当市场条件发生变化时，零售订单流的预测模式是否有所不同？我们 3.5 年的样本周期为不同的市场条件提供了一个很好的设置。2016 年的市场回报率为 -8.4%，2017 年为 12.1%，2018 年为 -22.2%，2019 年为 21。由于有两个正回报和负回报，我们的样本周期提供了市场上涨和下跌的设置。为了检验不同年份订单流的预测模式，我们修改了方程(2)中的 Fama 和 MacBeth (1973)规范，并将订单流变量与年份假人相互作用。交互项上的系数提供了关于预测模式是否每年变化的信息。

[Insert Table IX about here]
[在这里插入表 IX]

From Table IX Panel A, the negative predictive pattern of order flow from RT1-RT4 for the next-day return, as observed in Table II, is robust for different years. Moreover, it is interesting to notice that the negative magnitudes are generally larger for 2016 and 2018, indicating that the negative predictive pattern is even larger in a downward market. For the largest retail investor, RT5, the positive predictive pattern is also larger during 2016 and 2018, indicating that their information advantage, if any, may also be stronger when the market is down. Similar patterns are observed for institutional investors.

从表 IX 面板 a 看, rt1-rt4 订单流对次日回报的负预测模式, 如表 II 所观察到的, 对不同年份是稳健的。此外, 值得注意的是, 2016 年和 2018 年的负值通常较大, 表明在市场下跌时, 负值预测模式甚至更大。对于最大的散户投资者 rt5 而言, 2016 年和 2018 年的正预测模式也更大, 表明当市场下跌时, 他们的信息优势(如果有的话)也可能更强。机构投资者也观察到类似的模式。

²⁷Previous literature (Kaniel et al., 2008; Barrot et al., 2016) argue that retail investors provide liquidity to the market and are compensated for their liquidity provision. We further decompose retail investors' performances from the perspective of liquidity provision, and provide the results in Appendix Table VII. Both small and large retail investors' performance are significantly negative if the trades demand liquidity. But if they provide liquidity, small retail investors' performance become insignificantly different from zero, and large retail investors are compensated for liquidity provision, which generally supports the hypothesis that liquidity provision is compensated.

以前的文献(Kaniel 等, 2008; Barrot 等, 2016)认为, 散户投资者为市场提供流动性, 并因其提供的流动性而得到补偿。我们进一步从流动性供给的角度分解散户投资者的表现, 并在附录表 VII 中提供结果。如果交易需要流动性, 小型和大型散户投资者的表现都是显著的负面。但如果它们提供流动性, 小型散户投资者的表现与零相差无几, 大型散户投资者因提供流动性而得到补偿, 这通常支持流动性提供得到补偿的假设。

B. Subgroups: Predicting Patterns Across Firms with Different Characteristics

子组: 预测具有不同特征的企业模式

Previous studies show that stock returns can be significantly affected by firm and stock characteristics, such as size, EP ratio, and liquidity. Do the predictive patterns of retail order flows differ across firms with different characteristics? To answer this question, we modify the Fama and MacBeth (1973) specification in equation (2) and allow different coefficients for firms with different characteristics, by including interactions with characteristic dummies. Consider the size as an example. We first divide all firms on day d into three groups based on the previous month-end firm market capitalization. The dummy variable, $\text{size}_{i,t}$, takes the value 1 if firm i belongs to the smallest 1/3 of firms and zero otherwise; $\text{size}_{i,t}^{\text{medium}}$, takes the value 1 if firm i belongs to the medium 1/3 of firms and zero otherwise; and $\text{size}_{i,t}^{\text{large}}$, takes the value 1 if

以往的研究表明，股票收益可以显著影响公司和股票的特点，如规模，收益率和流动性。零售订单流的预测模式在具有不同特征的公司之间是否存在差异？为了回答这个问题，我们修改了方程(2)中的 Fama 和 MacBeth (1973)规范，允许不同特征的公司采用不同的系数，包括与特征假人的相互作用。以规模为例。我们首先根据前一个月末的公司市值将所有在 d 天的公司分为三组。假变量 $\text{size}_{i,t}$ 取值 1，如果公司 i 属于最小的 1/3 的公司，否则为 0; $\text{size}_{i,t}^{\text{medium}}$ 取值 1，如果公司 i 属于中等的 1/3 的公司，否则为 0; 和 $\text{size}_{i,t}^{\text{large}}$ 取值 1，如果

firm i belongs to the largest 1/3 of firms and zero otherwise. The coefficients provide information on whether the predictive pattern changes for firms of different sizes.

公司 i 属于最大的 1/3 的公司，否则为 0。这些系数提供了关于不同规模公司的预测模式是否发生变化的信息。

Panel B of Table IX reports the results. In the first three rows, we separate the firms based on their market capitalization. The negative predictive pattern of order flow from RT1-RT4 for the next-day return, as shown in Table II, is quite robust for firms with different sizes. However, it is interesting to notice that the magnitudes generally decrease from the smallest to the largest firms, indicating that the negative predictive pattern is the strongest for smaller firms. For large retail investors, RT5, the positive predictive pattern remains for small- and medium-sized firms, but not for large firms, indicating that their information about future returns, if any, might be concentrated in smaller firms. In comparison, order flows from institutions significantly predict next-day returns in all three rows, and more so for large firms, suggesting that their information about future returns, if

any, might be more prominent for larger firms. When we separate firms by EP, turnover, and stock prices, we observe similar patterns. In other words, the predictive patterns in Table II are

表九 b 小组报告了结果。在前三行中，我们根据公司的市值来区分这些公司。Rt1-rt4 订单流对于第二天收益的负面预测模式，如表二所示，对于不同规模的公司是相当稳健的。然而，值得注意的是，规模一般从最小的企业减少到最大的企业，这表明，负预测模式是最强的较小的企业。对于大型散户投资者，RT5，正面预测模式仍然适用于中小型公司，但不适用于大型公司，这表明它们关于未来回报的信息，如果有的话，可能集中在较小的公司。相比之下，来自机构的订单流显著地预测了所有三行的第二天收益，对于大公司来说更是如此，这表明它们关于未来收益的信息(如果有的话)对于大公司来说可能更为突出。当我们将公司按照收益、营业额和股票价格分类时，我们观察到了类似的模式。换句话说，表 2 中的预测模式是

generally robust across firms with different characteristics, and the negative (positive) predictive power of smaller (larger) retail investors is stronger for small, low-EP, and higher-turnover firms, while the positive predictive power of institutional investors is stronger for large and high-EP firms.²⁸

对于具有不同特征的公司，一般来说都是强有力的，较小(较大)散户投资者的负(正)预测能力对于小型、低收益率和高收益率公司更强，而机构投资者的正预测能力对于大型和高收益率公司更强

C. Different Filters **不同的过滤器**

We consider three different sets of filters in the robustness check. In the main results, we discard stocks with fewer than 15 days of trading during the most recent month. In addition to this filter, Liu et al. (2019) also eliminate stocks that have become public within the past six months, stocks with less than 120 days of trading during the past 12 months, and the smallest 30% of all firms listed in China A-share market. We add all these additional filters and check the robustness of our results. In Panel C of Table IX, the order imbalance prediction directions are similar to those in Table II and the economic magnitude of institutions remains large. Thus, our main results are robust to the stricter filters from Liu et al. (2019).

在鲁棒性检查中，我们考虑了三组不同的滤波器。在主要结果中，我们抛弃了最近一个月交易天数少于 15 天的股票。除了这个过滤器，Liu 等人(2019)还排除了过去 6 个月内上市的股票，过去 12 个月内交易日不足 120 天的股票，以及在中国 a 股市场上市的所有公司中最小的 30% 的股票。我们添加了所有这些额外的过滤器，并检查我们的结果的稳健性。在表 9 的 c 组中，顺序不平衡预测方向与表 2 中的相似，并且机构的经济规模仍然很大。因此，我们的主要结果对 Liu 等人(2019)更严格的过滤器是稳健的。

Our second set of filters relates to leverage trading. Our trade-level data identify investors' margin buys, short sales, and collateral trades. Leveraged trading may differ from non-leverage trading. Each day, margin buys account for 10% of the trading volume, short sales account for 0.2%, and collateral trading accounts for 15% during our sample period. We exclude leverage trades and re-estimate equation (2). Panel D of Table IX show that the order imbalance prediction directions are similar to the results in Table II and the economic magnitudes are quantitatively similar. That is, our results are also robust to whether or not we include these leverage trades.

我们的第二组过滤器与杠杆交易有关。我们的交易水平数据识别了投资者的保证金买入、卖空和抵押品交易。杠杆交易可能不同于非杠杆交易。每天，融资融券交易占交易量的10%，卖空交易占0.2%，抵押品交易占15%。我们排除了杠杆交易和重新估计方程(2)。表IX的图d显示，订单不平衡预测方向与表II的结果相似，经济规模在数量上相似。也就是说，我们的结果对于是否包括这些杠杆交易也是稳健的。

²⁸We obtain a three-month sample from January 2019 to March 2019 with investors' gender and age information, and examine whether there are significant differences between male/female and young/old. The results are reported in Appendix Table VI. Across all gender-age groups, older male investors trade the most. For return prediction, the male investors across all ages significantly and negatively predict returns, especially for older males, while they are insignificant and different from zero for female retail investors. These patterns are mostly consistent with previous findings in Barber and Odean (2001).

我们收集了2019年1月至2019年3月三个月的投资者性别和年龄信息样本，并检验了男性/女性和年轻/老年人之间是否存在显著差异。结果见附录表六。在所有性别年龄组中，年龄较大的男性投资者交易最多。在回报预测方面，所有年龄段的男性投资者对回报的预测都是显著和负面的，特别是对于年龄较大的男性，而对于女性散户投资者来说，这种预测微不足道，不同于零。这些模式与Barber和Odean (2001)的研究结果基本一致。

Finally, we consider the price limits. One institutional feature of the Chinese market is price limit restrictions. That is, investors can buy and sell stocks freely when the stock's price is within $\pm 10\%$ of the previous day's closing price. If the price moves out of the $\pm 10\%$ range, trading stops until the next day opens. Chen et al. (2019) focus on price limit days and find that large investors tend to trade differently on these days. Here, we examine whether our results hold when we exclude price limit days. We re-estimate equation (2) without the price limit days and present the results in Panel E of Table IX. Our results are robust to whether or not we include these price limit days.

最后，我们考虑价格限制。中国市场的一个制度特征是价格限制。也就是说，当股票价格在前一天收盘价的 $\pm 10\%$ 以内时，投资者可以自由买卖股票。如果股价超出 $\pm 10\%$ 的范围，交易就会停止，直到第二天开盘。Chen 等人(2019)关注价格限制日，发现大投资者在这些日子交易倾向不同。在这里，我们检查我们的结果是否成立时，我们排除价格限制天。我们重新估计方程(2)没有价格限制天数，并在表 IX 的 e 小组中呈现结果。我们的结果对于我们是否包括这些价格限制日是稳健的。

D. Alternative Portfolio Forming Method **另类投资组合形成方法**

Our main results on return predictive patterns are based on the Fama and MacBeth (1973) regressions, which assume linear relations between future returns and order flow variables. In this section, we adopt an alternative portfolio approach and examine whether our results still hold. More specifically, we sort firms into five groups each day based on the previous day's order imbalance from a particular investor group, buying and selling 20% of stocks with the highest and lowest order imbalance measures for that particular investor group. We report the risk-adjusted returns (alphas) on this long-short strategy for one to 60 days, where we conduct risk adjustment using the Liu et al. (2019) three factor model.

我们关于回报预测模式的主要结果是基于 Fama 和 MacBeth (1973)回归，它假定了未来回报和订单流变量之间的线性关系。在这一部分，我们采用了另一种投资组合方法，并检查我们的结果是否仍然成立。更具体地说，我们每天根据某个特定投资者群体前一天的订单不平衡情况，将公司分为五组，按照该特定投资者群体的最高和最低订单不平衡情况，买卖 20% 的股票。我们使用 Liu 等人(2019)的三因素模型进行风险调整，报告了 1 至 60 天这种多空策略的风险调整回报率(alpha)。

From Panel F of Table IX, the one-day long-short portfolio alpha, using the previous day order imbalance from RT1, is -0.0042 and highly significant. From one week to 12 weeks, the cumulative alphas for the long-short portfolio decrease from -0.0089 to -0.0183, and they are all highly significant. That is, the cumulative alphas using *OibRT1* are consistently negative and significant, and there are no signs of reversal within 12 weeks, which echoes our earlier results in Tables II and V. Similar patterns exist for RT2 and RT3. For RT4, the one-day alpha is negative at -0.0007, but quickly becomes insignificant when we extend the holding horizon to one week,

从表 IX 的面板 f 来看，使用 rt1 的前一天订单不平衡的一天多空投资组合 alpha 是 -0.0042，并且非常显著。从 1 周到 12 周，多空投资组合的累积阿尔法值从 -0.0089 下降到 -0.0183，且都具有高度显著性。也就是说，使用 oibrt1 的累积阿尔法值始终是负值和显著的，并且在 12 周内没有逆转的迹象，这与我们在表 II 和表 v 中的早期结果相呼应。rt2 和 rt3 也存在类似的模式。对于 RT4，一天 α 在 -0.0007 处为负值，但当我们将其持有期限延长到一周时，它很快变得不重要，

indicating that RT4 for horizons longer than one day. For the largest retail investors in RT5, the one-day alpha is 0.0017, which is positive and significant. The 12-week cumulative alpha is 0.0057, still positive and significant, confirming the results in Tables II and V that RT5 has both short- and long-term predictive power for future returns. The results for *OibINST* are similar to those for *OibRT5*.

表明 rt4 的视野超过一天。对于 rt5 最大的散户投资者来说，一天 alpha 是 0.0017，这是正值和显著的。12 周累积 α 值为 0.0057，仍为正值且显著，证实了表二和表五的结果，即 rt5 对未来回报具有短期和长期预测能力。*OibINST* 的结果与 *oibrt5* 的结果相似。

V. Conclusion

结论

Using comprehensive retail trading and holding data from 2016 to 2019, we divide tens of millions of retail investors into five groups based on their account sizes and examine their roles in the price discovery process in terms of return predictive power and the driving forces of these predictive patterns. Retail investors with account sizes of less than 3 mil CNY buy and sell stocks in the wrong direction. The prices of stocks they buy experience negative returns the next day, whereas those they sell experience positive returns. For retail investors with large account balances, trading predicts returns correctly. By tracing their differences in predicting future returns, we provide evidence that the negative predictive power of retail investors with smaller account sizes is mostly related to their order persistence, daily momentum trading, behavioral biases, and failures in processing earnings news. By contrast, the positive predictive power of large retail investors is mostly associated with order persistence, contrarian trading, trading against behavioral biases, and advantages in processing earnings news. Following BLLO (2009), we construct net buy and sell portfolios for each group of investors, and track their trading flows. This performance generates results consistent with the predictive patterns, in the sense that

smaller retail investors have low stock selection skills and market timing skills, while large retail investors have better stock selection skills.

利用 2016 年至 2019 年的综合零售交易和持有数据，我们根据账户规模将数千万散户投资者分为五组，并根据回报预测能力和这些预测模式的驱动力，研究他们在价格发现过程中的作用。账户规模小于 300 万元人民币的散户投资者在错误的方向上买卖股票。他们购买的股票第二天的价格经历了负回报，而他们卖出的股票经历了正回报。对于拥有巨额账户余额的散户投资者来说，交易预测的回报率是正确的。通过追踪他们在预测未来收益方面的差异，我们提供了证据表明，账户规模较小的散户投资者的负面预测能力主要与他们的订单持续性、每日动量交易、行为偏差和处理盈利新闻的失败有关。相比之下，大型散户投资者的正向预测能力主要与指令持续性、反向交易、反对行为偏差的交易以及处理收益信息的优势有关。在 BLLO (2009)之后，我们为每组投资者构建净买卖投资组合，并跟踪他们的交易流。这种表现产生的结果与预测模式一致，即小型散户投资者的选股技能和市场时机选择技能较低，而大型散户投资者的选股技能较好。

Our study of the heterogeneous trading behavior of Chinese retail investors provides many unique insights into this large group of investors. In addition, the exchange acknowledges the heterogeneity of retail investors and focuses on adopting policies on investor education and suitability that restrict some kinds of trading for the smallest accounts. For example, retail investors are required to have at least 500k CNY holdings of stocks for at least 20 trading days to open a leverage trading account or to trade on a riskier Science and Technology Innovation Board (or STAR Market). These policies effectively exclude the smallest retail investors from leverage trading and trading on riskier start-ups, which could help protect them from even worse losses. Our study also raises many interesting questions. For example, why do retail investors dominate trading in the Chinese stock market? What role do institutional investors play? We leave these interesting and important questions to future research.

我们对中国散户投资者异质性交易行为的研究为我们提供了许多独特的见解。此外，该交易所承认散户投资者的多样性，并侧重于采取投资者教育和适用性政策，限制最小账户的某些类型的交易。例如，散户投资者必须持有至少 50 万元人民币股票至少 20 个交易日，才能开立杠杆交易账户，或在风险较高的科技创新板(STAR Market)进行交易。这些政策实际上将最小的散户投资者排除在杠杆交易和风险较高的初创企业的交易之外，这可能有助于保护他们免受更严重的损失。我们的研究也提出了许多有趣的问题。例如，为什么散户投资者在中国股市的交易中占据主导地位？机构投资者扮演什么角色？我们把这些有趣而重要的问题留给未来的研究。

References

参考文献

- An, Li, Dong Lou, and Donghui Shi, 2022, Wealth redistribution in bubbles and crashes. 安、李东楼、石东辉, 2022, 泡沫与崩溃中的财富再分配。 *Journal of Monetary Economics*, 126(3), 134-153. 货币经济学杂志, 126(3), 134-153。
- Anagol, Santosh, Vimal Balasubramaniam, and Tarun Ramadorai, 2021, Learning from noise: Anagol, Santosh, Vimal Balasubramaniam, and Tarun Ramadorai, 2021, Learning from noise: Evidence from India's IPO lotteries. *Journal of Financial Economics*, 140(3), 965-986. 来自印度 IPO 彩票的证据。《金融经济学杂志》, 140(3), 965-986。
- Bach, Laurent, Laurent E. Calvet, and Paolo Sodini, 2020, Rich pickings? Risk, return, and skill in household wealth. *American Economic Review*, 110(9), 2703-47. 巴赫, 劳伦特, 劳伦特 e 卡尔维特和保罗索蒂尼, 2020 年, 丰富的选择? 风险、回报和家庭财富的技能。美国经济评论, 110(9), 2703-47。
- Balasubramaniam, Vimal, John Y. Campbell, Tarun Ramadorai, and Benjamin Ranish, 2023, Who owns what? A factor model for direct stockholding. *The Journal of Finance*, 78(3), 1545-1591. Balasubramaniam, Vimal, John y. Campbell, Tarun Ramadorai, and Benjamin Ranish, 2023, 谁拥有什么? 直接持股的因素模型。The Journal of Finance, 78(3), 1545-1591 金融期刊, 78(3), 1545-1591。
- Bali, Turan G., Nusret Cakici, and Robert F. Whitelaw, 2011, Maxing out: Stocks as lotteries and the cross-section of expected returns, *Journal of Financial Economics*, 99(2), 427-446. 巴厘, 图兰 g., Nusret Cakici, 和罗伯特 f. 怀特洛, 2011, 最大化: 股票作为彩票和预期收益的横截面, 金融经济学杂志, 99(2), 427-446。
- Barber, Brad M., Xing Huang, Terrance Odean, and Christopher Schwarz, 2022, Attention induced trading and returns: Evidence from robinhood users. *Journal of Finance*, 77(6), 3141-3190. Barber, Brad m., Xing Huang, Terrance Odean, and Christopher Schwarz, 2022, Attention induced trading and returns: Evidence from robinhood users。Journal of Finance, 77(6), 3141-3190 金融期刊, 77(6), 3141-3190。
- Barber, Brad M., Yi-Tsung Lee, Yu-Jane Liu, and Terrance Odean, 2009, Just how much do individual investors lose by trading?. *The Review of Financial Studies*, 22(2), 609-632. 李一宗、刘裕珍和 Terrance Odean, 2009, 个人投资者在交易中究竟损失了多少? .The Review of Financial Studies, 22(2), 609-632 金融研究评论, 22(2), 609-632。
- Barber, Brad M., Yi-Tsung Lee, Yu-Jane Liu, and Terrance Odean, 2014, The cross-section of speculator skill: Evidence from day trading. *Journal of Financial Markets*, 18, 1-24. 投机者技巧的横截面: 来自日内交易的证据。金融市场杂志, 18,1-24。
- Barber, Brad M., Shengle Lin, and Terrance Odean, 2023, Resolving a paradox: Retail trades positively predict returns but are not profitable. *Journal of Financial and Quantitative Analysis*, forthcoming. 解决一个悖论: 零售交易正向预测回报, 但不盈利。金融与定量分析杂志, 即将出版。
- Barber, Brad M., and Terrance Odean, 2000, Trading is hazardous to your wealth: The common stock investment performance of individual investors, *Journal of Finance* 55, 773-806. 交易对你的财富是有害的: 个人投资者的普通股投资业绩, 金融期刊 55,773-806。

Barber, Brad M., and Terrance Odean, 2001, Boys will be boys: Gender, overconfidence, and common stock investment, *The Quarterly Journal of Economics*, 116(1), 261-292.

男孩就是男孩: 性别、过度自信和普通股投资, 《经济学季刊》, 116(1), 261-292。

Barber, Brad M., and Terrance Odean, 2008, All that glitters: The effect of attention and news on the buying behavior of individual and institutional investors, *Review of Financial Studies* 21, 785-818.

巴伯, 布拉德 m. 和特伦斯奥迪恩, 2008, 所有闪光: 关注和新闻对个人和机构投资者购买行为的影响, 金融研究评论 21, 785-818。

Barber, Brad M., Terrance Odean, and Ning Zhu, 2009, Do retail trades move markets? *Review of Financial Studies*, 22, 151-186.

2009 年, 零售交易是否会影响市场? 金融研究回顾, 22, 151-186。

Barrot, Jean-Noel, Ron Kaniel, and David Alexandre Sraer, 2016, Are retail traders compensated for providing liquidity? *Journal of Financial Economics*, 120, 146-168.

Barrot, Jean-Noel, Ron Kaniel, and David Alexandre Sraer, 2016, 零售商是否因提供流动性而得到补偿? 金融经济学杂志, 120, 146-168。

Black, Fischer, 1986, Noise, *Journal of Finance*, 41(3), 528-543.

布莱克, 费舍尔, 1986, 噪音, 金融杂志, 41(3), 528-543。

Blume, Marshall E. and Robert F. Stambaugh, 1983, Biases in computed returns: An application to the size effect, *Journal of Financial Economics*, 12, 387-404.

Blume, Marshall e. and Robert f. Stambaugh, 1983, 计算回报的偏差: 对规模效应的应用, 《金融经济学杂志》, 12, 387-404。

Boehmer, Ekkehart, Charles M. Jones, Xiaoyan Zhang, and Xinran Zhang, 2021, Tracking retail investor activity, *Journal of Finance*, 76(5), 2249-2305.

Boehmer, Ekkehart, Charles m. Jones, Xiaoyan Zhang, and Xinran Zhang, 2021, Tracking retail investor activity, *Journal of Finance*, 76(5), 2249-2305.

Boehmer, Ekkehart, Bo, Sang, and Zhe Zhang, 2020, Can Retail Investors Learn from Insiders? Working Paper.

Boehmer, Ekkehart, Bo, Sang, and Zhe Zhang, 2020, Can Retail Investors Learn from Insiders.

Chen, Ting, Zhenyu Gao, Jibao He, Wenxi Jiang, and Wei Xiong, 2019, Daily price limits and destructive market behavior, *Journal of Econometrics*, 208(1), 249-264.

陈婷, 高振宇, 何季宝, 江文喜, 魏雄, 2019, 每日价格限制与破坏性市场行为, 计量经济学杂志, 208(1), 249-264。

Chordia, Tarun, and Avanidhar Subrahmanyam, 2004, Order imbalance and stock returns:

Chordia, Tarun, and Avanidhar Subrahmanyam, 2004, 订单失衡与股票回报: Theory and evidence, *Journal of Financial Economics*, 72, 485-518.
理论与证据, 金融经济学杂志, 72, 485-518。

Dorn, Daniel, Gur Huberman, and Paul Sengmueller, 2008, Correlated trading and returns. *The Journal of Finance*, 63(2), 885-920.

丹尼尔·古尔·休伯曼和保罗·森穆勒, 2008, 《关联交易和回报》, 《金融期刊》, 63(2), 885-920。

Eaton, Gregory W., T. Clifton Green, Brian Roseman, and Yanbin Wu, 2022, Retail trader sophistication and stock market quality: Evidence from brokerage outages. *Journal of Financial Economics*, 146(2), 502-528.

格里高利·W·克利夫顿·格林, 布莱恩·罗斯曼, 吴彦斌, 2022, 零售交易员的成熟度和股票市场质量: 来自经纪业务中断的证据。金融经济学杂志, 146(2), 502-528。

Fama, Eugene F., and James D. MacBeth, 1973, Risk, return, and equilibrium: Empirical tests, *Journal of Political Economy*, 81, 607-636.

《风险、回报和均衡: 实证检验》, 《政治经济学杂志》, 81,607-636。

Fong, Kingsley YL, David R. Gallagher, and Adrian D. Lee, 2014, Individual investors and broker types. *Journal of Financial and Quantitative Analysis*, 49(2), 431-451.

Fong, Kingsley YL, David R. Gallagher, and Adrian D. Lee, 2014, 散户投资者和经纪人类别。金融与定量分析杂志, 49(2), 431-451。

Gervais, Simon, Ron Kaniel, and Dan H. Mingelgrin, 2001, The high-volume return premium.

Gervais, Simon, Ron Kaniel, and Dan H. Mingelgrin, 2001, The high-volume return premium.

The Journal of Finance, 56(3), 877-919.

The Journal of Finance, 56(3), 877-919 《金融期刊》, 56(3), 877-919。

Grinblatt, Mark, and Matti Keloharju, 2000, The investment behavior and performance of various investor types: a study of Finland's unique data set. *Journal of Financial Economics*, 55(1), 43-67.

格林布拉特, 马克, 马蒂克洛哈朱, 2000, 投资行为和表现的不同类型的投资者: 研究芬兰的独特数据集。金融经济学杂志, 55(1), 43-67。

Grinblatt, Mark, Matti Keloharju, and Juhani T. Linnainmaa, 2012, IQ, trading behavior, and performance. *Journal of Financial Economics*, 104(2), 339-362.

马克·格林布拉特, 马蒂·克洛哈朱, 朱哈尼·林南马, 2012, 智商, 交易行为和绩效。金融经济学杂志, 104(2), 339-362。

Hu, Conghui, Yu-Jane Liu, and Xin Xu, 2021, The valuation effect of stock dividends or splits:

Hu, Conghui, Yu-Jane Liu, and Xin Xu, 2021, 股票分红或拆分的估值效应:

Evidence from a catering perspective, *Journal of Empirical Finance*, 61, 163-179.

来自餐饮业的证据, *Journal of Empirical Finance*, 61,163-179。

Jiang, Lei, Jinyu Liu, Lin Peng, and Baolian Wang, 2022, Investor attention and asset pricing anomalies, *Review of Finance*, 26(3), 563-593.

江, 雷, 刘金玉, 林鹏, 王宝莲, 2022, 投资者关注与资产定价异常, 《金融评论》, 26(3), 563-593。

Kaniel, Ron, Saar Gideon, and Titman, Sheridan, 2008, Individual investor sentiment and stock returns, *Journal of Finance*, 63, 273-310.

卡尼尔, 罗恩, 萨尔吉迪恩, 和提特曼, 谢里丹, 2008 年, 个人投资者情绪和股票回报, 金融期刊, 63,273-310。

Kaniel, Ron, Liu, Shuming, Saar, Gideon, and Titman, Sheridan, 2012, Individual investor trading and return patterns around earnings announcements, *Journal of Finance*, 67, 639-680.

Kaniel, Ron, Liu, Shuming, Saar, Gideon, and Titman, Sheridan, 2012, 散户投资者的交易和收益模式围绕收益公告, *Journal of Finance*, 67,639-680。

Kelley, Eric K. and Paul C. Tetlock, 2013, How wise are crowds? Insights from retail orders and stock returns, *Journal of Finance*, 68, 1229-1265.

Kelley, Eric K. and Paul C. Tetlock, 2013, 群体有多聪明? 来自零售订单和股票回报的见解, *Journal of Finance*, 68,1229-1265。

Kyle, Albert S., 1985, Continuous auctions and insider trading, *Econometrica: Journal of the Econometric Society*, 1315-1335.

《连续拍卖和内幕交易》，《计量经济学会杂志》，1315-1335。

Li, Xindan, Ziyang Geng, Avanidhar Subrahmanyam, Honghai Yu, 2017, Do wealthy investors have an informational advantage? Evidence based on account classifications of individual investors, *Journal of Empirical Finance* 44, 1-18.

Li, Xindan, Ziyang 耿, Avanidhar Subrahmanyam, Honghai Yu, 2017, 富有的投资者有信息优势吗？基于个人投资者账户分类的证据, *Journal of Empirical Finance* 44,1-18。

Liao, Jingchi, Cameron Peng, and Ning Zhu, 2022, Extrapolative bubbles and trading volume, *Review of Financial Studies*, , 35(4), 1682-1722.

廖景池(音译), 彭金龙(音译), 朱宁(音译), 2022, 推断泡沫与交易量, 《金融研究评论》, 35(4), 1682-1722。

Linnainmaa, Juhani T., 2010, Do limit orders alter inferences about investor performance and behavior?. *The Journal of Finance*, 65(4), 1473-1506.

Linnainmaa, Juhani t., 2010, 限制订单是否会改变投资者表现和行为的推论。

Liu, Jianan, Robert F. Stambaugh, and Yu Yuan, 2019, Size and value in China, *Journal of Financial Economics*, 134(1), 48-69.

Liu, Jianan, Robert f. Stambaugh, and Yu Yuan, 2019, Size and value in China, *Journal of Financial Economics*, 134(1), 48-69.

Liu, Hongqi, Cameron Peng, Wei A. Xiong, and Wei Xiong, 2021, Taming the bias zoo, *Journal of Financial Economics*, 143(2), 716-741.

刘红琦, 彭金龙, 魏 a 熊, 魏雄, 2021, 驯服偏见动物园, 《金融经济学杂志》, 143(2), 716-741。

Liu, Yang, Guofu, Zhou, and Yingzi Zhu, 2021, Trend factor in China: The role of large individual trading. Working Paper.

刘, 杨, 国富, 周, 朱英子, 2021, 《中国的趋势因素: 大型个人交易的作用》, 工作文件。

Newey, Whitney K., and Kenneth D. West, 1987, A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix, *Econometrica*, 55, 703-708.

Newey, Whitney k., Kenneth d. West, 1987, 一个简单的, 正半定的, 异方差和自相关一致的协方差矩阵, *Econometrica*, 55,703-708。

Newey, Whitney K., and Kenneth D. West, 1994, Automatic lag selection in covariance matrix estimation, *The Review of Economic Studies*, 61.4, 631-653.

协方差矩阵估计中的自动滞后选择, 《经济研究评论》, 61.4,631-653。

Ozik, Gideon, Ronnie Sadka, and Siyi Shen, 2021, Flattening the illiquidity curve: Retail trading during the COVID-19 lockdown, *Journal of Financial and Quantitative Analysis*, 56(7), 2356-2388.

Ozik, Gideon, Ronnie Sadka, and Siyi Shen, 2021, flat the illiquidity curve: Retail trading during the covid-19 lockdown, *Journal of Financial and Quantitative Analysis*, 56(7), 2356-2388,2019 冠状病毒疾病封锁期间的零售交易。

Stoffman, Noah, 2014, Who trades with whom? Individuals, institutions, and returns. *Journal of Financial Markets*, 21, 50-75.

斯托夫曼, 诺亚, 2014, 谁与谁交易? 个人, 机构和回报。《金融市场杂志》, 21,50-75。

Titman, Sheridan, Chishen Wei, and Bin Zhao, 2022, Corporate actions and the manipulation of retail investors in China: An analysis of stock splits. *Journal of Financial Economics*, 145(3), 762-787.

Titman, Sheridan, Chishen Wei, and Bin Zhao, 2022, 企业行为与中国散户投资者的操纵: 股票分割的分析。金融经济学杂志, 145(3), 762-787。

Van Kervel, Vincent, and Albert J. Menkveld, 2019, High frequency trading around large institutional orders. *Journal of Finance*, 74(3), 1091-1137.

范克维尔, 文森特和阿尔伯特 j. 门克维尔, 2019, 围绕大型机构订单的高频交易。

Welch, Ivo, 2022, The wisdom of the Robinhood crowd, *Journal of Finance*, 77(3), 1489-1527.

韦尔奇, Ivo, 2022, 《罗宾汉群体的智慧》, 《金融期刊》, 77(3), 1489-1527。

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Table I. Summary Statistics**表一. 统计摘要**

This table reports summary statistics for trading and holdings by different investor groups. Our sample period is January 2016 to June 2019, and our sample firms are A-share stocks listed on a major stock exchange with at least 15 trading days during the previous month. Panel A reports the number of accounts, aggregate trading and holdings by different types of investors. Panel B reports the time series average of the cross-sectional distribution of stock characteristics. Panel C presents the time series average of the cross-sectional statistics of order imbalances for each investor group. Order imbalances (*Oib*) are computed as the buy share volume minus sell share volume divided by buy share volume plus sell share volume for each investor group, as specified in equation (1).

本表汇报了不同投资者群体的交易和持股总结统计数据。我们的样本期间是 2016 年 1 月至 2019 年 6 月，我们的样本公司是在一个主要证券交易所上市的 a 股股票，在前一个月至少有 15 个交易日。面板 a 报告了不同类型投资者的账户数量、总交易量和持有量。B 组报告了股票特征横截面分布的时间序列平均值。面板 c 展示了每个投资者群体订单不平衡的横截面统计的时间序列平均值。订单不平衡(*Oib*)的计算方法是：每个投资者群体的买入股票数量减去卖出股票数量除以买入股票数量加卖出股票数量，如方程(1)所示。

Panel A. Number of accounts, trading, and holdings by different types of investors

不同类型投资者的账户、交易和持股数量

	RT1 RT1	RT2 RT2	RT3 RT3	RT4 RT4 (3M,10M) (3 米, 10 米)	RT5 RT5 >10M CNY 人民币超过一 千万	INST 国家科学 研究所	CORP 公司名称
Account value 帐户价值	<100K < 100K	(100K,500K) (100K, 500k)	(500K,3M) (500K, 3 米)				
Number of Accounts (thousands) 帐户数目(千)	31,410 31,410	15,282 15,282	5,827 5,827	735	235	40	47
Aggregate Trading Volume (Bil. CNY) 总成交量(人民币)	9	35	54	27	37	35	3
Aggregate Trading Volume (% of total) 总成交量(占总成交量的百分比)	5% 5%	17% 17%	27% 27%	13% 13%	19% 19%	17% 17%	2% 百分之二
Aggregate Holdings Value (Bil CNY) 总持有价值(人民币)	336	951	1,566	840	1,794	4,201	15,547
Aggregate Holdings Value (% of total) 总持有价值(占总值的百分比)	1% 1%	4% 4%	6% 6%	3% 3%	7% 7%	17% 17%	62% 62%

Panel B. Stock characteristics

股票特征

	Mean	Std	P25	P50	P75
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	刻薄	性病	P25	P50	P75
Market Capitalization (Billion CNY)					
市值(十亿元人民币)	20.1	80.3	2.9	5.6	12.1
Earnings to Price Ratio					
市盈率	0.0075	0.0155	0.0018	0.0060	0.0122
Daily Stock Return	-0.01%	2.17%	-1.09%	-0.22%	0.77%
每日股票回报	-0.01%	2.17%	-1.09%	-0.22%	0.77%
Daily Turnover (of tradable A shares)	2.45%	4.37%	0.61%	1.15%	2.40%
(可交易 a 股的)每日成交量	2.45%	4.37%	0.61%	1.15%	2.40%

Panel C. Order imbalance in the cross section by different types of investors
不同类型投资者在截面上的顺序不平衡

	Mean 刻薄	Std 性病	AR1 AR1	Correlations 相关性							
				OibRT1 OibRT1	OibRT2 OibRT2	OibRT3 OibRT3	OibRT4 OibRT4	OibRT5 OibRT5	OibINST 奥宾斯特	OibCORP OibCORP	
OibRT1	-0.021										
OibRT1	-0.021	0.187	0.243	1							
OibRT2	-0.011										
OibRT2	-0.011	0.171	0.259	0.802	1						
OibRT3	-0.006										
OibRT3	-0.006	0.166	0.216	0.610	0.710	1					
OibRT4											
OibRT4	0.002	0.250	0.059	0.194	0.244	0.256	1				
OibRT5				-0.151	-0.158	-0.163	-0.091				
OibRT5	0.019	0.352	0.102	-0.151	—0.158	-0.163	—0.091	1			
OibINST	-0.011			-0.315	-0.365	-0.380	-0.263	-0.188			
奥宾斯特	-0.011	0.455	0.205	-0.315	—0.365	-0.380	—0.263	-0.188	1		
OibCORP	-0.004						-0.007	-0.043	-0.044		
OibCORP	-0.004	0.720	0.088	0.022	0.029	0.021	-0.007	-0.043	—0.044	1	

Table II. Predicting Next Day Stock Returns Using Order Imbalances from Different Investor Groups

表二: 利用来自不同投资者群体的订单不平衡预测第二天的股票收益

This table reports the estimation results of whether trading activity by different investor groups can predict the cross-section of future stock returns. The coefficients are estimated from Fama-MacBeth (1973) regressions, as specified in equation (2). The dependent variable is the next-day stock return (*Ret*) and the independent variable is the previous day order imbalance *Oib*(-1). The control variables are the previous day return *Ret*(-1), previous week return *Ret*(-6,-2), previous month return *Ret*(-27,-7), previous month log market cap (*Size*), earnings to price ratio (*EP*), and monthly turnover (*Turnover*). The interquartile range for the relevant explanatory order imbalance is to compute the difference in predicted future returns for the interquartile range (*Interquartile return*). To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

本表报告了不同投资者群体的交易活动是否能够预测未来股票收益的截面的估计结果。系数是由 Fama-MacBeth (1973)回归估计的, 如方程(2)所示。因变量是次日股票收益率(*Ret*), 自变量是前一天的订单不平衡 *Oib* (- 1)。控制变量为前一天收益率(- 1)、前一周收益率(- 6,-2)、前一个月收益率(- 27,-7)、前一个月对数市值(*Size*)、盈利价格比(*EP*)和月成交量(*Turnover*)。相关解释性顺序不平衡的四分差范围是计算四分差范围(四分差收益)的预测未来收益的差异。为了考虑系数的序列相关性, 采用 Newey 和 West (1987)方法对时间序列的标准误差进行调整, 并按照 Newey 和 West (1994)方法选择最佳带宽。***, ** 和 * 分别表示 1%, 5% 和 10% 水平的显着性。

Dep.var		Ret	Ret	Ret	Ret	Ret	Ret
风险管理局		休息	休息	休息	休息	休息	休息
Oib.var		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5	OibINST
Oib.var		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5	奥宾斯特
Oib(-1)	Estimate	-0.0093***	-0.0091***	-0.0065***	-0.0009***	0.0012***	0.0016***
Oib (- 1)	估计	-0.0093 * * *	-0.0091 * * *	-0.0065 * * *	-0.0009 * * *	0.0012 * *	0.0016 * * *
	[t-stat]	[-17.36]	[-17.10]	[-15.74]	[-6.95]	[10.48]	[18.06]
	[t-stat]	[-17.36]	[-17.10]	[-15.74]	[-6.95]	[10.48]	[18.06]
Ret(-1)	Estimate	-0.0027	-0.0091**		0.0189***	0.0190***	0.0132**
Ret (- 1)	估计	-0.0027	-0.0091 * *	0.0006	0.0189 * * *	0.0190 * * *	0132 * *
	[t-stat]	[-0.61]	[-2.09]	[0.12]	[3.68]	[3.74]	[2.49]
	[t-stat]	[-0.61]	[-2.09]	[0.12]	[3.68]	[3.74]	[2.49]
Ret(-6,-2)	Estimate	-0.0149***	-0.0132***	-0.0124***	-0.0120***	-0.0115***	-0.0113***
Ret (- 6, -2)	估计	-0.0149 * * *	-0.0132 * * *	-0.0124 * * *	-0.0120 * * *	-0.0115 * *	-0.0113 * * *
	[t-stat]	[-8.17]	[-7.15]	[-6.66]	[-6.41]	[-6.17]	[-6.08]
	[t-stat]	[-8.17]	[-7.15]	[-6.66]	[-6.41]	[-6.17]	[-6.08]
Ret(-27,-7)	Estimate	-0.0039***	-0.0036***	-0.0034***	-0.0033***	-0.0032***	-0.0034***
Ret (- 27,-7)	估计	-0.0039 * * *	-0.0036 * * *	-0.0034 * * *	-0.0033 * * *	-0.0032 * * *	-0.0034 * * *
	[t-stat]	[-4.35]	[-4.08]	[-3.84]	[-3.70]	[-3.61]	[-3.84]

	[t-stat]	[-4.35]	[-4.08]	[-3.84]	[-3.70]	[-3.61]	[-3.84]
Size	Estimate						
大小	估计	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	[t-stat]	[0.36]	[0.17]	[-0.16]	[-0.32]	[-0.18]	[-0.21]
	[t-stat]	[0.36]	[0.17]	[-0.16]	[-0.32]	[-0.18]	[-0.21]
EP	Estimate	0.0147***	0.0150***	0.0145***	0.0144***	0.0146***	0.0140***
EP	估计	0.0147 * *	0.0150 美元	0.0145	0.0144 * *	0.0146 * * *	0.0140 美元
	[t-stat]	[2.88]	[2.89]	[2.75]	[2.77]	[2.82]	[2.72]
	[t-stat]	[2.88]	[2.89]	[2.75]	[2.77]	[2.82]	[2.72]
Turnover	Estimate	-0.0007***	-0.0007***	-0.0007***	-0.0007***	-0.0007***	-0.0007***
营业额	估计	—0.0007 * * *	—0.0007 * * *	—0.0007 * * *	—0.0007 * * *	—0.0007 * * *	—0.0007 * * *
	[t-stat]	[-3.50]	[-3.65]	[-3.64]	[-3.79]	[-3.79]	[-3.79]
	[t-stat]	[-3.50]	[-3.65]	[-3.64]	[-3.79]	[-3.79]	[-3.79]
Intercept	Estimate	-0.0012	-0.0006				
拦截	估计	—0.0012	-0.0006	0.0002	0.0005	0.0001	0.0002
	[t-stat]	[-0.47]	[-0.25]	[0.06]	[0.20]	[0.06]	[0.10]
	[t-stat]	[-0.47]	[-0.25]	[0.06]	[0.20]	[0.06]	[0.10]
Adj.R2		8.83%	8.68%	8.43%	8.10%	8.11%	8.25%
R2		8.83%	8.68%	8.43%	8.10%	8.11%	8.25%
Interquartile oib							
四分位数		0.2222	0.1827	0.1678	0.2868	0.4536	0.6740
Interquartile return		-0.2062%	-0.1668%	-0.1089%	-0.0247%	0.0523%	0.1046%
四分位间回归		-0.2062%	-0.1668%	-0.1089%	-0.0247%	0.0523%	0.1046%

Table III. Predictive Patterns for Next Day Returns with Different Counterparties
表三. 不同交易对手第二天报表的预测模式

This table reports the estimation results for the predictive patterns of next-day returns with different counterparties. For the counterparties, we put RT1 to RT4 together as one group and RT5 and INST as two separate groups. With the three groups of investors and two sides of trades (buy or sell), we divide the remaining trades into six bins: BBS, BSB, BSS, SBB, SBS, and SSB. The first letter indicates the trade direction of RT1-RT4, the second indicates the trade direction of RT5, and the third indicates INST. Panel A reports the coverage of each bin in the total sample. Panel B reports the coefficients estimated from the Fama-MacBeth (1973) regressions, as specified in equation (3). The control variables are the same as those in Table II and are not reported for abbreviations. To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

该表报告了不同交易对手的次日收益预测模式的估计结果。对于交易对手，我们将 rt1 至 rt4 作为一个组，rt5 和 INST 作为两个单独的组。通过三组投资者和两边交易(买卖)，我们将剩余的交易划分为六个区域: BBS、BSB、BSS、SBB、SBS 和 SSB。第一个字母表示 rt1-rt4 的交易方向，第二个字母表示 rt5 的交易方向，第三个字母表示 INST。面板 a 报告总样本中每个仓的覆盖率。图 b 报告了从方程(3)中指定的 Fama-MacBeth (1973)回归估计的系数。控制变量与表 II 中的相同，并且没有报告缩写。为了考虑系数的序列相关性，采用 Newey 和 West (1987)方法对时间序列的标准误差进行调整，并按照 Newey 和 West (1994)方法选择最佳带宽。***，** 和 * 分别表示 1%，5% 和 10% 水平的显着性。

Panel A. Distribution of counterparties
A. 交易对手的分布

	RT1-4 RT1-4	RT5 RT5	INST 国家科学研 究所	Proportions of Total Sample 总样本的比例
BBS	Buy	Buy	Sell	17%
BBS	买入	买入	卖掉	17%
BSB	Buy	Sell	Buy	11%
BSB	买入	卖掉	买入	11%
BSS	Buy	Sell	Sell	21%
BSS	买入	卖掉	卖掉	21%
SBB	Sell	Buy	Buy	22%
SBB	卖掉	买入	买入	22%
SBS				
SBS 苏格	Sell	Buy	Sell	13%
兰银行	卖掉	买入	卖掉	13%
SSB	Sell	Sell	Buy	17%
SSB	卖掉	卖掉	买入	17%

Panel B. Predictive patterns with different counterparties
面板 b: 不同交易对手的预测模式

	I Ret 休息	II 二 Ret 休息	III 三 Ret 休息
Dep.var 风险管理局			

1 ,		OibRT1-4	OibRT5	OibINST
1、 ,		oibRT1-4	OibRT5	奥宾斯特
Oib(-1)*BBS	Estimate	-0.0020***	-0.0007***	0.0021***
Oib (- 1) * BBS	估计	-0.0020	—0.0007 * * *	0.0021
	[t-stat]	[-2.85]	[-3.21]	[11.39]
	[t-stat]	[-2.85]	[-3.21]	[11.39]
Oib(-1)*BSB	Estimate	-0.0041***	0.0004**	-0.0014***
Oib (- 1) * BSB	估计	-0.0041 * * *	0.0004 * *	-0.0014 * * *
	[t-stat]	[-3.24]	[2.26]	[-4.53]
	[t-stat]	[-3.24]	[2.26]	[-4.53]
Oib(-1)*BSS	Estimate	-0.0079***	0.0036***	0.0036***
Oib (- 1) * BSS	估计	-0.0079 * * *	0.0036 * * *	0.0036 * * *
	[t-stat]	[-12.10]	[13.91]	[15.11]
	[t-stat]	[-12.10]	[13.91]	[15.11]
Oib(-1)*SBB	Estimate	-0.0145***	0.0032***	0.0019***
Oib (- 1) * SBB	估计	-0.0145 * * *	0.0032 * * *	0.0019 * * *
	[t-stat]	[-17.23]	[12.42]	[8.88]
	[t-stat]	[-17.23]	[12.42]	[8.88]
Oib(-1)*SBS	Estimate	-0.0255***	0.0023***	-0.0013***
Oib (- 1) * SBS	估计	-0.0255 * * *	0.0023 * *	-0.0013 * * *
	[t-stat]	[-10.57]	[7.59]	[-3.33]
	[t-stat]	[-10.57]	[7.59]	[-3.33]
Oib(-1)*SSB	Estimate	-0.0093***	-0.0027***	0.0015***
Oib (- 1) * SSB	估计	-0.0093 * * *	—0.0027 * * *	0.0015 * * *
	[t-stat]	[-13.20]	[-11.11]	[10.10]
	[t-stat]	[-13.20]	[-11.11]	[10.10]
Controls		Yes	Yes	Yes
控制器		是的	是的	是的
Adj.R2		9.18%	8.41%	8.60%
R2		9.18%	8.41%	8.60%

Table IV. Predictive Patterns for Next Day Return for Different Holding Horizons

表四：不同持股期限的第二天回报预测模式

This table reports the estimation results on the predictive patterns for the next day return for different holding horizons. We measure the holding days using the days-to-cover ratio (*DTCR*), defined as the total shares held by investor group *G* divided by daily shares traded on stock *i* for this investor group. Panel A reports the cross-sectional distribution of each investor group's holding horizons. Panel B reports the coefficients estimated from the Fama-MacBeth (1973) regressions, as specified in equation (4). We control the relative trading volume (*RVol*), defined as the daily trading volume divided by the average daily trading volume from previous 49 days. Other control variables are the same as those in Table II, and are not reported for brevity. To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

这个表格报告了对不同持仓时间段的第二天收益预测模式的估计结果。我们使用日覆盖率(DTCR)来度量持有天数，定义为投资者 *g* 集团所持有的总股份除以该投资者集团在 *i* 股票上交易的日股份。面板 a 报告了每个投资者集团的持股范围的横截面分布。小组 b 报告了方程(4)中指定的 Fama-MacBeth (1973)回归估计的系数。我们控制相对交易量(RVol)，定义为日交易量除以前 49 天的平均日交易量。其他控制变量与表 II 中的相同，并且为了简洁起见没有报告。为了考虑系数的序列相关性，采用 Newey 和 West (1987)方法对时间序列的标准误差进行调整，并按照 Newey 和 West (1994)方法选择最佳带宽。***，** 和 * 分别表示 1%，5% 和 10% 水平的显著性。

Panel A. Distribution of holding horizon (days-to-cover-ratio)

面板 a. 持有期限的分布(天数对覆盖率)

	Mean 刻薄	Std 性病	P5 P5	P25 P25	P50 P50	P75 P75	P95 P95
RT1							
RT1	57	691	4	11	24	49	116
RT2							
RT2	43	402	4	10	20	36	74
RT3							
RT3	46	458	4	12	22	38	73
RT4							
RT4	64	833	4	13	26	48	114
RT5		4,479					
RT5	251	4,479	4	17	41	95	420
INST							
国家科学研究 所	199	3,750 3750	3	12	31	72	265

Panel B. Predictive patterns for different holding horizons

面板 b: 不同持有视野的预测模式

Dep.var 风险管理局		Ret 休息 OibRT1 OibRT1	Ret 休息 OibRT2 OibRT2	Ret 休息 OibRT3 OibRT3	Ret 休息 OibRT4 OibRT4	Ret 休息 OibRT5 OibRT5	Ret 休息 OibINST 奥宾斯特
Oib(-1)*(0,10]Days Oib (- 1) * (0,10) Days (0,10)天	Estimate 估计 [t-stat] [t-stat]	-0.0171*** -0.0171 * * * [-29.84] [-29.84]	-0.0191*** -0.0191 * * * [-28.77] [-28.77]	-0.0172*** -0.0172 * * * [-23.05] [-23.05]	-0.0023*** -0.0023 * * * [-4.28] [-4.28]	0.0056*** 0.0056 * * [13.68] [13.68]	0.0017*** 0.0017 [12.51] [12.51]
Oib(-1)*(10,20]Days Oib (- 1) * (10,20)天	Estimate 估计 [t-stat] [t-stat]	-0.0086*** -0.0086 * * * [-21.46] [-21.46]	-0.0078*** -0.0078 * * * [-19.74] [-19.74]	-0.0054*** -0.0054 * * [-12.75] [-12.75]	-0.0006** -0.0006 [-2.43] [-2.43]	0.0015*** 0.0015 * * * [7.07] [7.07]	0.0012*** 0.0012 * * [9.62] [9.62]
Oib(-1)*(20,60]Days Oib (- 1) * (20,60)天	Estimate 估计 [t-stat] [t-stat]	-0.0064*** -0.0064 * * * [-18.24] [-18.24]	-0.0052*** -0.0052 * * [-15.42] [-15.42]	-0.0033*** -0.0033 * * * [-11.82] [-11.82]	-0.0004*** -0.0004 * * * [-2.81] [-2.81]	0.0007*** 0.0007 * * * [6.37] [6.37]	0.0013*** 0.0013 * * [11.49] [11.49]
Oib(-1)*Above60Days Oib (- 1) * 超过 60 天	Estimate 估计 [t-stat] [t-stat]	-0.0074*** -0.0074 * * * [-6.26] [-6.26]	-0.0007 —0.0007 [-0.12] [-0.12]	-0.0086* -0.0086 * [-1.72] [-1.72]	0.0008 0.0008 [0.35] [0.35]	0.0002 0.0002 [1.52] [1.52]	0.0008*** 0.0008 * * [7.80] [7.80]
Rvol(-1) Rvol (- 1)	Estimate 估计 [t-stat] [t-stat]	-0.0007*** —0.0007 * * * [-5.91] [-5.91]	-0.0008*** -0.0008 * * * [-6.97] [-6.97]	-0.0011*** -0.0011 * * * [-9.03] [-9.03]	-0.0014*** -0.0014 * * * [-11.31] [-11.31]	-0.0013*** -0.0013 * * * [-10.82] [-10.82]	-0.0013*** -0.0013 * * * [-10.88] [-10.88]
Controls 控制器 Adj.R2 R2		Yes 是的 10.03% 10.03%	Yes 是的 9.94% 9.94%	Yes 是的 9.72% 9.72%	Yes 是的 9.24% 9.24%	Yes 是的 9.31% 9.31%	Yes 是的 9.26% 9.26%

Table V. Predict Returns over the Next 12 Weeks**预测未来 12 周的回报**

This table reports the estimation results of whether trading activity by different investor groups can predict the cross-section of returns over the next 12 weeks. We report the coefficient estimates from the Fama-MacBeth (1973) regressions specified in equation (5). The main independent variables are the previous day's order imbalance, *Oib*(-1). The control variables are the same as those in Table II, and are not reported for brevity. To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

本表报告了不同投资者群体的交易活动是否能够预测未来 12 周的回报横截面的估计结果。我们报告的系数估计从 Fama-MacBeth (1973)回归中指定的方程(5)。主要的自变量是前一天的订单不平衡, *Oib* (- 1)。控制变量与表 II 中的相同, 并且为了简洁起见没有报告。为了考虑系数的序列相关性, 采用 Newey 和 West (1987)方法对时间序列的标准误差进行调整, 并按照 Newey 和 West (1994)方法选择最佳带宽。***, ** 和 * 分别表示 1%, 5% 和 10% 水平的显着性。

wth weeks 几个星期了		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5	OibINST
		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5	奥宾斯特
1	Estimate	-0.0226***	-0.0220***	-0.0144***	-0.0019***	0.0027***	0.0044***
	估计	-0.0226 * * *	-0.0220 * * *	-0.0144 * * *	-0.0019 * * *	0.0027 美元	0.0044 * * *
	[t-stat]	[-18.84]	[-16.98]	[-13.43]	[-6.70]	[10.17]	[17.98]
	[t-stat]	[-18.84]	[-16.98]	[-13.43]	[-6.70]	[10.17]	[17.98]
2	Estimate	-0.0065***	-0.0060***	-0.0037***		0.0010***	0.0012***
	估计	-0.0065 * * *	-0.0060 * * *	-0.0037 * * *	0.0001	0.0010 * * *	0.0012 * *
	[t-stat]	[-9.83]	[-8.56]	[-5.89]	[0.50]	[6.01]	[5.64]
	[t-stat]	[-9.83]	[-8.56]	[-5.89]	[0.50]	[6.01]	[5.64]
3	Estimate	-0.0038***	-0.0031***	-0.0015**		0.0007***	0.0007***
	估计	-0.0038 * * *	-0.0031 * * *	-0.0015 * *	0.0001	0.0007 * * *	0.0007 * * *
	[t-stat]	[-5.37]	[-4.09]	[-2.31]	[0.45]	[3.77]	[3.17]
	[t-stat]	[-5.37]	[-4.09]	[-2.31]	[0.45]	[3.77]	[3.17]
4	Estimate	-0.0024***	-0.0021**	-0.0012	-0.0001	0.0007***	0.0005**
	估计	-0.0024 * * *	-0.0021 * *	-0.0012	-0.0001	0.0007 * * *	0.0005
	[t-stat]	[-3.46]	[-2.57]	[-1.63]	[-0.37]	[3.94]	[2.09]
	[t-stat]	[-3.46]	[-2.57]	[-1.63]	[-0.37]	[3.94]	[2.09]
5	Estimate	-0.0014***	-0.0014**	-0.0011**	-0.0005**	0.0004**	
	估计	-0.0014 * * *	-0.0014 * *	-0.0011 * *	-0.0005 * *	0.0004 * *	0.0002
	[t-stat]	[-2.69]	[-2.44]	[-2.07]	[-2.00]	[2.25]	[1.39]
	[t-stat]	[-2.69]	[-2.44]	[-2.07]	[-2.00]	[2.25]	[1.39]
6	Estimate	-0.0029***	-0.0024***	-0.0016***	-0.0003		0.0005***
	估计	-0.0029 * * *	-0.0024 * * *	-0.0016 * * *	-0.0003	0.0001	0.0005 * * *

	[t-stat]	[-4.69]	[-3.55]	[-2.71]	[-1.22]	[0.72]	[2.66]
	[t-stat]	[-4.69]	[-3.55]	[-2.71]	[-1.22]	[0.72]	[2.66]
	Estimate	-0.0027***	-0.0025***	-0.0018***	-0.0002		0.0007***
7	估计	—0.0027 * * *	-0.0025 * * *	-0.0018 * * *	-0.0002	0.0001	0.0007 * * *
	[t-stat]	[-5.02]	[-4.43]	[-3.65]	[-0.68]	[0.39]	[4.25]
	[t-stat]	[-5.02]	[-4.43]	[-3.65]	[-0.68]	[0.39]	[4.25]
	Estimate	-0.0015**	-0.0010	-0.0007	-0.0002	0.0003**	0.0004**
8	估计	-0.0015 * *	-0.0010	—0.0007	-0.0002	0003 * *	0.0004 * *
	[t-stat]	[-2.57]	[-1.52]	[-1.08]	[-0.73]	[2.04]	[2.39]
	[t-stat]	[-2.57]	[-1.52]	[-1.08]	[-0.73]	[2.04]	[2.39]
	Estimate	-0.0010*	-0.0005	-0.0004		0.0004**	
9	估计	-0.0010 *	-0.0005	-0.0004	0.0002	0.0004 * *	0.0001
	[t-stat]	[-1.94]	[-1.00]	[-0.73]	[1.03]	[2.43]	[0.43]
	[t-stat]	[-1.94]	[-1.00]	[-0.73]	[1.03]	[2.43]	[0.43]
	Estimate	-0.0007	-0.0004	-0.0004		-0.0001	
10	估计	—0.0007	-0.0004	-0.0004	0.0002	—0.0001	0.0002
	[t-stat]	[-1.22]	[-0.60]	[-0.81]	[0.94]	[-0.44]	[0.90]
	[t-stat]	[-1.22]	[-0.60]	[-0.81]	[0.94]	[-0.44]	[0.90]
	Estimate	-0.0006	-0.0007	-0.0001		-0.0002	
11	估计	-0.0006	—0.0007	—0.0001	0.0000	-0.0002	0.0002
	[t-stat]	[-1.02]	[-1.01]	[-0.12]	[-0.04]	[-1.10]	[1.10]
	[t-stat]	[-1.02]	[-1.01]	[-0.12]	[-0.04]	[-1.10]	[1.10]
	Estimate	-0.0005	-0.0001				
12	估计	-0.0005	—0.0001	0.0005	0.0001	0.0001	0.0000
	[t-stat]	[-0.90]	[-0.12]	[0.76]	[0.41]	[0.39]	[0.05]
	[t-stat]	[-0.90]	[-0.12]	[0.76]	[0.41]	[0.39]	[0.05]

Table VI. Two Stage Decomposition for Understanding the Predictive Patterns of Retail Order Flows

表六: 理解零售订单流预测模式的两阶段分解

This table reports the estimation results of the decomposition of the predictive power of different investor groups' order imbalances for the cross-section of future stock returns. We estimate two-stage Fama-MacBeth (1973) regressions. Panel A reports the first-stage estimation results, where the order imbalance measures are decomposed into five components as specified in equation (A1). Variable *Overconf*(-1) is measured as the corresponding investor group's average turnover on the stock from the previous 20 days as a proxy for each investor's overconfidence. The variable *Gamble*(-1) is the maximum daily return from the previous 20 days as a proxy for gambling preference. Panel B reports the second-stage decomposition of the order imbalance's predictive power, as specified in equations (A2) and (A3). As an independent variable, the variable *Oib*(-1,*Persistence*) is estimated in the first stage using the past order imbalance and reflects price pressure. The variable *Oib*(-1,*Liquidity*) is estimated in the first stage using past returns over different horizons, and is connected to the liquidity provision or liquidity demand hypothesis. The variable *Oib*(-1,*Overconf*) is estimated in the first stage, reflecting overconfidence. The variable *Oib*(-1,*Gamble*) is estimated in the first stage using the maximum daily returns from the previous 20 days and reflects a preference for gambling. The residual part of the previous day order imbalance from the first-stage estimation is denoted "other," which can be attributed to private information about future returns. The control variables are the same as those in Table II, and are not reported for brevity. For each regression, we also report the difference in predicted day-ahead returns for observations at the two ends of the interquartile range (*Interquartile return*) in Panel B. To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

这个表格报告了不同投资者群体的订单不平衡对未来股票收益横截面的预测能力分解的估计结果。我们估计两阶段 Fama-MacBeth (1973)回归。面板 a 报告第一阶段的估计结果，其中顺序不平衡度量被分解成方程(A1)中指定的五个组分。变量 *Overconf* (- 1)是以相应投资者群体过去 20 天的股票平均成交量作为衡量每个投资者过度自信的指标。变量 *Gamble* (- 1)是过去 20 天的最大日回报率，作为赌博偏好的代表。面板 b 报告了顺序不平衡的预测能力的第二阶段分解，如方程(A2)和(A3)所指定的。作为一个自变量，变量 *Oib* (- 1, 持久性)在第一阶段使用过去的订单不平衡进行估计，并反映价格压力。变量 *Oib* (- 1, 流动性)是在第一阶段使用过去的回报在不同的视野估计，并与流动性提供或流动性需求假设。变量 *Oib* (- 1, *Overconf*)在第一阶段被估计，反映了过度自信。变量 *Oib* (- 1, *Gamble*)在第一阶段使用前 20 天的最大日收益进行估计，并反映出对赌博的偏好。来自第一阶段估计的前一天订单不平衡的剩余部分被称为“其他”，这可以归因于有关未来收益的私人信息。控制变量与表 II 中的控制变量相同，并且为了简洁起见没有报告。对于每个回归，我们还报告了 b 组中四分位间距(四分位间距)两端观测的预测日前收益的差异。为了说明系数的序列相关性，使用 Newey 和 West (1987)方法对时间序列的标准误差进行了调整，按照 Newey 和 West (1994)方法选择了最佳带宽。***，** 和 * 分别表示 1%，5% 和 10% 水平的显著性。

Panel A. First stage of projecting the order imbalance on persistence, past returns, overconfidence, and gambling proxies

第一阶段的预测顺序不平衡的持续性，过去的回报，过度自信，和赌博代理

Dep.var	OibRT1	OibRT2	OibRT3	OibRT4	OibRT5
风险管理局	OibRT1	OibRT2	OibRT3	OibRT4	OibRT5

Oib(-1)	Estimate	0.1870***	0.1967***	0.1711***	0.0499***	0.1036***
Oib (- 1)	估计	1870 年 * * *	1967 年 9 月 10 日	0.1711	0.0499 * * *	0.1036 * * *
	[t-stat]	[24.24]	[32.92]	[35.03]	[15.03]	[27.85]
	[t-stat]	[24.24]	[32.92]	[35.03]	[15.03]	[27.85]
Ret(-1)	Estimate	0.5159***	0.7269***	0.4482***	-0.2205***	-1.2968***
Ret (- 1)	估计	0.5159 * * *	0.7269 * * *	0.4482 美元	-0.2205 * *	-1.2968 * * *
	[t-stat]	[9.17]	[16.36]	[14.52]	[-7.57]	[-28.84]
	[t-stat]	[9.17]	[16.36]	[14.52]	[-7.57]	[-28.84]
Ret(-6,-2)	Estimate	-0.4214***	-0.2196***	-0.1063***	-0.0804***	-0.0485***
Ret (- 6, -2)	估计	-0.4214 * * *	-0.2196 * * *	-0.1063 * * *	-0.0804 * * *	-0.0485 * * *
	[t-stat]	[-19.83]	[-15.14]	[-7.71]	[-4.22]	[-2.74]
	[t-stat]	[-19.83]	[-15.14]	[-7.71]	[-4.22]	[-2.74]
Ret(-27,-7)	Estimate	-0.0350***	-0.0196***	-0.0235***	-0.0399***	-0.0203***
Ret (- 27,-7)	估计	-0.0350 * * * *	-0.0196 * * *	-0.0235 * * *	-0.0399 * * *	-0.0203 * * *
	[t-stat]	[-6.49]	[-4.89]	[-6.72]	[-8.36]	[-3.74]
	[t-stat]	[-6.49]	[-4.89]	[-6.72]	[-8.36]	[-3.74]
Overconf(-1)	Estimate	0.0894***	0.0611***	0.0657***	0.0418***	-0.0881***
Overconf (- 1)	估计	0.0894 * *	0.0611 * *	0.0657 * * *	0.0418 * * *	-0.0881 * * *
	[t-stat]	[4.53]	[4.57]	[7.28]	[3.64]	[-8.80]
	[t-stat]	[4.53]	[4.57]	[7.28]	[3.64]	[-8.80]
Gamble(-1)	Estimate		0.0784***	0.1731***	0.2423***	-0.0671**
Gamble (- 1)	估计	0.0330	0.0784 * * *	0.1731 * * *	0.2423 * *	-0.0671 * *
	[t-stat]	[1.49]	[5.06]	[11.69]	[11.90]	[-2.53]
	[t-stat]	[1.49]	[5.06]	[11.69]	[11.90]	[-2.53]
Intercept	Estimate	-0.0214***	-0.0120***	-0.0115***	-0.0078***	0.0231***
拦截	估计	-0.0214 * * *	-0.0120 * * *	-0.0115 * *	-0.0078 * * *	0.0231 * *

	[t-stat]	[-5.70]	[-5.09]	[-7.63]	[-4.00]	[9.41]
	[t-stat]	[-5.70]	[-5.09]	[-7.63]	[-4.00]	[9.41]
Adj.R2		7.08%	5.60%	3.89%	0.73%	2.00%
R2		7.08%	5.60%	3.89%	0.73%	2.00%

Panel B. Second stage decomposition of order imbalance's predictive power

面板 b: 顺序不平衡预测能力的第二阶段分解

Dep.var		Ret	Ret	Ret	Ret	Ret
风险管理局		休息	休息	休息	休息	休息
Oib.var		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5
Oib.var		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5
Oib(-1,Persistence)	Estimate	-0.0301***	-0.0276***	-0.0205***	-0.0086***	0.0060***
Oib (- 1, 持久性)	估计	-0.0301 * * *	-0.0276 * * *	-0.0205 * * *	-0.0086 * * *	0.0060 * *
	[t-stat]	[-16.73]	[-15.39]	[-12.22]	[-4.37]	[7.87]
	[t-stat]	[-16.73]	[-15.39]	[-12.22]	[-4.37]	[7.87]
Oib(-1,Liquidity)	Estimate	-0.0135***	-0.0205***	-0.0261***		0.0042*
离岸价格(- 1, 流动性)	估计	-0.0135 * * *	-0.0205 * * *	-0.0261 * * *	0.0140	0.0042 *
	[t-stat]	[-4.48]	[-6.18]	[-4.69]	[0.69]	[1.69]
	[t-stat]	[-4.48]	[-6.18]	[-4.69]	[0.69]	[1.69]
Oib(-1,Overconf)	Estimate	-0.1006***	-0.2158***	-0.2413***	-0.3730***	0.1134***
Oib (- 1, Overconf)	估计	-0.1006 * * *	-0.2158 * * *	-0.2413 * *	-0.3730 * * *	0.1134 * * *
	[t-stat]	[-4.00]	[-5.97]	[-6.05]	[-5.85]	[5.90]
	[t-stat]	[-4.00]	[-5.97]	[-6.05]	[-5.85]	[5.90]
Oib(-1,Gamble)	Estimate	-0.2877***	-0.0981***	-0.0474***	-0.0389***	0.1918***
Oib (- 1, Gamble)	估计	-0.2877 * * *	-0.0981 * * *	-0.0474 * * *	-0.0389 * * *	0.1918 * *
	[t-stat]	[-4.89]	[-4.08]	[-4.35]	[-4.98]	[6.35]
	[t-stat]	[-4.89]	[-4.08]	[-4.35]	[-4.98]	[6.35]
Oib(-1,Other)	Estimate	-0.0086***	-0.0084***	-0.0060***	-0.0008***	0.0011***
Oib (- 1, 其他)	估计	-0.0086 * * *	-0.0084 * * *	-0.0060 * * *	-0.0008 * * *	0.0011 * * *
	[t-stat]	[-15.54]	[-15.51]	[-14.16]	[-6.19]	[9.59]
	[t-stat]	[-15.54]	[-15.51]	[-14.16]	[-6.19]	[9.59]
Adj.R2		10.50%	10.35%	10.07%	9.59%	9.52%
R2		10.50%	10.35%	10.07%	9.59%	9.52%
Interquartile return						
四分位间回归						
Oib(-1,Persistence)		-0.1177%	-0.0962%	-0.0591%	-0.0125%	0.0281%

Oib (- 1, 持久性)	-0.1177%	-0.0962%	-0.0591%	-0.0125%	0.0281%
Oib(-1,Liquidity)					
离岸价格(- 1, 流动性)	-0.0290%	-0.0351%	-0.0261%	0.0091%	0.0097%
Oib(-1,Overconf)	-0.0337%	-0.0475%	-0.0484%	-0.0428%	0.0257%
Oib (- 1, Overconf)	-0.0337%	-0.0475%	-0.0484%	-0.0428%	0.0257% 0.0257%
Oib(-1,Gamble)	-0.0314%	-0.0254%	-0.0271%	-0.0312%	0.0425%
Oib (- 1, Gamble)	-0.0314%	-0.0254%	-0.0271%	-0.0312%	0.0425%
Oib(-1,Other)	-0.1778%	-0.1493%	-0.1008%	-0.0233%	0.0490%
Oib (- 1, 其他)	-0.1778%	-0.1493%	-0.1008%	-0.0233%	0.0490%

Table VII. A Closer Look at the Relation between Investor Order Flows and Earnings News

表七: 投资者订单流量与盈利新闻关系的近期分析

This table reports the estimation results on the relation between different investor groups' order flows and public earnings announcements. Panel A reports whether investor order flow can predict earnings surprises, as specified in equation (6). The dependent variable is the cumulative abnormal return from day $d-1$ to day d , $CAR[-1,0]$, and the independent variable is order imbalance from day $d-2$, $Oib(-2)$. Panel B reports whether trades from different retail groups can process contemporaneous news, as specified in equation (7). The dependent variable is the order imbalance $Oib(0)$, and the independent variable is the cumulative abnormal return from day $d-1$ to day d , $CAR[-1,0]$. Panel C reports the effect of earnings news days on the return predictability of different investor group trades, as specified in equation (8). The dependent variable is the return on day d , and the independent variables include the previous day's order imbalance $Oib(-1)$, the news dummy $Event(-1)$, and the interaction terms $Oib(-1)*Event(-1)$. The $Event(-1)$ dummy equals 1 if there is an earnings announcement for that firm day and zero otherwise. The other control variables are the same as those in Table II, and these coefficients are not reported. To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

本表报告了不同投资者群体的订单流量与公开收益公告之间关系的估计结果。面板 a 报告投资者订单流是否可以预测盈利意外, 如方程(6)所示。因变量是从第 $d-1$ 天到第 d 天的累积异常收益, $CAR [-1,0]$, 自变量是从第 $d-2$ 天的顺序不平衡, $Oib (-2)$ 。小组 b 报告来自不同零售集团的交易是否可以处理同时的新闻, 如方程(7)所述。因变量是顺序不平衡的 $Oib (0)$, 自变量是从第 $d-1$ 天到第 d 天累积的异常收益, $CAR [-1,0]$ 。面板 c 报告盈利新闻日对不同投资者集团交易的回报可预测性的影响, 如方程(8)所述。因变量是第 d 天的返回值, 自变量包括前一天的订单不平衡 $Oib (-1)$ 、新闻假人 $Event (-1)$ 和交互项 $Oib (-1) * Event (-1)$ 。如果该公司当天有收益公告, 则 $Event (-1)$ 假设值等于 1, 否则为 0。其他控制变量与表二中的相同, 这些系数没有报告。为了考虑系数的序列相关性, 采用 Newey 和 West (1987)方法对时间序列的标准误差进行调整, 并按照 Newey 和 West (1994)方法选择最佳带宽。***, ** 和 * 分别表示 1%, 5% 和 10% 水平的显著性。

Panel A. Investor order flow predicting future earnings announcement news events

投资者订单流预测未来收益公告新闻事件

Dep.var		CAR[-1,0]	CAR[-1,0]	CAR[-1,0]	CAR[-1,0]	CAR[-1,0]
风险管理局		CAR [-1,0]	CAR [-1,0]	CAR [-1,0]	CAR [-1,0]	CAR [-1,0]
Oib.var		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5
Oib.var		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5
Oib(-2)	Estimate	-0.0251***	-0.0234***	-0.0166***	-0.0003	0.0023***
Oib (- 2)	估计	-0.0251 * * *	-0.0234 * * *	-0.0166 * * *	-0.0003	0.0023 * *
	[t-stat]	[-8.12]	[-5.81]	[-4.56]	[-0.50]	[3.67]
	[t-stat]	[-8.12]	[-5.81]	[-4.56]	[-0.50]	[3.67]
Adj.R2		6.33%	5.98%	5.57%	5.19%	5.15%
R2		6.33%	5.98%	5.57%	5.19%	5.15%

Panel B. Investor order flow regressed on contemporaneous earnings announcement news events

面板 b: 投资者订单流量因同期收益公告新闻事件而出现回归

Dep.var		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5
风险管理局		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5
CAR[-1,0]	Estimate	-1.9225***	-1.8291***	-1.4349***	-0.8781***	0.1583**
CAR [-1,0]	估计	-1.9225 * *	—1.8291 * * *	-1.4349 * * *	-0.8781 * * *	0.1583 * *
	[t-stat]	[-22.65]	[-40.09]	[-14.53]	[-5.40]	[2.12]
	[t-stat]	[-22.65]	[-40.09]	[-14.53]	[-5.40]	[2.12]
Adj.R2		13.66%	14.60%	10.14%	1.85%	0.60%
R2		13.66%	14.60%	10.14%	1.85%	0.60%

Panel C. Return predictive power of investor order flow interacted with earnings announcement news events

面板 c 投资者订单流与收益公告新闻事件相互作用的预测能力

Dep.var		Ret	Ret	Ret	Ret	Ret
风险管理局		休息	休息	休息	休息	休息
Oib.var		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5
Oib.var		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5
Oib(-1)	Estimate	-0.0079***	-0.0070***	-0.0038***	-0.0008*	0.0005**
Oib (- 1)	估计	-0.0079 * * *	-0.0070 * * * *	-0.0038 * * *	—0.0008 *	0.0005
	[t-stat]	[-9.33]	[-7.67]	[-5.55]	[-1.93]	[3.08]
	[t-stat]	[-9.33]	[-7.67]	[-5.55]	[-1.93]	[3.08]
Oib(-1)*Event(-1)						
Oib (- 1) * 事件(-1)	Estimate	-0.0080***	-0.0093***	-0.0071**	-0.0007	0.0014**
	估计	-0.0080 * * *	-0.0093 * * *	-0.0071 * *	—0.0007	0.0014 * *

	[t-stat]	[-4.22]	[-3.90]	[-2.52]	[-1.54]	[2.07]
	[t-stat]	[-4.22]	[-3.90]	[-2.52]	[-1.54]	[2.07]
Event(-1)	Estimate	0.0011***	0.0008****	0.0006***	0.0005**	0.0005**
事件(- 1)	估计	0.0011 * * *	0.0008 美元	0.0006 * *	0.0005	0.0005
	[t-stat]	[3.94]	[2.89]	[2.80]	[2.55]	[2.40]
	[t-stat]	[3.94]	[2.89]	[2.80]	[2.55]	[2.40]
Adj.R2		7.36%	7.16%	6.82%	6.38%	6.35%
R2		7.36%	7.16%	6.82%	6.38%	6.35%

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Table VIII. Annual Performance Using Tracking Portfolios of Retail Trading, from Stock Selection and Market Timing
表八: 利用跟踪零售交易组合的年度业绩, 来自股票选择和市场时机

This table reports the annual performance for different retail investor groups using Barber, Lee, Liu, and Odean's (2009) approach between January 2016 and June 2019. The net buy and sell portfolios are constructed each day, as specified in equation (A4), and the average construction prices are defined in equation (A6). The daily portfolio's total performance of holding n days is the portfolio cumulative excess return minus the trading cost, as specified in equation (A5), and we choose n equals to 140 days following Barber et al. (2009). We then annualize the daily portfolio's total performance by timing the 240 trading days on each year and dividing by aggregate holding value in Table I Panel A to present the investor group's annual performance in percentage. The total performance could be decomposed into stock selection, market timing, and trading cost, as specified in equation (A7)-(A10). Panel A presents the total annual performance. Panel B reports annual performance conditional on the momentum and contrarian. If the investor group G 's order imbalance times previous day return, $(\sum_{i=1}^n \Delta R_{i,t}) * (\sum_{i=1}^n \Delta R_{i,t-1}) > 0$, we classify the stock i on day d for investor group G to be momentum; if $(\sum_{i=1}^n \Delta R_{i,t}) * (\sum_{i=1}^n \Delta R_{i,t-1}) \leq 0$, we classify the stock i on day d for investor group G to be contrarian. Panel C reports the annual performance of the earnings announcements. Earnings announcement events include the stock on announcement day d and the previous day $d-1$. To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

这个表格使用 Barber, Lee, Liu 和 Odean's (2009)的方法报告了 2016 年 1 月至 2019 年 6 月期间不同散户投资者群体的年度业绩。按照方程(A4)的规定, 每天构造净买卖投资组合, 平均建筑价格在方程(A6)中定义。每日投资组合持有 n 天的总表现是投资组合累积超额收益减去交易成本, 如方程(A5)所述, 我们选择 n 等于 Barber 等人(2009)之后的 140 天。然后, 我们通过计算每年 240 个交易日的时间, 除以表 i 面板 a 中的总持有价值, 将每日投资组合的总表现年化, 以百分比表示投资者集团的年度表现。总表现可以分解为股票选择, 市场时机, 和交易成本, 如方程(A7)-(A10)所指定。面板 a 显示了年度总体业绩。小组 b 报告的年度业绩取决于势头和反转。如果投资者群体 g 的顺序失衡乘以前一天的回报, $(\sum_{i=1}^n \Delta R_{i,t}) * (\sum_{i=1}^n \Delta R_{i,t-1}) > 0$, 我们将 d 天的股票 i 归类为投资者群体 g 的动量; 如果 $(\sum_{i=1}^n \Delta R_{i,t}) * (\sum_{i=1}^n \Delta R_{i,t-1}) \leq 0$, 我们将 d 天的股票 i 归类为投资者群体 g 的反向投资。面板 c 报告盈利公告的年度表现。收益公告事件包括公告日 d 和前一天 $d-1$ 的股票。为了考虑系数的序列相关性, 采用 Newey 和 West (1987)方法对时间序列的标准误差进行调整, 并按照 Newey 和 West (1994)方法选择最佳带宽。***, ** 和 * 分别表示 1%, 5% 和 10% 水平的显着性。

Panel A. Annual trading performance using tracking portfolios
使用跟踪投资组合的年度交易业绩

		Total 总计	Stock Selection 股票选择	Market Timing 市场时机	Trading Cost 交易成本
RT1	Performance	-5.61%***	-4.03%***	-0.19%	-1.38%***
RT1	表演	-5.61%***	-4.03%**	-0.19%	-1.38%**
	[t-stat]	[-10.29]	[-8.39]	[-0.56]	[-23.47]
	[t-stat]	[-10.29]	[-8.39]	[-0.56]	[-23.47]
RT2	Performance	-4.61%***	-2.55%***	-0.27%	-1.80%***
RT2	表演	-4.61%***	-2.55%***	-0.27%	-1.80%***
	[t-stat]	[-8.96]	[-6.21]	[-0.93]	[-23.07]
	[t-stat]	[-8.96]	[-6.21]	[-0.93]	[-23.07]
RT3	Performance	-3.60%***	-1.60%***	-0.29%*	-1.72%***
RT3	表演	-3.60%**	-1.60%**	-0.29%*	-1.72%

	[t-stat]	[-9.49]	[-5.51]	[-1.71]	[-20.89]
	[t-stat]	[-9.49]	[-5.51]	[-1.71]	[-20.89]
RT4	Performance	-2.80%***	-0.90%***	-0.33%*	-1.58%***
RT4	表演	-2.80% * * *	-0.90%	-0.33%	-1.58% * * *
	[t-stat]	[-8.51]	[-3.98]	[-1.78]	[-19.53]
	[t-stat]	[-8.51]	[-3.98]	[-1.78]	[-19.53]
RT5	Performance	-0.29%	1.05%***	-0.30%	-1.03%***
RT5	表演	-0.29%	1.05%	-0.30%	-1.03%
	[t-stat]	[-0.92]	[5.27]	[-1.39]	[-18.84]
	[t-stat]	[-0.92]	[5.27]	[-1.39]	[-18.84]
INST					
国家科					
学研究	Performance	1.15%***	1.38%***	0.19%	-0.41%***
所	表演	1.15% * *	1.38% * * *	0.19%	-0.41% * *
	[t-stat]	[3.57]	[4.85]	[0.93]	[-20.49]
	[t-stat]	[3.57]	[4.85]	[0.93]	[-20.49]

Panel B. Annual trading performance using tracking portfolios: separate momentum and contrarian trading
 面板 b 使用跟踪投资组合的年度交易业绩: 单独的动量和反向交易

		Total 总计		StockSelection 股票选择		MarketTiming 市场时机		TradingCost 交易成本	
		Contrarian 反向投资	Momentum 动量	Contrarian 反向投资	Momentum 动量	Contrarian 反向投资	Momentum 动量	Contrarian 反向投资	Momentum 动量
RT1	Performance	-1.04%**	-4.57%***	0.07%	-4.10%***	-0.41%	0.22%	-0.70%***	-0.68%***
RT1	表演	-1.04% * *	-4.57% * *	0.07%	-4.10%	-0.41%	0.22%	-0.70% * *	-0.68%
	[t-stat]	[-2.57]	[-8.47]	[0.25]	[-8.51]	[-1.39]	[1.03]	[-23.33]	[-21.29]
	[t-stat]	[-2.57]	[-8.47]	[0.25]	[-8.51]	[-1.39]	[1.03]	[-23.33]	[-21.29]
RT2	Performance	-0.95%**	-3.67%***	0.43%	-2.97%***	-0.46%*	0.19%	-0.91%***	-0.89%***
RT2	表演	-0.95% * *	-3.67% * * *	0.43%	-2.97% * *	-0.46%	0.19%	-0.91% * *	-0.89% * *
	[t-stat]	[-2.29]	[-7.88]	[1.46]	[-7.63]	[-1.79]	[1.12]	[-23.61]	[-21.15]
	[t-stat]	[-2.29]	[-7.88]	[1.46]	[-7.63]	[-1.79]	[1.12]	[-23.61]	[-21.15]
RT3	Performance	-1.14%***	-2.47%***	0.17%	-1.77%***	-0.42%**	0.14%	-0.88%***	-0.83%***
RT3	表演	-1.14% * *	-2.47% * *	0.17%	-1.77% * *	-0.42% * *	0.14%	-0.88% * *	-0.83% * *
	[t-stat]	[-4.04]	[-8.00]	[0.85]	[-7.48]	[-2.47]	[1.02]	[-21.06]	[-19.65]
	[t-stat]	[-4.04]	[-8.00]	[0.85]	[-7.48]	[-2.47]	[1.02]	[-21.06]	[-19.65]
RT4	Performance	-0.95%***	-1.85%***	0.33%**	-1.23%***	-0.44%***	0.12%	-0.84%***	-0.74%***
RT4	表演	-0.95% * *	-1.85% * * *	0.33% * *	-1.23% * * *	-0.44% * *	0.12%	-0.84% * *	-0.74% * *
	[t-stat]	[-4.09]	[-7.23]	[2.01]	[-6.66]	[-3.32]	[0.65]	[-19.59]	[-18.95]
	[t-stat]	[-4.09]	[-7.23]	[2.01]	[-6.66]	[-3.32]	[0.65]	[-19.59]	[-18.95]
RT5	Performance	1.32%***	-1.61%***	2.45%***	-1.40%***	-0.56%***	0.26%	-0.57%***	-0.46%***
RT5	表演	1.32% * * *	-1.61% * * *	2.45%	-1.40% * *	-0.56% * *	0.26%	-0.57% * *	-0.46%
	[t-stat]	[3.76]	[-4.13]	[7.96]	[-5.32]	[-3.03]	[1.18]	[-18.58]	[-18.66]
	[t-stat]	[3.76]	[-4.13]	[7.96]	[-5.32]	[-3.03]	[1.18]	[-18.58]	[-18.66]
INST									
国家									
科学									
研究	Performance	0.64%***	0.51%**	0.89%***	0.49%***	-0.05%	0.23%	-0.19%***	-0.22%***
所	表演	0.64%	0.51% * *	0.89% * *	0.49% * *	-0.05%	0.23%	-0.19% * *	-0.22% * *
	[t-stat]	[4.49]	[2.04]	[6.71]	[3.04]	[-0.62]	[1.32]	[-18.23]	[-19.52]
	[t-stat]	[4.49]	[2.04]	[6.71]	[3.04]	[-0.62]	[1.32]	[-18.23]	[-19.52]

Panel C. Annual trading performance using tracking portfolios: around earnings announcements
 面板 c 使用跟踪投资组合的年度交易业绩: 围绕盈利公告

		Total	Stock Selection	Market Timing	Trading Cost
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		总计		股票选择		市场时机		交易成本	
		EA 电子艺 界	EA/all days 电子艺界/全 天候	EA 电子艺 界	EA/all days 电子艺界/全 天候	EA 电子 艺界	EA/all days 电子艺界/全 天候	EA 电子艺 界	EA/all days 电子艺界/全天 候
RT1	Performance	-0.39%***	6.89%	-0.36%***	8.96%	0.02%	-8.63%	-0.04%***	3.06%
RT1	表演	-0.39% * *	6.89%	-0.36% * * *	8.96%	0.02%	-8.63%	-0.04% * *	3.06%
	[t-stat]	[-3.68]		[-4.14]		[0.57]		[-5.91]	
	[t-stat]	[-3.68]		[-4.14]		[0.57]		[-5.91]	
RT2	Performance	-0.36%***	7.71%	-0.31%***	12.27%	0.01%	-4.85%	-0.06%***	3.12%
RT2	表演	-0.36% * * *	7.71%	-0.31% * *	12.27%	0.01%	-4.85%	-0.06% * *	3.12%
	[t-stat]	[-4.08]		[-3.51]		[0.47]		[-5.90]	
	[t-stat]	[-4.08]		[-3.51]		[0.47]		[-5.90]	
RT3	Performance	-0.24%***	6.61%	-0.18%***	11.46%	0.00%	0.03%	-0.05%***	3.18%
RT3	表演	-0.24% * *	6.61%	-0.18% * *	11.46%	0.00%	0.03%	-0.05% * *	3.18%
	[t-stat]	[-4.24]		[-2.95]		[-0.01]		[-5.86]	
	[t-stat]	[-4.24]		[-2.95]		- 0.01		[-5.86]	
RT4	Performance	-0.14%***	5.02%	-0.07%	8.36%	-0.01%	4.32%	-0.05%***	3.26%
RT4	表演	-0.14% * * *	5.02%	-0.07%	8.36%	-0.01%	4.32%	-0.05% * *	3.26%
	[t-stat]	[-2.70]		[-1.10]		[-1.32]		[-5.86]	
	[t-stat]	[-2.70]		[-1.10]		[-1.32]		[-5.86]	
RT5	Performance	0.10%	-33.28%	0.14%**	13.81%	-0.01%	4.77%	-0.03%***	3.37%
RT5	表演	0.10%	-33.28%	0.14% * *	13.81%	-0.01%	4.77%	-0.03% * *	3.37%
	[t-stat]	[0.91]		[2.56]		[-1.30]		[-5.87]	
	[t-stat]	[0.91]		[2.56]		[-1.30]		[-5.87]	
INST									
国家科									
学研究									
所									
	Performance	0.11%*	9.80%	0.12%**	9.01%	0.00%	1.92%	-0.01%***	3.56%
	表演	0.11% *	9.80%	0.12% * *	9.01%	0.00%	1.92%	-0.01% * *	3.56%
	[t-stat]	[1.81]		[2.01]		[0.26]		[-5.86]	
	[t-stat]	[1.81]		[2.01]		[0.26]		[-5.86]	

Table IX. Robustness

表九. 稳健性

This table reports the robustness results. The sample period covers January 2016 to June 2019, and the sample firms are A-share stocks merged with the proprietary data and have at least 15 trading days in the previous month. Panel A reports the return prediction on each year. Panel B reports the return prediction by different stock characteristics. Panel C reports return predictions by including all filters in Liu, Stambaugh and Yuan (2019). Panel D reports return predictions by excluding leveraged trading, which includes margin buys, short sales, and collateral trading. Panel E reports return predictions by excluding days that hit the price limits (+10% and -10%). Panel F reports the alternative method using Liu, Stambaugh, and Yuan's (2019) three-factor adjusted alphas for long-short portfolios. To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

这个表格报告了健壮性结果。样本期间涵盖 2016 年 1 月至 2019 年 6 月，样本公司是 a 股股票与专有数据合并，在前一个月至少有 15 个交易日。面板 a 报告了每年的收益预测。面板 b 报告不同股票特征的收益预测。面板 c 报告通过包括 Liu, Stambaugh 和 Yuan 的所有过滤器返回预测(2019)。小组 d 报告通过排除杠杆交易(包括保证金买入、卖空交易和抵押品交易)返回预测。Panel e 报告通过排除达到价格上限的天数(+ 10% 和 -10%)返回预测。Panel f 使用 Liu, Stambaugh 和 Yuan (2019) 的三因素调整阿尔法来计算多空投资组合。为了考虑系数的序列相关性，采用 Newey 和 West (1987)方法对时间序列的标准误差进行调整，并按照 Newey 和 West (1994)方法选择最佳带宽。***，** 和 * 分别表示 1%，5% 和 10% 水平的显著性。

Panel A. Cross sectional return predictions by different years

不同年份的横截面回报预测

Year 年份	Market 市场	Dep.var 风险管理局		Ret 休息	Ret 休息	Ret 休息	Ret 休息	Ret 休息	Ret 休息
	Return 返回	Oib.var Oib.var		OibRT1 OibRT1	OibRT2 OibRT2	OibRT3 OibRT3	OibRT4 OibRT4	OibRT5 OibRT5	OibINST 奥宾斯特
				-0.0027***					
2016	-8.4%	Oib(-1)*Y2016	Estimate	-0.0027***	-0.0025***	-0.0020***	-0.0004***	0.0004***	0.0003***
	-8.4%	Oib (- 1) * Y2016	估计	*	-0.0025***	-0.0020	-0.0004***	0.0004***	0.0003***
			[t-stat]	[-3.46]	[-3.41]	[-3.45]	[-3.37]	[3.43]	[3.26]
			[t-stat]	[-3.46]	[-3.41]	[-3.45]	[-3.37]	[3.43]	[3.26]
2017	12.1%	Oib(-1)*Y2017	Estimate	-0.0023***	-0.0022***	-0.0015***	-0.0002***	0.0003***	0.0004***
	12.1%	Oib (- 1) * Y2017	估计	-0.0023***	-0.0022***	-0.0015***	-0.0002***	0.0003***	0.0004***
			[t-stat]	[-3.83]	[-3.70]	[-3.71]	[-3.11]	[3.39]	[3.87]
			[t-stat]	[-3.83]	[-3.70]	[-3.71]	[-3.11]	[3.39]	[3.87]
				-0.0027***					
2018	-22.2%	Oib(-1)*Y2018	Estimate	-0.0026***	-0.0027***	-0.0018***	-0.0002***	0.0002***	0.0006***
	-22.2%	Oib (- 1) * Y2018	估计	-0.0026***	*	-0.0018***	-0.0002***	0.0002***	0.0006***

2019 Jan-Jun 一月至六月	21.2%	Oib(-1)*Y2019	[t-stat]	[-3.75]	[-3.80]	[-3.70]	[-2.17]	[3.45]	[3.81]
			[t-stat]	[-3.75]	[-3.80]	[-3.70]	[-2.17]	[3.45]	[3.81]
	21.2%	Oib (- 1) * Y2019	Estimate	-0.0018**	-0.0018**	-0.0013**	-0.0001*	0.0003**	0.0003**
			估计	-0.0018 **	-0.0018 **	-0.0013 **	-0.0001 *	0003 **	0003 **
	Control 控制		[t-stat]	[-2.25]	[-2.26]	[-2.29]	[-1.84]	[2.13]	[2.35]
			[t-stat]	[-2.25]	[-2.26]	[-2.29]	[-1.84]	[2.13]	[2.35]
			Yes	Yes	Yes	Yes	Yes	Yes	Yes
			是的	是的	是的	是的	是的	是的	是的
			InterquartilereturnY2016	-0.0569%	-0.0434%	-0.0289%	-0.0088%	0.0136%	0.0248%
			四分位间回归 2016	-0.0569%	-0.0434%	-0.0289%	-0.0088%	0.0136%	0.0248%
			InterquartilereturnY2017	-0.0478%	-0.0377%	-0.0240%	-0.0053%	0.0127%	0.0255%
			四分位数回归 2017	-0.0478%	-0.0377%	-0.0240%	-0.0053%	0.0127%	0.0255%
								0.0129%	
			InterquartilereturnY2018	-0.0603%	-0.0527%	-0.0341%	-0.0060%	0.0129%	0.0375%
			四分位间回归 2018	-0.0603%	-0.0527%	-0.0341%	-0.0060%	0.0129%	0.0375%
			InterquartilereturnY2019	-0.0414%	-0.0339%	-0.0215%	-0.0033%	0.0123%	0.0160%
			四分之一回归 2019	-0.0414%	-0.0339%	-0.0215%	-0.0033%	0.0123%	0.0160%

Panel B. Cross-sectional return predictions by different stock characteristics

面板 b: 不同股票特征的横截面回报预测

Oib.var	OibRT1	OibRT2	OibRT3	OibRT4	OibRT5	OibINST
Oib.var	OibRT1	OibRT2	OibRT3	OibRT4	OibRT5	奥宾斯特
Oib(-1)*Small_size	-0.0119***	-0.0131***	-0.0090***	-0.0006***	0.0020***	0.0014***
Oib (- 1) * 小尺寸	-0.0119 * * *	-0.0131 * * *	-0.0090 * * *	-0.0006 * * *	0.0020 美元 * *	0.0014 * * *
Oib(-1)*Median_size	-0.0096***	-0.0093***	-0.0066***	-0.0011***	0.0009***	0.0015***
Oib (- 1) * 中位数 _ 大小	-0.0096 * * *	-0.0093 * * *	-0.0066 * * *	-0.0011 * * *	0.0009 * * *	0.0015 * * *
Oib(-1)*Large_size	-0.0067***	-0.0061***	-0.0044***	-0.0012***		0.0021***
Oib (- 1) * Large _ size	-0.0067 * * *	-0.0061 * * *	-0.0044 * * *	-0.0012 * * *	0.0001	0.0021
Oib(-1)*Low_EP	-0.0122***	-0.0131***	-0.0095***	-0.0010***	0.0020***	0.0015***
Oib (- 1) * Low _ ep	-0.0122 * * *	-0.0131 * * *	-0.0095 * * *	-0.0010 * * *	0.0020 美元 * *	0.0015 * * *
Oib(-1)*Median_EP	-0.0097***	-0.0098***	-0.0069***	-0.0009***	0.0012***	0.0015***
Oib (- 1) * 中位数 _ ep	-0.0097 * * *	-0.0098 * * *	-0.0069 * * *	-0.0009 * * *	0.0012 * *	0.0015 * * *
Oib(-1)*High_EP	-0.0062***	-0.0056***	-0.0039***	-0.0007***		0.0017***
Oib (- 1) * High _ ep	-0.0062 * * *	-0.0056 * * *	-0.0039 * * *	—0.0007 * * *	0.0001	0.0017
Oib(-1)*Low_turnover	-0.0061***	-0.0057***	-0.0041***	-0.0007***	0.0003***	0.0013***
Oib (- 1) * 营业额偏低	-0.0061 * * *	-0.0057 * * *	-0.0041 * * *	—0.0007 * * *	0.0003 * * *	0.0013 * *
Oib(-1)*Median_turnover	-0.0091***	-0.0092***	-0.0066***	-0.0010***	0.0011***	0.0014***
Oib (- 1) * 营业额中位数	—0.0091 * * *	-0.0092 * * *	-0.0066 * * *	-0.0010 * * *	0.0011 * * *	0.0014 * * *
Oib(-1)*High_turnover	-0.0159***	-0.0176***	-0.0128***	-0.0009***	0.0026***	0.0019***
Oib (- 1) * 高成交量	-0.0159 * * *	-0.0176 * * *	-0.0128 * * *	-0.0009 * * *	0.0026 * * *	0.0019 * * *
Oib(-1)*Low_price	-0.0088***	-0.0086***	-0.0067***	-0.0012***	0.0009***	0.0014***
Oib (- 1) * 低价	-0.0088 * * *	-0.0086 * * *	-0.0067 * * *	-0.0012 * * *	0.0009 * * *	0.0014 * * *
Oib(-1)*Median_price	-0.0098***	-0.0095***	-0.0068***	-0.0007***	0.0013***	0.0016***
Oib (- 1) * 股价中位数	-0.0098 * * *	-0.0095 * * *	-0.0068 * * *	—0.0007 * * *	0.0013 * *	0.0016 * * *
Oib(-1)*High_price	-0.0094***	-0.0094***	-0.0060***	-0.0007***	0.0014***	0.0016***
Oib (- 1) * 高 _ 价	-0.0094 * * *	-0.0094 * * *	-0.0060 * * *	—0.0007 * * *	0.0014 * * *	0.0016 * * *

Panel C. Cross-sectional return predictions, including all filters from Liu Stambaugh and Yuan (2019)

面板 c 横截面回报预测, 包括刘斯坦博和袁(2019)的所有过滤器

Dep.var						
风险管理	Ret	Ret	Ret	Ret	Ret	Ret
局	休息	休息	休息	休息	休息	休息
Oib.var	OibRT1	OibRT2	OibRT3	OibRT4	OibRT5	OibINST
Oib.var	OibRT1	OibRT2	OibRT3	OibRT4	OibRT5	奥宾斯特

Oib(-1)	Estimate	-0.0075***	-0.0071***	-0.0052***	-0.0011***	0.0004***	0.0017***
Oib (- 1)	估计	-0.0075 * * *	-0.0071 * * *	-0.0052 * *	-0.0011 * * *	0.0004 * * *	0.0017
	[t-stat]	[-21.47]	[-19.65]	[-16.85]	[-8.09]	[5.07]	[16.98]
	[t-stat]	[-21.47]	[-19.65]	[-16.85]	[-8.09]	[5.07]	[16.98]
	Interquartile return	-0.17%	-0.14%	-0.09%	-0.03%	0.02%	0.11%
	四分位间回归	-0.17%	-0.14%	-0.09%	-0.03%	0.02%	0.11%

Panel D. Cross-sectional return predictions, excluding leveraged trading

横截面回报预测，不包括杠杆交易

Dep.var							
风险管理		Ret	Ret	Ret	Ret	Ret	Ret
局		休息	休息	休息	休息	休息	休息
Oib.var		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5	OibINST
Oib.var		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5	奥宾斯特
Oib(-1)	Estimate	-0.0092***	-0.0088***	-0.0056***	-0.0003***	0.0008***	0.0016***
Oib (- 1)	估计	-0.0092 * * *	-0.0088 * * *	-0.0056 * * *	-0.0003 * * *	0.0008 * *	0.0016 * * *
	[t-stat]	[-17.26]	[-17.16]	[-15.54]	[-3.37]	[10.09]	[18.28]
	[t-stat]	[-17.26]	[-17.16]	[-15.54]	[-3.37]	[10.09]	[18.28]
	Interquartile return	-0.20%	-0.16%	-0.11%	-0.01%	0.05%	0.11%
	四分位间回归	-0.20%	-0.16%	-0.11%	-0.01%	0.05%	0.11%

Panel E. Cross-sectional return predictions, excluding days that hitting the price limits (+10%, and -10%)
横截面回报预测，不包括触及价格上限的日子(+ 10% 和 -10%)

Dep.var 风险管理 局		Ret 休息	Ret 休息	Ret 休息	Ret 休息	Ret 休息	Ret 休息
Oib.var Oib.var		OibRT1 OibRT1	OibRT2 OibRT2	OibRT3 OibRT3	OibRT4 OibRT4	OibRT5 OibRT5	OibINST 奥宾斯特
Oib(-1)	Estimate	-0.0087***	-0.0087***	-0.0064***	-0.0010***	0.0009***	0.0017***
Oib (- 1)	估计	-0.0087 * * *	-0.0087 * * *	-0.0064 * * *	-0.0010 * * *	0.0009 * * *	0.0017
	[t-stat]	[-19.60]	[-19.57]	[-17.95]	[-8.38]	[10.41]	[18.10]
	[t-stat]	[-19.60]	[-19.57]	[-17.95]	[-8.38]	[10.41]	[18.10]
	Interquartile return	-0.19%	-0.16%	-0.11%	-0.03%	0.04%	0.11%
	四分位间回归	-0.19%	-0.16%	-0.11%	-0.03%	0.04%	0.11%

Panel F. Risk adjusted alphas for long-short portfolios over the next 12 weeks
在接下来的 12 周里，风险调整了多空投资组合的阿尔法值

	OibRT1 OibRT1 Alpha 阿尔法		OibRT2 OibRT2 Alpha 阿尔法		OibRT3 OibRT3 Alpha 阿尔法		OibRT4 OibRT4 Alpha 阿尔法		OibRT5 OibRT5 Alpha 阿尔法		OibINST 奥宾斯特 Alpha 阿尔法	
	t-stat T-stat		t-stat T-stat		t-stat T-stat		t-stat T-stat		t-stat T-stat		t-stat T-stat	
					-0.0027***		-0.0007***					
1 day	-0.0042***	-20.55	-0.0036***	-17.53	-0.0027 * *	-15.13	-0.0007 * *	-6.71	0.0017***		0.0025***	
1 天	-0.0042	-20.55	-0.0036 * * *	-17.53	*	-15.13	*	-6.71	0.0017	12.71	0.0025 * * *	19.28
1 week	-0.0089***	-15.34	-0.0068***	-12.97	-0.0038***	-9.22	-0.0004	-1.63	0.0034***		0.0056***	
一星期	-0.0089 * * *	-15.34	-0.0068 * * *	-12.97	-0.0038 * * *	-9.22	-0.0004	-1.63	0.0034 * *	11.29	0.0056 * *	13.67
			-0.0091***									
2 weeks	-0.0115***	-11.96	-0.0091 * *	-10.50	-0.0048***	-6.84	-0.0001	-0.16	0.0046***		0.0075***	
两个星期	-0.0115 * *	-11.96	*	-10.50	-0.0048 * * *	-6.84	-0.0001	-0.16	0.0046 * *	9.08	0.0075 美元	11.53
3 weeks	-0.0130***	-10.52	-0.0103***	-9.37	-0.0051***	-5.68			0.0054***		0.0086***	
三星期	-0.0130 * * *	-10.52	-0.0103 * * *	-9.37	-0.0051 * * *	-5.68	0.0001	0.26	0.0054 * * *	8.71	0.0086 * *	9.73
4 weeks	-0.0130***	-8.64	-0.0104***	-7.94	-0.0052***	-4.94			0.0056***		0.0098***	
四个星期	-0.0130 * * *	-8.64	-0.0104 * * *	-7.94	-0.0052 * *	-4.94	0.0003	0.81	0.0056 * *	7.87	0.0098 * *	9.19
5 weeks	-0.0130***	-7.45	-0.0099***	-6.53	-0.0050***	-4.01			0.0062***		0.0098***	
5 周	-0.0130 * * *	-7.45	-0.0099 * * *	-6.53	-0.0050 * * *	-4.01	0.0006	1.13	0.0062 * * *	7.51	0.0098 * *	8.39
6 weeks	-0.0148***	-7.84	-0.0110***	-6.49	-0.0056***	-3.73			0.0064***		0.0103***	
六个星期	-0.0148 * *	-7.84	-0.0110 * *	-6.49	-0.0056 * * *	-3.73	0.0005	0.98	0.0064 * * *	7.29	0.0103 * *	7.99

7 weeks	-0.0173***	-8.76	-0.0130***	-6.74	-0.0068***	-3.61			0.0064***		0.0119***	
7 周	-0.0173 * * *	—8.76	-0.0130 * * *	—6.74	-0.0068 * * *	—3.61	0.0001	0.49	0.0064 * * *	7.25	0.0119 * *	8.29
8 weeks	-0.0183***	-8.43	-0.0136***	-6.37	-0.0069***	-3.36			0.0063***		0.0128***	
8 周	-0.0183 * * *	-8.43	-0.0136 * * *	-6.37	-0.0069 * * *	-3.36	0.0001	0.50	0.0063 * * *	6.63	0.0128 * * *	6.95
					-0.0070***							
9 weeks	-0.0191***	-7.62	-0.0142***	-5.59	-0.0070 * * *	-2.97			0.0068***		0.0135***	
9 周	-0.0191 * * *	—7.62	-0.0142 * * *	-5.59	*	-2.97	0.0003	0.64	0.0068 美元	7.45	0.0135 * *	6.18
						-2.98						
10 weeks	-0.0187***	-6.98	-0.0136***	-5.27	-0.0063***	-2.98 美			0.0065***		0.0133***	
10 周	-0.0187 * * *	-6.98	-0.0136 * * *	-5.27	-0.0063 * * *	元	0.0005	0.87	0.0065 * *	7.68	0.0133 * * *	6.38
					-0.0070***							
11 weeks	-0.0187***	-6.24	-0.0141***	-4.66	-0.0070 * * *	-2.77	-0.0005		0.0061***		0.0138***	
十一周	-0.0187 * * *	—6.24	-0.0141 * * *	—4.66	*	-2.77	-0.0005	0.27	0.0061 * *	6.83	0.0138 * * *	6.00
12 weeks	-0.0183***	-5.75	-0.0139***	-4.52	-0.0072***	-2.89	-0.0010		0.0057***		0.0137***	
12 周	-0.0183 * * *	—5.75	-0.0139 * * *	—4.52	-0.0072 * * *	—2.89	-0.0010	0.03	0.0057	5.78	0.0137 美元	6.03

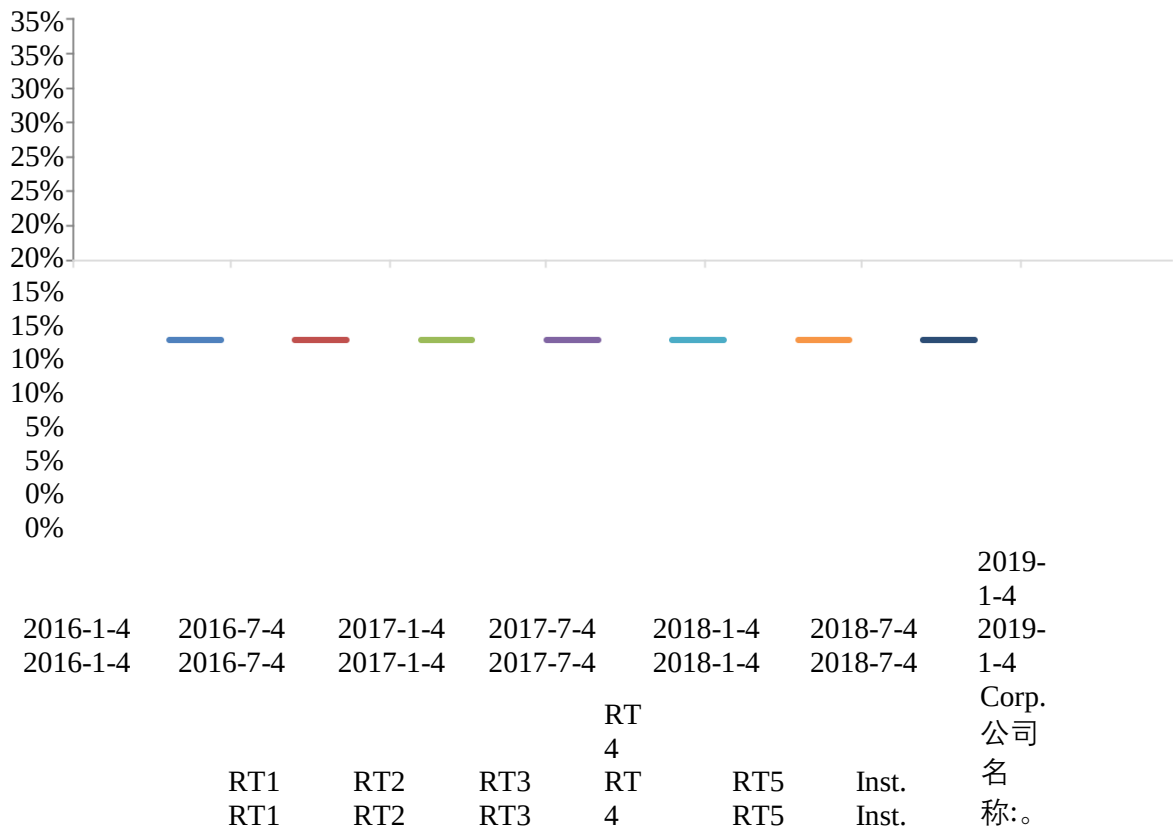
Figure I. Different Investor Type Order Flows between Jan 2016 and Jun 2019
2016 年 1 月至 2019 年 6 月期间不同投资者类型的订单流量

The figures report the time-series plot of the different types of investor trading activity from January 2016 to June 2019. Our sample firms are A-share stocks listed on a major stock exchange with at least 15 trading days during the previous month. Panel A presents time-series plot of the volume percentage for each investor type, reported as cross-sectional mean. Panel B shows the time-series plot of shares held by investor type, aggregated at market level. Panel C presents the time-series plot of each investor type’s trading volume, scaled by the shares held by that investor type, and reported as the cross-sectional mean.

这些数据反映了 2016 年 1 月至 2019 年 6 月不同类型投资者交易活动的时间序列图。我们的样本公司是在主要证券交易所上市的 a 股股票，在上个月至少有 15 个交易日。面板 a 呈现了每个投资者类型的成交量百分比的时间序列图，以横截面平均值报告。面板 b 显示了按投资者类型持有的股票的时间序列图，在市场水平上汇总。面板 c 展示了每种类型投资者的交易量的时间序列图，按该类型投资者所持有的股份进行比例，并以横截面平均数报告。

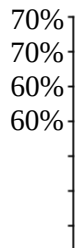
Panel A. Share volume (%), cross-sectional mean for different investor types

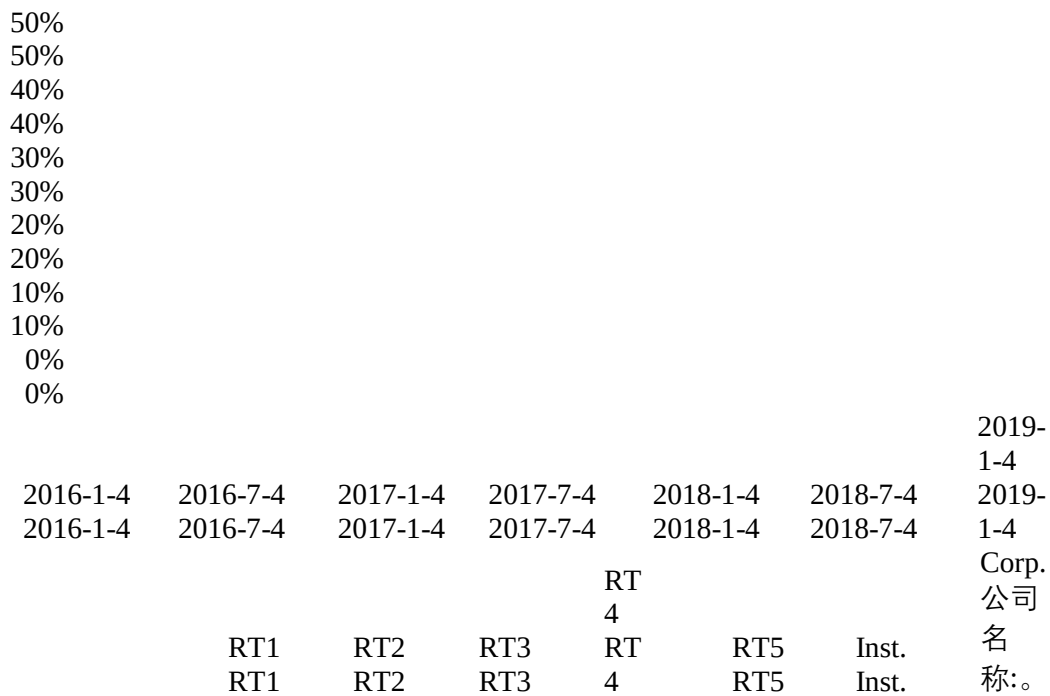
面板 a: 不同类型投资者的股票成交量(%), 截面平均值



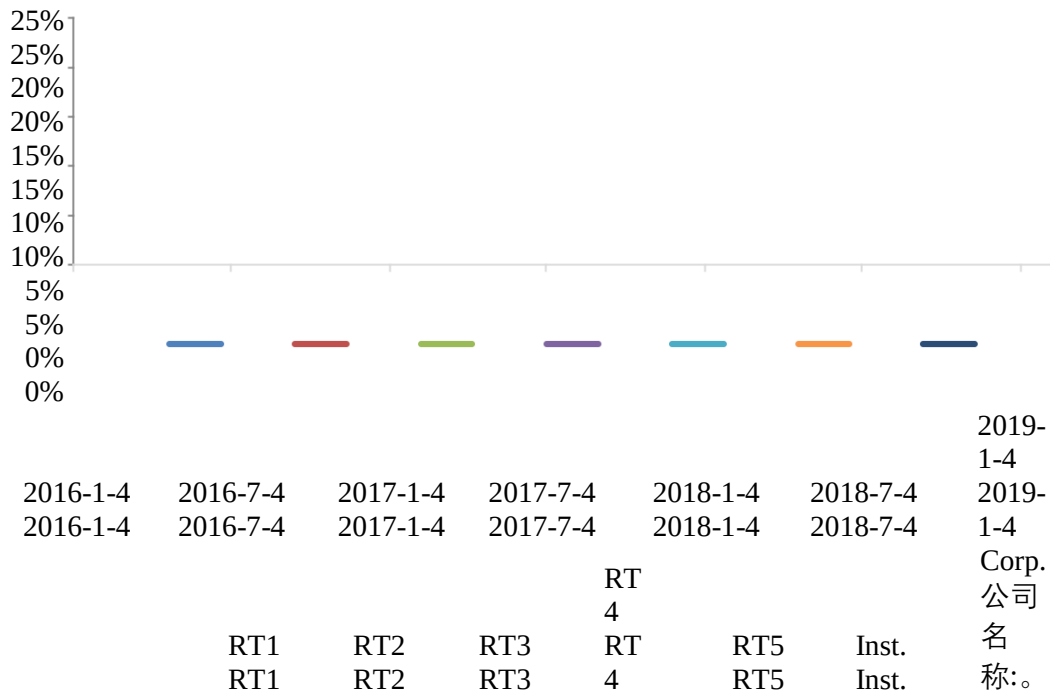
Panel B. Shares held (%), aggregated at market level for different investor types

面板 b: 不同类型投资者的持股比例(%), 按市场水平计算





Panel C. Share volume scaled by shares held (%), cross-sectional mean for different investor types
 股票成交量按所持股份比例(%), 不同投资者类型的截面平均值



Boehmer 等人(2021)的观点, 两阶段分解方法的优点在于, 它包括了在一个统一的、内部一致的经验框架中来自

不同假设的各种组成部分。这种方法的一个注意事项是，在为不同的假设选择代理时，有必要做出经验假设。虽然这些假设看起来是合理的，但我们必须谨慎，因为结果仍然取决于经验假设的有效性。

Appendix B. Barber et al. (2009) Methods for Trading Performances

附录 b. Barber 等人(2009)的交易表现方法

With access to all trading records from the Taiwan Stock Exchange, Barber, Lee, Liu, and Odean (2009, BLLO hereafter) designs a simple and intuitive method to compute the trading performances of different types of investors in four steps. First, BLLO separate all investors into “individuals”, “corporations”, “dealers”, “foreigners”, and “mutual funds”, using identities provided by the exchange. Second, within each investor group, BLLO compute the aggregate daily net buy and net sell positions and create daily matching trading portfolios, the “net buy” and “net sell” portfolios. Specifically, the net buy volumes and sell volumes for stock i on day d of group G are summed over investor j in group G :

利用台湾证券交易所的所有交易记录，Barber、Lee、Liu 和 Odean (2009, BLLO 以后)设计了一个简单直观的方法，分四个步骤计算不同类型投资者的交易表现。首先，BLLO 使用交易所提供的身份将所有投资者分为“个人”、“公司”、“交易商”、“外国人”和“共同基金”。其次，在每个投资者群体中，BLLO 计算每日净买入和净卖出头寸的总和，并创建每日匹配的交易组合，即“净买入”和“净卖出”组合。具体来说， g 组在 d 天的股票 i 的净买入量和卖出量相加于 g 组的投资者 j :

(A4)

(A4)

¹ All estimations in this study are estimated within each investor group. Beginning in this section, we omit the subscripts G to make the formula more readable.
本研究中的所有估计都是在每个投资者群体中进行估计的。从本节开始，我们省略下标 g 以使公式更加可读。

$$\begin{aligned}
& \text{,,} = (\sum_{i \in G} \text{,,} - \sum_{i \in S} \text{,,}, 0), \\
& = (\sum_{i \in G} \text{,,} - \sum_{i \in S} \text{,,}, 0)
\end{aligned}$$

where the ,, and ,, are defined in equation (1). If net buy volume for stock i on day d in investor group G is positive, it belongs to the net buy portfolio, ,, ; if the net buy volume is negative (thus the net sell volume is positive), it belongs to the net sell portfolio, ,, . 其中, 和, 在方程(1)中定义。如果投资者组 g 中股票 i 在 d 日的净买入量为正, 则它属于净买入组合, 如果净买入量为负(因此净卖出量为正), 则它属于净卖出组合。

Third, BLLO track these “net buy” and “net sell” portfolios over a holding horizon of n days, and compute cumulative cash flows and returns from this tracking strategy, and treat them as proxies for trading performances of different investor groups. Specifically, BLLO computes the total performance of the net buy and net sell portfolios as:

第三, BLLO 在 n 天的持有期内跟踪这些“净买入”和“净卖出”投资组合, 计算这种跟踪策略的累积现金流和收益, 并将其视为不同投资者群体交易业绩的代理。具体来说, BLLO 计算净买入和净卖出投资组合的总体表现如下:

$$\begin{aligned}
& \sum_{d=1}^n \sum_{i \in G} \text{,,} \times \text{,,} - \sum_{d=1}^n \sum_{i \in S} \text{,,} \times \text{,,} - \sum_{d=1}^n \sum_{i \in G} \text{,,} \times \text{,,} - \sum_{d=1}^n \sum_{i \in S} \text{,,} \times \text{,,} \\
& \sum_{d=1}^n \sum_{i \in G} \text{,,} \times \text{,,} + \sum_{d=1}^n \sum_{i \in S} \text{,,} \times \text{,,}
\end{aligned} \tag{A5}$$

The tracking net buy and sell portfolios are rebalanced once a day, so we follow BLLO and use 跟踪净买卖投资组合每天重新平衡一次, 所以我们遵循 BLLO 并使用 the average prices, defined as 平均价格定义为

$$\begin{aligned}
& \max (\sum_{i \in G} \text{,,} - \sum_{i \in S} \text{,,}, 0) \\
& \text{Max} (\sum_{i \in G} \text{,,} - \sum_{i \in S} \text{,,}, 0)
\end{aligned} \tag{A6}$$

$$\begin{aligned}
& \text{,,} = \frac{\sum_{d=1}^n \sum_{i \in G} \text{,,} \times \text{,,} - \sum_{d=1}^n \sum_{i \in S} \text{,,} \times \text{,,}}{\sum_{d=1}^n \sum_{i \in G} \text{,,} \times \text{,,} + \sum_{d=1}^n \sum_{i \in S} \text{,,} \times \text{,,}} \\
& \text{,,} = \frac{\max (\sum_{i \in G} \text{,,} - \sum_{i \in S} \text{,,}, 0)}{\text{Max} (\sum_{i \in G} \text{,,} - \sum_{i \in S} \text{,,}, 0)} \\
& \text{,,} = \frac{\sum_{d=1}^n \sum_{i \in G} \text{,,} \times \text{,,} - \sum_{d=1}^n \sum_{i \in S} \text{,,} \times \text{,,}}{\sum_{d=1}^n \sum_{i \in G} \text{,,} \times \text{,,} + \sum_{d=1}^n \sum_{i \in S} \text{,,} \times \text{,,}} \\
& \text{,,} = \frac{\max (\sum_{i \in G} \text{,,} - \sum_{i \in S} \text{,,}, 0)}{\text{Max} (\sum_{i \in G} \text{,,} - \sum_{i \in S} \text{,,}, 0)}
\end{aligned}$$

Notice the

注意, 分子中的, 和, 是现金交易量, 交易股票交易量和交易价格的产品, 而不是股票交易量。Variable, $+1$, $+$ 和 $+1$, $+$ 分别是从 $+1$ 天到 $+1$ 天的股票 i 的累积收益率和无风险率。交易成本计算如下:

$$\begin{aligned}
& \text{,,} = \sum \sum 1 \times \text{,,} + \sum \sum 2 \times \text{,,} \\
& \text{,,} = \sum \sum 1 \times \text{,,} + \sum \sum 2 \times \text{,,}
\end{aligned} \tag{A7}$$

$$\begin{aligned}
& \in \\
& \in \\
& \in
\end{aligned}$$

Variables 1 and 2 capture the proportional trading costs relative to cash volumes. According to China's policies and practices, there are commissions of 0.05% of cash volumes on both the buy and sell sides, a stamp tax of 0.10% on cash volumes of sales, and a transfer fee of 0.002% on cash volumes on both the buy and sell sides during our sample period. In other words, $1 = 0.05\% + 0.002\%$, $2 = 0.05\% + 0.10\% + 0.002\%$. 变量 1 和变量 2 反映了相对于现金流量的成比例交易成本。根据中国的政策和做法, 在我们的样本期间, 买卖双方都有 0.05% 的现金交易佣金, 销售现金交易印花税 0.10%, 买卖双方现金交易转移费 0.002%。换句话说, $1 = 0.05\% + 0.002\%$, $2 = 0.05\% + 0.10\% + 0.002\%$ 。

Three additional considerations are added in the third step. First, BLLO assumes that the net buy and sell portfolios are held for n days and sets n to different horizons. As the main discussion of BLLO focuses on the holding horizon of 140 days ($n=140$), we choose the same horizon for ease of comparison. Second, BLLO assumes that the net buy (sell) portfolios are independent for each day d , and annualizes the performance by aggregating the daily net buy and sell portfolios over each year.² Third, to have a perspective of total performance as a percentage of the total investment of these tracking portfolios, which is similar to the idea of return on investments, BLLO measures total investment as the aggregate holding value for each group of investors. For comparison, we choose the aggregate holding values in Panel A of Table I and present the return on investment as the total performance over aggregate holding.

第三步增加了三个额外的考虑因素。首先, BLLO 假设净买卖投资组合持有 n 天, 并将 n 设置为不同的视野。由于 BLLO 的主要讨论集中在 140 天的持有期限($n = 140$), 我们选择相同的期限以便于比较。第二, BLLO 假设每天的净买卖投资组合都是独立的, 并通过将每年的净买卖投资组合合计, 将业绩年化。第三, 从总业绩占这些跟踪投资组合总投资的百分比的角度来看, 这类似于投资回报的概念, BLLO 将总投资作为每组投资者的总持有价值来衡量。为了进行比较, 我们在表 i 的面板 a 中选择了总持有价值, 并将投资回报表示为总表现相对于总持有的表现。

²We compute the standard error estimates using the Newey and West (1987) method, with the optimal bandwidth chosen by using Newey and West (1994) procedure. We conduct robustness check using optimal lag numbers selected by the Bayesian Information Criterion (BIC). The results are similar and available on request.

我们使用 Newey 和 West (1987)方法计算标准误差估计, 并使用 Newey 和 West (1994)方法选择最佳带宽。我们使用贝斯信息量准则(BIC)选择的最佳滞后数进行鲁棒性检查。结果是相似的, 并可根据要求提供。

After computing the total performance, BLLO decomposes the total into three intuitive components,

在计算总体性能之后，BLLO 将总体性能分解为三个直观的组成部分，

$$R_{g,d} = R_{g,d}^{(1)} + R_{g,d}^{(2)} + R_{g,d}^{(3)} \quad (A8)$$

$$R_{g,d}^{(1)} = R_{g,d}^{(1)} + R_{g,d}^{(2)} + R_{g,d}^{(3)} \quad (A8)$$

Here, the stock selection component on day d for investor group G is computed as:

在这里，投资者集团 g 在 d 日的选股部分计算如下：

$$R_{g,d}^{(1)} = \sum_{i \in G} w_{i,d} (R_{i,d} - R_{m,d}) \quad (A9)$$

$$R_{g,d}^{(2)} = \sum_{i \in G} w_{i,d} (R_{i,d} - R_{m,d}) \quad (A9)$$

In other words, BLLO compares each stock's return, $R_{i,d}$, with the contemporaneous market return, $R_{m,d}$, to determine whether investors are capable of choosing stocks that beat the market. The second component, the market timing component, is computed as:
换句话说，BLLO 将每只股票的收益， $R_{i,d}$ ，与同期市场收益， $R_{m,d}$ ，进行比较，以确定投资者是否有能力选择比市场更好的股票。第二个组成部分，市场时机组成部分，计算如下：

$$R_{g,d}^{(2)} = \sum_{i \in G} w_{i,d} (R_{i,d} - R_{m,d}) \quad (A10)$$

$$R_{g,d}^{(3)} = \sum_{i \in G} w_{i,d} (R_{i,d} - R_{m,d}) \quad (A10)$$

With the difference between the market portfolio and interest rate, the BLLO method aims to capture whether the investor can choose the allocation between stocks (market portfolio) and bonds (interest rate), and thus, time the market. The trading cost term is defined in equation (A7). 基于市场投资组合与利率之间的差异，BLLO 方法的目的是获取投资者是否可以选股票(市场投资组合)和债券(利率)之间的配置，从而确定市场的时间。交易成本定义在方程(A7)中。

Appendix Table I. Previous Studies on Retail Investors in Different Markets
附录表一以往有关不同市场一般投资者的研究

Publication Details 发布细节	Data 资料	Research Questions and Main Findings 研究问题和主要发现
Barber and Odean, Journal of 巴伯和奥迪恩 Finance, 2000. 金融, 2000. “Trading is hazardous to ”交易有害于 your wealth: The common 你的财富: 普通人 stock investment 股票投资 performance of individual 个人表现 investors?” 投资者?”	U.S. data, 66,465 数据, 66,465 households over 1991 一九九一年的住户数 目 to 1996 from a large 对 1996 年的大规模 discount brokerage 折扣经纪	Research Question: Return performance of equities held directly by households. 研究问题: 家庭直接持有的股票的回报表现。 Findings: 结论: (1) Individual investors who hold common stocks directly pay a tremendous (1) 直接持有普通股的个人投资者支付巨额 performance penalty for active trading. 对活跃交易的业绩处罚。 (2) Overconfidence can explain high trading levels and the resulting poor performance (2) 过度自信可以解释高水平的交易和由此导致的糟糕表现 of individual investors. 个人投资者。
Kaniel, Saar, and Titman, 卡尼尔, 萨尔和泰特曼, Journal of Finance, 2008. 金融杂志, 2008 年。 “Individual investor trading ”个人投资者交易 and stock returns.” 以及股票回报。”	U.S. data, all retail 所有零售数据 orders from NYSE 纽约证交所的订单 CAUD file from year 年份的 CAUD 档案 2000 to 2003 2000 年至 2003 年	Research Question: The relation between net individual investor trading and short- 研究问题: 净个人投资者交易与卖空的关系 horizon cross-sectional stock returns. 水平横截面股票回报。 Findings: 结论: (1) Individuals buy stocks following declines in the previous month and sell following (1) 个人在前一个月股市下跌后买入股票, 随后卖出 price increases. 物价上涨。 (2) Positive excess returns in the month following intense buying by individuals and (2) 在个人和个人大量买入后的一个月实现正超额回报 negative excess returns after individuals sell. 个人出售后的负超额回报。
Barber, Odean, and Zhu, 巴伯, 奥迪恩, 还有朱老师, Review of Financial Studies, 金融研究综述,	U.S. data, tick-by- 美国数据, 滴答作响 tick trades from TAQ 来自 TAQ 的股票交 易	Research Question: The trading of individual investors and stock returns. 研究问题: 个人投资者的交易与股票回报。 Findings: 结论:

2008. “Do retail trades move 零售交易是否会变动 markets?” 市场?”	and ISSM over 1983 以及 1983 年的 ISSM to 2001, small trades 至 2001 年, 小型贸易 proxy for retail 零售代理人	(1) Small trade order imbalance correlates well with order imbalance based on trades (1) 小额交易订单不平衡与基于交易的订单不平衡有很好的相关性 from retail brokers. 零售经纪公司。 (2) Retail investors herd. (2) 散户投资者蜂拥。 (3) Small trade order imbalance negatively forecasts future annual returns. (3) 小额贸易订单不平衡对未来年度收益的预测是负面的。 (4) Small trade order imbalance positively predicts future week returns. (4) 小额交易订单不平衡正面预测未来一周的回报。
Kaniel, Liu, Saar, and 卡尼尔, 刘, 萨尔, 和 Titman, Journal of Finance, Titman, 金融杂志, 2012. “Individual investor trading ”个人投资者交易 and return patterns around 并返回周围的模式 earnings announcements.” 收益公告。”	U.S. data, all retail 所有零售数据 orders from NYSE 纽约证交所的订单 CAUD file from year 年份的 CAUD 档案 2000 to 2003 2000 年至 2003 年	Research Question: The informed trading by individual investors around earnings 研究问题: 个人投资者围绕收益进行的知情交易 announcements 公告 Findings: 结论: (1) The intense aggregate individual investor buying (selling) predicts large positive (1) 强烈的个人投资者买入(卖出)预示着巨大的正面影响 (negative) abnormal returns on and after earnings announcement dates. (负)收益公布日期和之后的异常收益。 (2) Decompose abnormal returns following the event into information and liquidity (2) 将事件发生后的异常收益分解为信息和流动性 provision components, and show that about half of the returns can be attributed to 准备金组成部分, 并显示大约一半的回报可归因于 private information. 私人资料。

Publication Details 发布细节	Data 资料	Research Questions and Main Findings 研究问题和主要发现
Kelley and Tetlock, Journal of Finance, 2013. 凯莉和泰特洛克, 日记金融, 2013 年。 “How wise are crowds? “群众有多聪明? Insights from retail orders 来自零售订单的见解 and stock returns.” 以及股票回报。”	U.S. data, retail 美国数据, 零售 orders from dozens of 来自几十个 retail brokerages over 零售经纪公司 year 2003 to 2007 2003 年至 2007 年	Research question: The role of retail investors in stock pricing by separately examining 研究问题: 通过分别考察散户投资者在股票定价中的作用 aggressive (market) and passive (limit) orders. 激进(市场)指令和被动(限制)指令。 Findings: 结论: (1) Both market and limit order imbalance positively predict firms' monthly stock (1) 无论是市场失衡还是限价订单失衡都能正向预测企业的月库存量 returns. 回报。 (2) Market orders correctly predict firm news, including earnings surprises. (2) 市场指令正确地预测了企业的利好消息, 包括意外收益。 (3) Limit orders following negative returns, consistent with traders providing liquidity. (3) 限制负回报后的指令, 与提供流动性的交易员相一致。
Boehmer, Jones, Zhang and 博默说, 琼斯, 张 以及 of Zhang, Journal of Finance, 张, 日记 期 财经、, 2021. retail “Tracking 零售 investor 追踪 业 投资者 activity.” 活动。”	U.S. data, an 美国的数据、 algorithm to identify 识别的算法 marketable retail 可销售的零售 trades from TAQ 来自 TAQ 的贸易 over year 2010 to 2010 年至 2010 年 2015	Research Question: Provide an algorithm to identify marketable retail purchases and 研究问题: 提供一个算法, 以确定市场零售购买和 sales using TAQ. 使用 TAQ 的销售。 Findings: 结论: (1) Provide and validate the algorithm. (1) 提供并验证算法。 (2) Marketable retail order imbalance positive predict future weekly stock return. (2) 市场零售订单不平衡正面预测未来每周股票回报。 (3) Predictive power of marketable retail order imbalance could be attributable to order (3)可销售零售订单失衡的预测能力可归因于订单 flow persistence, contrarian trading, public news sentiment and unexplained part. 流量持续性、反向交易、公众新闻情绪及不明原因部分。
Barber, Lin, and 理发 小 以 Odean, 师, 玲, 及 欧迪安,	U.S. data, retail 美国数据, 零售	Research Question: Retail order imbalance positively predicts returns, but in aggregate 研究问题: 零售订单不平衡正向预测回报, 但总体而言

of Journal 日 Financial and 日记 期 财务 以及 Quantitative Analysis, 量化 分析, forthcoming . 即将到来。 “Resolving a paradox: Retail 解决一个悖论: 零售 trades positively predict 贸易 积极的 预测 returns but are not profitable.” 投资回报率较低, 但无利可 图。”	trades identified by 认定的行业 Boehmer et al. (2021) boehmer 等(2021) from TAQ over year 过去一年从 TAQ 2010-2019 2010-2019	retail investor trades lose money. 散户投资者交易亏损。 Findings: 结论: (1) Order imbalance tests equally weight stocks, but retail purchases concentrate in (1) 订单不平衡测试股票权重, 但零售购买集中在 stocks that subsequently underperform. 表现不佳的股票。 (2) Trades by retail investors with less knowledge, experience, and wealth are more (2) 知识、经验和财富较少的散户投资者进行的交易较多 likely to underperform. 可能表现不佳。
Welch, of 韦尔, Journal 日 Finance, 奇, 日记 期 财经、, 2022. of “The wisdom 日 the ”他说 智慧 期 是的 Robinhood crowd.” 罗宾汉的人群	U.S. data, Robinhood 美国数据, 罗宾汉 investors holding 投资者持有 from May 2018 to 由 2018 年 5 月至 August 2020 二零二零年八月	Research Question: Robinhood investors trading behavior. 研究问题: 投资者的交易行为。 Findings: 结论: (1) Robinhood investors increased their holdings in the March 2020 COVID bear (1) 罗宾汉的投资者增持了 2020 年 3 月的冠状病毒疾病熊 market. 市场。 (2) Robinhood investors tend to buy stocks with high past share volume and dollar- (2) 投资者倾向于购买过去成交量和美元价值较高的股票 trading volume. 交易量。 (3) From mid-2018 to mid-2020, an aggregated Robinhood portfolio had both good (3) 从 2018 年中期到 2020 年中期, 一个综合的 Robinhood 投资组合都很好 timing and good alpha. 时机和良好的阿尔法。

Publication Details 发布细节	Data 资料	Research Questions and Main Findings 研究问题和主要发现
Ozik, and 奥兹, Sadka, 以 Shen, 克, 萨德卡, 及 小沈、, of Journal 日 Financial and 日记 期 财务 以及 Quantitative Analysis, 2021. 定量分析, 2021 年。 illiquidity “Flattening the 流动性不 “扁平化 是的 足 curve: Retail trading during 曲线: 零售交易 期间 the COVID-19 lockdown.” 2019 冠状病毒毒疾病。”	U.S. data, Robinhood 美国数据, 罗宾汉 investors holding 投资者持有 from May 2018 to 由 2018 年 5 月至 August 2020 二零二零年八月	Research Question: The impact of retail investors on stock liquidity during the 研究问题: 零售投资者对股票流动性的影响 Coronavirus pandemic lockdown. 冠状病毒大流行封锁。 Findings: 结论: (1) Retail trading exhibits a sharp increase during Covid. (1)在冠状病毒疾病期间, 零售交易呈现急剧增长。 (2) Retail trading attenuated the rise in illiquidity by roughly 40%, but less so for high- (2)零售交易将流动性不足的上升幅度减少了约 40%, 但对于高流动性的上升幅度较小 media-attention stocks. 媒体关注的股票。
Barber , Odean, 理发 Huang, 欧迪 and 师, 黄长官, 安, 以及 Schwarz, Journal of Finance, 施瓦茨, 《金融期刊》, 2022. “Attention- Induced Trading ”引起注意 贸易 and Returns: Evidence from 返回: 证据 来自 Robinhood Users.” Robinhood 用 户。”	U.S. data, Robinhood 美国数据, 罗宾汉 investors holding 投资者持有 from May 2018 to 由 2018 年 5 月至 August 2020 二零二零年八月	Research Question: The influence of financial innovation by fintech brokerages on 研究问题: 金融科技经纪公司的金融创新对 individual investors' trading and stock prices. 个人投资者的交易和股票价格。 Findings: 结论: (1) Robinhood investors engage in more attention-induced trading than other retail (1)与其他零售商相比, Robinhood 投资者参与更多的注意力诱导交易 investors. 投资者。 (2) Intense buying by Robinhood users forecasts negative returns. Average 20-day (2) Robinhood 用户的大量购买预示着负回报。平均 20 天 abnormal returns are -4.7% for the top stocks purchased each day. 每天购买的最高股票的异常回报率为 -4.7% 。
Grinblatt and Keloharju,	Finnish data, daily	Research Question: The past-return-based behavior and the performance of various

格林布拉特和克洛哈朱, 芬兰每日数据 Journal of Financial Economics, 2000. 日记 期 财务 “The investment behavior and “投资行为和 performanc e of various 表现 期 各种 investor types: a study of 投资者类型: 对... 的研究 Finland's unique data set.” 芬兰独特的数据集。”	retail trades and 零售业及 holdings from 控股公司 Finnish Central 芬兰中部 Securities Depository 证券托管处 (FCSD) over 1994 to (FCSD)1994 年至 1996	研究问题: 基于过去回报的行为和各种表现 investor types. 投资者类型。 Findings: 结论: (1) Foreign investors tend to be momentum, while domestic investors, particularly (1)外国投资者往往势头强劲, 而国内投资者尤其如此 households, tend to be contrarians. 家庭, 往往是逆向投资者。 (2) The portfolios of foreign investors seem to outperform the portfolios of households, (2)外国投资者的投资组合表现似乎优于家庭投资组合, even after controlling for behavior differences. 即使在控制了行为差异之后。
Linnainmaa, Journal of Linnainmaa, 日记 日期 Finance, 2011. 财经, 2011。 “Do limit orders alter 他说 限额 命令 改变 inferences about investor 推论 关于 投资者 performance and behavior?” 表现和行为?”	Finnish data, daily 芬兰每日数据 retail trades and 零售业及 holdings from 控股公司 Finnish Central 芬兰中部 Securities Depository 证券托管处 (FCSD) over 1995 to 一九九五年至今 2002	Research Question: Individual investors' trading behaviors could be explained by 研究问题: 个人投资者的交易行为可以用 investors' use of limit orders. 投资者对限价指令的使用。 Findings: 结论: These patterns arise because limit orders are price-contingent and suffer from adverse 这些模式之所以出现, 是因为限价订单与价格有关, 而且受到不利影响 selection. 选择。

Publication Details 发布细节	Data 资料	Research Questions and Main Findings 研究问题和主要发现
Grinblatt, Keloharju, 格林布拉特, 克洛哈朱, Linnainmaa, Journal of Linnainmaa, 日记 日期 Financial Economics, 2012. 金融经济学, 2012. “IQ, trading behavior, and 智商, 交易 行为, 以及 performance.” 业绩。”	Finnish data, daily 芬兰每日数据 retail trades and 零售业及 holdings from 控股公司 Finnish Central 芬兰中部 Securities Depository 证券托管处 (FCSD) over 1995 to 一九九五年至今 2002 and intelligence 2002 年和情报局 (IQ) test administered (智商)测试执行 to Finnish male 芬兰男性	Research Question: Whether IQ influences trading behavior, performance, and 研究问题: 智商是否会影响交易行为、业绩和 transaction costs. 交易成本。 Findings: 结论: (1) High-IQ investors are less subject to disposition effect, more aggressive about tax- (1)高智商投资者较少受处置效应的影响, 对税收更积极主动 loss trading, and more likely to supply liquidity when stock experience a one-month 亏损交易, 更有可能提供流动性时, 股票经历了一个月 high. 高。 (2) High-IQ investors exhibit superior market timing, stock-picking skill, and trade (2)高智商投资者表现出优越的市场时机、选股技巧和交易能力 execution. 执行。
Bach, Calvet and Sodini, 巴赫, Calvet 以及 索迪尼, American Economic Review, 美国经济评论, 2020. “Rich pickings? Risk, return, 丰厚的回报, 风险, 回报, and skill in household 以及 熟练掌握 家务 wealth.” 财富。”	Sweden data, annual 瑞典年度数据 administrative panel 行政委员会 that reports the full 报告全部 balance sheet of 资产负债表 Swedish residents 瑞典居民 between 2000 and 在 2000 年至 2000 年 2007	Research Question: Examine the risk and return characteristics of household wealth. 研究问题: 考察家庭财富的风险和收益特征。 Findings: 结论: (1) The expected return on household net wealth increases with net worth. (1)家庭净资产的预期回报率随着净资产的增加而增加。 (2) The expected wealth return is driven by systematic risk-taking and exhibits strong (2)预期财富回报受系统性冒险行为驱动, 且表现强劲 persistence. Idiosyncratic risk is transitory but sufficiently large to generate substantial 特殊风险是短暂的, 但足够大, 足以产生实质性的影响 long-term dispersion in returns. 收益的长期分散。 (3) Wealth returns explain most of the historical increase in top wealth shares. (3)财富回报率解释了历史上最高财富份额的大部分增长。
Dorn, Huberman and 多恩, 休伯曼 以及	German data, 37,000 德国数据, 37,000	Research Question: Investors correlated trading and stock returns. 研究问题: 投资者关联交易和股票回报。

Sengmueller, Journal of retail clients at one of Sengmueller, 日记 日期 零售客户之一 Finance, 2008. 财经, 2008 年。 “Correlated trading and 相关性 交易 以及 returns.” 返回。”	the three largest 最大的三家 German discount 德国折扣 brokers from 1998 to 经纪从 1998 年至 1999 年 2000	Findings: 结论: (1) Investors tend to be on the same side of the market. (1)投资者往往站在市场的同一边。 (2) Correlated market orders lead returns due to persistent speculative price pressure. (2)由于持续的投机价格压力, 相关的市场指令领先回报。 (3) Correlated limit orders also predict subsequent returns, consistent with liquidity (3)相关限价指令也预测随后的回报, 与流动性一致 demands. 需求。
Barrot, Kaniel, and Sraer, French data, from a 巴罗, 卡尼尔, 以及 Sraer, 法国的数据, 从一个 of Journal 日 Financial 日记 期 财务 Economics, 2016. 经济学, 2016。 “Are retail traders ”是 零售业 交易员 compensated for providing 补偿 为 提供 liquidity?” 流动性?”	leading European 领先的欧洲 online broker 网上经纪人 between 2002 and 在 2002 年至 2002 年 间 2010	Research Question: Whether individual investors provide liquidity to the stock market 研究问题: 个人投资者是否为股市提供了流动性 and whether they are compensated for doing so. 以及他们是否因此得到补偿。 Findings: 结论: (1) The ability of aggregate retail order imbalances to predict short-term future returns (1)总体零售订单失衡预测短期未来回报的能力 is significantly enhanced during times of market stress. 在市场紧张时期显著增加。 (2) Individual investors do not reap the rewards from liquidity provision because they (2)个人投资者没有从提供流动资金中获得回报, 因为他们 experience a negative return on the trading day and reverse their trades for too long after 在交易日经历负回报, 并在交易日之后很长时间内逆转交易 the liquidity provision dissipated. 流动性供应消失。

Publication Details 发布细节	Data 资料	Research Questions and Main Findings 研究问题和主要发现
Fong, Gallagher, and Lee, Fong Gallagher, 以及 李, of Journal of Financial and Quantitative Analysis, 2014. “Individual investors and 「个别投资者以及 broker types」 经纪人类型	Australian data, 澳大利亚数据, transaction data from 的交易数据 Australian Securities Exchange (ASX) 交易所(ASX) SIRCA with SIRCA 与 identification of the 识别 broker between 1995 经纪公司 1995 年间 and 2007 以及 2007 年	Research Question: Examine the informativeness of trades via discount and full-service 研究问题: 通过折扣和全方位服务来检验交易的信息性 retail brokers. 零售经纪。 Findings: 结论: (1) Trades via full-service retail brokers are more informative than are trades via (1)透过提供全面服务的零售经纪进行的交易比透过 discount retail brokers. 折扣零售经纪。 (2) Past returns, volatility, and news announcements could explain the net volume of (2)过去的回报率、波动率和新闻公告可以解释 discount retail brokers, but could not explain the net volume of full-service retail brokers. 折扣零售经纪, 但无法解释全面服务零售经纪的净额。
Barber, Liu, 理发 Lee, 小 and 师, 李, 刘, 以及 Odean, Review of Financial 金融评论 Studies, 2009. 研究, 2009. “Just howmuch do 只是 多少钱 做 individual investors lose by 个人投资者损失 trading?” 交易?”	Chinese Taiwan data, 中国台湾数据, transaction data in the 的交易数据 Taiwan stock exchange 台湾证券交易所 between 1995 and 1999 在 1995 年至 1999 年间	Research Question: How much do individual investors lose by trading? 研究问题: 个人投资者在交易中损失了多少? Findings: 结论: (1) The aggregate portfolio of individuals suffers an annual performance penalty of (1)个人总投资组合每年的业绩损失为 3.8%. 3.8% . (2) Nearly all individual trading losses can be traced to their aggressive orders. (2)几乎所有的个人交易损失都可以追溯到他们咄咄逼人的指令。
Barber, Liu, 理发 Lee, 小 and 师, 李, 刘, 以及	Chinese Taiwan data, 中国台湾数据,	Research Question: Examine the cross-sectional differences of returns earned by 研究问题: 检验收益的横截面差异

Odean, Journal of Financial Markets, 2014. 奥迪恩, 金融杂志市场, 2014。 “The cross-section of skill: Evidence from day trading”他说 截面图 日期 投机者 技巧: 证据 来自即日交易	day traders' transaction 交易员的交易 data in the Taiwan stock exchange 台湾的数据 证交所 between 1992 and 2006 从 1992 年到 2006 年	speculative day traders. 投机性的短线交易员。 Findings: Less than 1% of the day trader population can predictably and reliably earn positive abnormal returns net of fees. 研究结果: 只有不到 1% 的即日交易者可以预测和可靠地赚钱扣除费用后的正异常回报。
Balasubramaniam, Campbell, Ramadorai and Ranish, Journal of Finance, 2023. 巴拉苏布拉曼尼亚姆, 坎贝尔, 拉马多拉 以及 拉尼什, 金融杂志, 2023. “Who owns what? A factor model for direct stock holding”谁拥有什么? 一个因素模型为直接存货持有	Indian data, 10 million retail investors holdings data on August 2011 from Indian stock market 印度数据, 1000 万一般投资者持有的股份的数据 2011 年印度股市市场	Research Question: Build a cross-sectional factor model for retail investors' direct stock holdings. 研究问题: 构建散户直接投资股票的横截面因素模型持有。 Findings: 结论: (1) Stock characteristics such as firm age and share price have strong investor clienteles. (1) 公司年龄、股价等股票特征有较强的投资者客户群。 (2) Account attributes such as account age, account size, and extreme under diversification are associated with particular characteristic preferences. (2) 帐户属性, 如帐户年龄、帐户大小和极限多样化与特定的特征偏好有关。 (2) Coheld stocks have higher return covariance. (2) 共同持有的股票有较高的回报协方差。

Publication Details 发布细节	Data 资料	Research Questions and Main Findings 研究问题和主要发现
Balasubramaniam, Anagol, 巴拉苏布拉曼尼亚姆, of and Ramadorai, Journal 拉马多拉日报 Financial Economics, 2021. 金融经济学, 2021。 “Learning from noise: 从中学习” Evidence from India’s IPO 来自印度首次公开募股的 证据 lotteries” 彩票	Indian data, 1.5 million 印度数据, 150 万 investors participate in allocation lotteries for 54 IPO stocks between 54 只首次公开发行股票 2007 and 2012 2007 年和 2012 年	Research Question: Retail investors participate in allocation lotteries for Indian IPO 研究问题: 散户投资者参与印度 IPO 的配售彩票 stocks. 股票。 Findings: 结论: Investors who wind the IPO lottery and obtain IPO stocks that rise in value increase 投资者通过发行 IPO 彩票获得 IPO 股票, 股票价值增加 portfolio trading volume in non-IPO stocks relative to lottery losers. A learning model 非 ipo 股票相对于彩票输家的投资组合交易量。一个学习模型 could explain the results. 可以解释结果。
Titman, Wei, and Zhao, 泰特曼, 阿伟, 以及 赵老师, of Journal 日记 Financial Economics, 2022. 经济学, 2022。 “Corporate actions and the 企业行为和 manipulation 操纵 of retail investors in China: An analysis of stock splits” 股票分割分析”	Chinese data, account 中国数据、账户 trading data from a major stock exchange in China between 1999 1999 年在中国 and 2015 及 2015 年	Research Question: Corporate actions and the manipulation of retail investors in the 研究问题: 公司行为与操纵散户投资者的行为 stock splits events. 股票拆分事件。 Findings: 结论: (1) Share prices temporarily increase after stock splits, and subsequently decline below (1)股票价格在股票分割后暂时上升, 随后下降到低于 their presplit levels. 他们分裂前的水平。 (2) Small retail investors buy shares in firms initiating suspicious splits, while more (2)小型散户投资者购买可疑分拆公司的股票 sophisticated investors buy before suspicious split announcements and sell in the 复杂的投资者在可疑的分拆公告发布前买入, 然后在 postsplit period. 分手后的时期。

		(3) Insiders sell large blocks of shares and obtain loans using company stock as collateral (3)内部人士出售大量股份，并以公司股票作为抵押获得贷款 before the suspicious splits. 在可疑分裂之前。
An, Lou, and Shi, Journal of Monetary Economics, 2022. 安，娄，石，日记 货币经济学，2022。 “Wealth redistribution in bubbles and crashes.” 财富再分配 in 泡沫和崩溃。”	Chinese data, account trading from a major stock exchange in China, 2014-15 bubbles 中国数据、账户从一个主要的贸易证券交易所 China, 2014-15 bubbles 中国，2014-15 泡沫 and crashes episode 以及崩溃事件	Research Question: What are the social-economic consequences of financial market bubbles and crashes? 研究问题：金融市场的社会经济后果是什么 泡沫和崩溃？ Findings: 结论： The largest 0.5% households in the equity market gain, while the bottom 85% lose, 250B 股市中最大的 0.5% 的家庭获利，而最底层的 85% 家庭损失了 2500 亿美元 RMB through active trading in this period. 人民币在此期间通过积极的交易。
Liao, Peng, Zhu, Review of Financial Studies, 2021. 廖，彭，朱，回顾 金融研究，2021。 “Extrapolative bubbles and trading volume.” 外推泡沫和 交易量。”	Chinese data, account-level transaction data from one of the largest brokerage firms between 2014 and 2015 中文资料，帐户 - 水平交易数据 from one of the largest 其中一个最大的 brokerage firms 经纪公司 between 2014 and 2015 2014 年至 2015 年	Research Questions: Propose an extrapolative model to explain the sharp rise in prices and volume during financial bubbles. 研究问题：提出一个外推模型来解释价格的急剧上涨 以及金融泡沫期间的交易量。 Findings: 结论： (1) The model proposes a novel mechanism for volume: because of extrapolative beliefs and disposition effects, investors are quick to buy assets with positive past returns and sell them if good returns continue. (1)模型提出了一种新的体积机制：由于外推信念 以及处置效应，投资者会迅速购买过去回报率为正的资产 如果回报持续良好，则出售这些资产。 (2) Use Chinese account-level data to confirm the model's predictions. (2)使用中国账户级别的数据来证实模型的预测。

Publication Details 发布细节	Data 资料	Research Questions and Main Findings 研究问题和主要发现
Chen, Gao, He, Jiang, and Xiong, Journal of 陈, 高, 何, 江, 和阿雄, 日记 日期 Econometrics, 2019. 计量经济学, 2019. “Daily price limits and 日涨跌幅限制 以及 destructive market 破坏性的 市场 behavior.” 行为。”	Chinese data, account 中国数据、账户 trading from a major 从一个主要的贸易 stock exchange in China 中国证券交易所 from between 2012 and 由 2012 年至 2012 年 2015	Research Question: Daily price limits and investors trading behavior. 研究问题: 每日价格限制与投资者的交易行为。 Findings: 结论: Large investors tend to buy on the day when a stock hits the 10% upper price limit and 大型投资者倾向于在股价触及 10% 上限的那一天买入 then sell on the next day; and their net buying on the limit-hitting day predicts stronger 然后在第二天卖出; 他们在触及限额当天的净买入预计会更强劲 long-run price reversal. 长期价格逆转。
Li, Geng, Subrahmanyam 李, 耿, 阿耿, subrahmanyam and Yu, Journal of 以 余先, 日记 日期 Empirical Finance, 2017. Empirical Finance, 2017. “Do wealthy investors have 富有的投资者 an informational advantage? 信息优势? Evidence based on account 基于帐户的证据 classifications of individual 个体的分类 investors” 投资者	Chinese data, a 中国的数据 brokerage firm providing 经纪公司提供 one million investors’ 100 万投资者 trading records between 之间的交易记录 January 2007 and 2009 2007 年 1 月和 2009 年 1 月	Research Question: Do wealthy investors have an informational advantage? 研究问题: 富有的投资者是否拥有信息优势? Findings: 结论: (1) Wealthy investors with portfolio values above the 99.5th percentile (“super” (1)投资组合价值高于第 99.5 个百分位的富有投资者(「超级」) investors) outperform all other investors. 投资者)跑赢所有其他投资者。 (2) Part of their excess returns could be explained by informational advantages. (2)他们的部分超额回报可以用信息优势来解释。
Liu, Peng, Xiong, and 刘, 彭, 阿雄, 以及	Chinese data, Combine 中国数据, 联合	Research Question: Many biases offer observationally similar predictions for a 研究问题: 许多偏见提供了观察上相似的预测

Xiong, Journal of Financial 熊先生，金融期刊 Economics, 2022. 经济学，2022。 “Taming the bias zoo.” ”驯服偏见动物园”	subjective survey 主观调查 responses in 2018 with 2018 年的回应 account-level transaction 帐户级别的交易 data from a major stock 来自主要股票的数据 exchange in China 中国的交易	targeted financial anomaly. This study combines subjective survey responses with 这项研究结合了主观调查的答复和 observational data to tame the bias zoo. 观测数据，以驯服偏见动物园。 Findings: In cross-sectional regressions of respondents’ actual turnover on survey- 结果：在横断面回归的受访者的实际营业额调查 - based trading motives, perceived information advantage, and gambling preference 基于交易动机、感知信息优势和赌博偏好 dominate other motives. 支配其他动机。
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Appendix Table II. Comparing Main Findings of This Study with the Results from Previous Literature
附录表二：本研究主要结果与以往文献结果的比较

Our main findings 我们的主要发现	Findings from previous literature 以往文献的研究结果
Return Predictability: 返回可预测性: Retail investors with smaller account sizes negatively predict future returns, whereas retail investors with larger account sizes positively predict future returns. 账户规模较小的散户投资者表现不佳预测未来回报，而散户投资者的回报率较高账户规模正面预测未来回报。	Retail trading negatively predicts future returns: Barber and Odean 零售交易负面预测未来收益: Barber 和 Odean (2000), Grinblatt and Keloharju (2000), Barber, Huang, Odean, and (2000) , Grinblatt and Keloharju (2000) , Barber, Huang, Odean, and Schwarz (2022) 施瓦茨(2022) Retail trading positively predicts future returns: Kaniel, Saar, and 零售交易积极预测未来的回报: Titman (2008), Barber, Odean, and Zhu (2008), Dorn, Huberman and Titman (2008) , Barber, Odean, and Zhu (2008) , Dorn, Huberman and Sengmueller (2008), Grinblatt, Keloharju, Linnainmaa (2012), Kelley Sengmueller (2008) , Grinblatt, Keloharju, Linnainmaa (2012) , Kelley and Tetlock (2013), Fong, Gallagher, and Lee (2014), Barber, Lee, Liu, and Tetlock (2013) , Fong, Gallagher, and Lee (2014) , Barber, Lee, Liu, and Odean (2014), Barrot, Kaniel, and Sraer (2016), Li, Geng, and Odean (2014) , Barrot, Kaniel, 和 Sraer (2016) , Li, Geng, Subrahmanyam, and Yu (2017), Boehmer, Jones, Zhang, and Zhang subrahmanyam, and Yu (2017) , Boehmer, Jones, Zhang, and Zhang (2021), Welch (2022) (2021)韦尔奇(2022)
Momentum/Contrarian: 势头/反向投资者: Retail investors with smaller account sizes display daily momentum and weekly contrarian patterns, whereas retail 账户规模较小的散户投资者每天都在显示势头和每周反转模式，而零售业 investors with larger account sizes display contrarian 账户规模较大的投资者则表现出反向操作 patterns. 图案。	Retail investors display momentum trading patterns: Kelley and Tetlock 散户投资者展示动量交易模式: Kelley 和 Tetlock (2013), Boehmer, Jones, Zhang, and Zhang (2021) (2013) , Boehmer, Jones, Zhang, and Zhang (2021) Retail investors display contrarian trading patterns: Grinblatt and 散户投资者表现出反向交易模式: Keloharju (2000), Kaniel, Saar, and Titman (2008), Linnainmaa (2011), Keloharju (2000) , Kaniel, Saar, and Titman (2008) , Linnainmaa (2011) , Grinblatt, Keloharju, Linnainmaa (2012), Barrot, Kaniel, and Sraer Grinblatt, Keloharju, Linnainmaa (2012) , Barrot, Kaniel, and Sraer (2016), Welch (2022)

	(2016) , Welch (2022)
Process Information: 过程信息: Retail investors with smaller account sizes fail to process public news, whereas retail investors with larger account sizes incorporate public news in trading. 账户规模较小的散户投资者无法处理公众新闻, 而一般投资者的帐户较大大小包括交易中的公共新闻。	Retail investors can process information: Kaniel, Liu, Saar, and Titman (2012), Kelley and Tetlock (2013), Boehmer, Jones, Zhang, and Zhang (2012) , Kelley and Tetlock (2013) , Boehmer, Jones, Zhang and Zhang (2021) (2021) Retail investors fail to process information: Li, Geng, Subrahmanyam, 散户投资者未能处理信息: Li, 耿, Subrahmanyam, and Yu (2017), Titman, Wei, and Zhao (2022) 余(2017)、提特曼、魏和赵(2022)
Behavior Biases: 行为偏差: Retail investors with smaller account sizes show behavioral patterns such as overconfidence and gambling preferences, while retail investors with larger account sizes could better predict returns in stocks more attractive to investors with behavioral biases. 账户规模较小的散户投资者表现出行为过度自信和赌博偏好等模式, 而拥有更大账户规模的散户投资者可以做得更好预测股票回报率对投资者更具吸引力行为偏见。	Retail investors display behavioral biases: Barber and Odean (2000), 散户投资者表现出行为偏差: Barber 和 Odean (2000) , Liu, Peng, Xiong, and Xiong (2022), Barber, Huang, Odean, and 刘, 彭, 熊, 熊(2022) , 理发师, 黄, 欧丁, 和 Schwarz (2022) 施瓦茨(2022) Some retail investors display behavioral biases, while others trade against behavioral biases: Chen, Gao, He, Jiang, and Xiong (2019) 一些散户投资者表现出行为偏差, 而另一些投资者则进行交易反对行为偏见: 陈, 高, 何, 江, 熊(2019)

Appendix Table III. Distributions of Investor Trading and Holding

附录表三投资者交易和持股的分布

This table reports the summary statistics for the trading and holding of different investor groups across different stock characteristics. Our sample period covers January 2016 to June 2019. Our sample firms are A-share stocks listed on a major stock exchange with at least 15 days of trade during the previous month. The trading and holding for different investor groups across different stock characteristics are reported in Panels A and B, which are the time-series averages of cross-sectional means. The bottom two and top two sectors, in terms of trading volume and holding within each investor group, are reported in Panels C and D. The sectors are classified according to the Datastream Level 4 sector classification. Panel E reports the trading volumes of different investor groups across different trade sizes, which are the time-series averages of cross-sectional means.

本表汇报了不同股票特征下不同投资者群体的交易和持有情况的汇总统计数据。我们的样本期涵盖 2016 年 1 月至 2019 年 6 月。我们的样本公司是在主要证券交易所上市的 a 股股票，在上个月至少有 15 天的交易。不同股票特征的不同投资者群体的交易和持有情况见图 a 和图 b，它们是截面平均值的时间序列平均值。底部的两个和顶部的两个部门，在每个投资者群体的交易量和持有量方面，在面板 c 和 d 中报告。各行业是按照「数据流四级行业分类」进行分类。面板 e 报告了不同交易规模的不同投资者群体的交易量，这是截面平均值的时间序列平均值。

Panel A. Trading volumes across different stock characteristics

面板 a: 不同股票特征的交易量

	RT1	RT2	RT3	RT4	RT5	INST 国家科学 研究所	CORP 公司名称
	RT1	RT2	RT3	RT4	RT5		
Small	6.3%	23.3%	32.5%	13.4%	14.0%	9.9%	0.7%
小型企业	6.3%	23.3%	32.5%	13.4%	14.0%	9.9%	0.7%
Medium	5.8%	21.2%	31.3%	13.3%	13.9%	13.7%	0.8%
中等	5.8%	21.2%	31.3%	13.3%	13.9%	13.7%	0.8%
Large	4.3%	16.3%	26.7%	12.9%	15.7%	22.4%	1.8%
大型	4.3%	16.3%	26.7%	12.9%	15.7%	22.4%	1.8%
Low EP	6.3%	22.0%	31.4%	13.7%	15.3%	10.5%	0.8%
低收入	6.3%	22.0%	31.4%	13.7%	15.3%	10.5%	0.8%
Medium EP	5.6%	21.2%	30.7%	12.9%	13.8%	14.9%	0.9%
中等收入	5.6%	21.2%	30.7%	12.9%	13.8%	14.9%	0.9%
High EP	4.3%	17.1%	28.0%	13.1%	14.7%	21.2%	1.6%
高电压	4.3%	17.1%	28.0%	13.1%	14.7%	21.2%	1.6%
Low Turnover	4.7%	17.5%	28.0%	12.9%	14.5%	20.7%	1.7%
低营业额	4.7%	17.5%	28.0%	12.9%	14.5%	20.7%	1.7%
Medium Turnover	5.2%	19.4%	29.9%	13.5%	15.0%	15.8%	1.1%
中等营业额	5.2%	19.4%	29.9%	13.5%	15.0%	15.8%	1.1%
High Turnover	6.3%	23.4%	32.1%	13.2%	14.2%	10.1%	0.6%
高营业额	6.3%	23.4%	32.1%	13.2%	14.2%	10.1%	0.6%

Panel B. Holding shares across different stock characteristics
 面板 b: 持有不同股票特征的股票

	RT1	RT2	RT3	RT4	RT5	INST 国家科学 研究所	CORP 公司名称
	RT1	RT2	RT3	RT4	RT5		
Small	4.1%	13.4%	20.2%	9.9%	17.4%	7.1%	27.9%
小型企业	4.1%	13.4%	20.2%	9.9%	17.4%	7.1%	27.9%
Medium	3.6%	10.8%	16.5%	8.0%	15.4%	11.1%	34.6%
中等	3.6%	10.8%	16.5%	8.0%	15.4%	11.1%	34.6%
Large	2.2%	6.6%	10.6%	5.3%	10.1%	16.7%	48.6%
大型	2.2%	6.6%	10.6%	5.3%	10.1%	16.7%	48.6%
Low EP	3.9%	11.2%	16.9%	8.5%	15.5%	8.2%	35.7%
低收入	3.9%	11.2%	16.9%	8.5%	15.5%	8.2%	35.7%
Medium EP	3.4%	11.3%	17.3%	8.2%	15.6%	12.6%	31.6%
中等收入	3.4%	11.3%	17.3%	8.2%	15.6%	12.6%	31.6%
High EP	2.4%	7.3%	12.0%	6.2%	11.8%	14.5%	45.8%
高电压	2.4%	7.3%	12.0%	6.2%	11.8%	14.5%	45.8%
Low Turnover	2.2%	5.8%	9.3%	4.9%	11.2%	11.9%	54.6%
低营业额	2.2%	5.8%	9.3%	4.9%	11.2%	11.9%	54.6%
Medium Turnover	3.1%	9.0%	14.7%	7.8%	15.5%	12.9%	37.0%
中等营业额	3.1%	9.0%	14.7%	7.8%	15.5%	12.9%	37.0%
High Turnover	4.3%	15.0%	22.2%	10.1%	16.2%	10.5%	21.6%
高营业额	4.3%	15.0%	22.2%	10.1%	16.2%	10.5%	21.6%

Panel C. Sectors with lowest and highest trading volumes
交易量最低和最高的板块

	RT1 RT1	RT2 RT2	RT3 RT3	RT4 RT4	RT5 RT5	INST 国家科学研究所	CORP 公司名称
Bottom1 Bottom1	Banks & Life 银行与amp;人寿保险 Insurance 保险业 2.4% 2.4%	Banks & Life 银行与amp;人寿保险 Insurance 保险业 10.5% 10.5%	Banks & Life 银行与amp;人寿保险 Insurance 保险业 20.4% 20.4%	Banks & Life 银行与amp;人寿保险 Insurance 保险业 11.9% 11.9%	Alternative 其他选择 Energy 能源 12.2% 12.2%	Alternative 其他选择 Energy 能源 6.5% 6.5%	Alternative 其他选择 Energy 能源 0.4% 0.4%
Bottom2 Bottom2	Financial 财务 Services 服务 3.6% 3.6%	Financial 财务 Services 服务 15.8% 15.8%	Beverages 饮料 27.4% 27.4%	Health Care 健康护理 Equipment & 设备及 Services 服务 11.9% 11.9%	Gas, Water and 煤气、水和 Multiutilities 多功能设备 12.3% 12.3%	Industrial 工业 Metals & 金属及 Mining 矿业 11.6% 11.6%	General 一般资料 Industrials 工业 0.7% 0.7%
Top2 第二名	Industrial 工业 Metals & 金属及 Mining 矿业 6.5% 6.5%	Industrial 工业 Engineering 工程学 22.3% 22.3%	Gas, Water and 煤气、水和 Multiutilities 多功能设备 32.0% 32.0%	Technology 科技 Hardware & 五金及 Equipment 设备 14.5% 14.5%	Software & 软件及 Computer 电脑 Services 服务 17.7% 17.7%	Travel & 旅游及 Leisure 休闲 21.3% 21.3%	Financial 财务 Services 服务 2.9% 2.9%
Top1 第一名	Alternative 其他选择 Energy 能源 7.6% 7.6%	Alternative 其他选择 Energy 能源 25.5% 25.5%	Alternative 其他选择 Energy 能源 34.2% 34.2%	Financial 财务 Services 服务 14.5% 14.5%	Banks & Life 银行与amp;人寿保险 Insurance 保险业 18.0% 18.0%	Banks & Life 银行与amp;人寿保险 Insurance 保险业 32.8% 32.8%	Banks & Life 银行与amp;人寿保险 Insurance 保险业 4.1% 4.1%

Panel D. Sectors with lowest and highest holding shares
面板 d 持股比例最低和最高的行业

	RT1 RT1	RT2 RT2	RT3 RT3	RT4 RT4	RT5 RT5	INST 国家科学研究所	CORP 公司名称
Bottom1 Bottom1	Banks & Life 银行与amp;人寿保	Banks & Life 银行与amp;人寿保险	Banks & Life 银行与amp;人寿保险	Banks & Life 银行与amp;人寿保险	Banks & Life 银行与amp;人寿保险	Alternative 其他选择	Household Goods & 家居用品及

Bottom2 Bottom2	险 Insurance 保险业 1.1% 1.1%	Insurance 保险业 4.0% 4.0%	Insurance 保险业 7.6% 7.6%	Insurance 保险业 4.2% 4.2%	Insurance 保险业 5.6% 5.6%	Energy 能源 2.9% 2.9%	Home Construction 家居建筑 19.2% 19.2%
	Aerospace & 航空航天 Defense 防守	Aerospace & 航空航天 Defense 防守	Food & Drug 食物及药物 Retailers 零售商	Industrial 工业 Transportation 交通工具	Electricity 电力	Mobile 移动电话 Telecommunicati 电讯 ons 网页	Health Care Equipment 卫生保健设备 & Services 及服务
	1.9% 1.9%	7.2% 7.2%	12.1% 12.1%	5.6% 5.6%	7.3% 7.3%	4.8% 4.8%	23.2% 23.2%
		Household 住户 Goods & 货品及服务 Home 家 Construction 建筑	Support 支持 Services 服务	Support 支持 Services 服务	Support 支持 Services 服务	Banks & Life 银行与人寿保险 Insurance 保险业	Banks & Life Insurance 银行及人寿保险
Top1 第一名	4.5% 4.5%	12.8% 12.8%	19.2% 19.2%	9.7% 9.7%	19.5% 19.5%	23.3% 23.3%	54.3% 54.3%
	Alternative & 其他选择 Energy 能源	Alternative 其他选择 Energy 能源	Alternative 其他选择 Energy 能源	Forestry & 林业及 Paper 纸	Software & 软件及 Computer 电脑 Services 服务	Health Care 健康护理 Equipment & 设备及 Services 服务	Mobile 移动电话 Telecommunications 电讯服务
	6.1% 6.1%	15.5% 15.5%	20.9% 20.9%	9.8% 9.8%	23.8% 23.8%	23.5% 23.5%	55.1% 55.1%

Panel E. Trading volumes across different trade sizes
 不同交易规模的交易量

Trade Size 交易规模	RT1 RT1	RT2 RT2	RT3 RT3	RT4 RT4	RT5 RT5	INST 国家科学研究 所	CORP 公司名称
<40,000 CNY	4.78%	10.86%	7.44%	1.22%	0.74%	5.35%	0.10%
< 40,000 人民币	4.78%	10.86%	7.44%	1.22%	0.74%	5.35%	0.10%
40,000-200,000 CNY	0.69%	8.47%	14.54%	5.04%	3.65%	4.36%	0.27%
40,000-200,000 元	0.69%	8.47%	14.54%	5.04%	3.65%	4.36%	0.27%
>200,000 CNY	0.00%	1.03%	8.19%	6.96%	10.30%	5.48%	0.73%
20 万元人民币以上	0.00%	1.03%	8.19%	6.96%	10.30%	5.48%	0.73%

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Appendix Table IV. Long-short Portfolios

附录表四. 多空投资组合

This table reports an alternative method for Tables III and IV using long-short portfolios. On each day, we first group stocks into different counterparty bins in Panel A and into different days-to-cover ratios in Panel B. Within each bin, we then sort stocks into three groups according to the order imbalances of different investor groups. We form a long-short portfolio by longing the stock with the highest 1/3 order imbalance measures and shorting those with the lowest 1/3 order imbalance measures. The long-short portfolios are held on the next day and the portfolio returns are adjusted by Liu, Stambaugh and Yuan (2019) three-factor model. To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

本表报告了表 III 和表 IV 使用多空投资组合的另一种方法。在每一天，我们首先将股票分为 a 组中不同的交易对手仓和 b 组中不同的日对日比率，然后根据不同投资者群体的顺序不平衡将股票分为三组。我们通过期望 1/3 阶失衡指标最高的股票和做空 1/3 阶失衡指标最低的股票来形成多空投资组合。多空投资组合在第二天进行持有，投资组合回报率由 Liu, Stambaugh 和 Yuan (2019)三因素模型进行调整。为了考虑系数的序列相关性，采用 Newey 和 West (1987)方法对时间序列的标准误差进行调整，并按照 Newey 和 West (1994)方法选择最佳带宽。***，** 和 * 分别表示 1%，5% 和 10% 水平的显着性。

Panel A. Daily alphas for long-short portfolios with different counterparties

面板 a: 不同交易对手的多空投资组合的每日阿尔法

RT1-4 RT1-4	RT5 RT5	INST 国家科学 研究所	Sample Coverage 样本覆盖范围		RT1-RT4 RT1-RT4	RT5 RT5	INST 国家科学研 究所
Buy 买入	Buy 买入	Sell 卖掉	17% 17%	Alpha 阿尔法	-0.0004** -0.0004 * *	0.0003** 0003 * *	0.0019*** 0.0019 * * *
				[t-stat] [t-stat]	[-2.16] [-2.16]	[2.32] [2.32]	[11.05] [11.05]
Buy 买入	Sell 卖掉	Buy 买入	11% 11%	Alpha 阿尔法	-0.0006** -0.0006	0.0002 0.0002	-0.0003* -0.0003 *
				[t-stat] [t-stat]	[-2.33] [-2.33]	[0.83] [0.83]	[-1.85] [-1.85]
Buy 买入	Sell 卖掉	Sell 卖掉	21% 21%	Alpha 阿尔法	-0.0005*** -0.0005 * * *	0.0011*** 0.0011 * * *	0.0020*** 0.0020 美元 * *
				[t-stat] [t-stat]	[-2.75] [-2.75]	[8.36] [8.36]	[11.30] [11.30]
Sell 卖掉	Buy 买入	Buy 买入	22% 22%	Alpha 阿尔法	-0.0028*** -0.0028 * * *	0.0011*** 0.0011 * * *	0.0009*** 0.0009 * * *
				[t-stat] [t-stat]	[-14.81] [-14.81]	[6.74] [6.74]	[6.11] [6.11]
Sell 卖掉	Buy 买入	Sell 卖掉	13% 13%	Alpha 阿尔法	-0.0033*** -0.0033 * * *	0.0023*** 0.0023 * *	-0.0004** -0.0004 * *

				[t-stat]	[-9.10]	[10.88]	[-1.99]
				[t-stat]	[-9.10]	[10.88]	[-1.99]
Sell	Sell	Buy	17%	Alpha	-0.0011***	-0.0003*	0.0007***
卖掉	卖掉	买入	17%	阿尔法	-0.0011 * * *	-0.0003 *	0.0007 * * *
				[t-stat]	[-6.60]	[-1.86]	[4.33]
				[t-stat]	[-6.60]	- 1.86	[4.33]

Panel B. Daily alphas for long-short portfolios with different counterparties

面板 b 不同交易对手的多空投资组合的每日阿尔法

Holding Horizon		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5	INST
占领地平线		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5	国家科学 研究所
(0,10]Days	Alpha	-0.0054***	-0.0048***	-0.0034***	-0.0003*	0.0039***	0.0027***
(0,10)天	阿尔法	-0.0054 * *	-0.0048 * * *	-0.0034 * * *	-0.0003 *	0.0039 * *	0.0027 美元
	[t-stat]	[-19.22]	[-17.82]	[-14.72]	[-1.67]	[12.40]	[13.30]
	[t-stat]	[-19.22]	[-17.82]	[-14.72]	[-1.67]	[12.40]	[13.30]
(10,20]Days	Alpha	-0.0028***	-0.0023***	-0.0016***	-0.0006***	0.0014***	0.0021***
(10,20)天	阿尔法	-0.0028 * * *	-0.0023 * * *	-0.0016 * * *	-0.0006 * * *	0.0014 * * *	0.0021
	[t-stat]	[-14.54]	[-11.45]	[-8.39]	[-4.63]	[7.80]	[9.85]
	[t-stat]	[-14.54]	[-11.45]	[-8.39]	[-4.63]	[7.80]	[9.85]
(20,60]Days	Alpha	-0.0020***	-0.0014***	-0.0011***	-0.0003***	0.0006***	0.0019***
(20,60)天	阿尔法	-0.0020	-0.0014 * * *	-0.0011 * * *	-0.0003 * * *	0.0006 * *	0.0019 * * *
	[t-stat]	[-14.70]	[-8.19]	[-7.25]	[-4.05]	[5.79]	[14.08]
	[t-stat]	[-14.70]	[-8.19]	[-7.25]	[-4.05]	[5.79]	[14.08]
Above60Days	Alpha	-0.0025***	-0.0049***	-0.0056***	-0.0017**	0.0003***	0.0015***
60 天以上	阿尔法	-0.0025 * * *	-0.0049 * * *	-0.0056 * * *	-0.0017 * *	0.0003 * * *	0.0015 * * *
	[t-stat]	[-4.56]	[-5.47]	[-4.49]	[-2.52]	[3.07]	[12.84]
	[t-stat]	[-4.56]	[-5.47]	[-4.49]	[-2.52]	[3.07]	[12.84]

Appendix Table V. Predict Returns over the Next 12 Weeks Using Previous Week Order Imbalances

预测未来 12 周的回报使用前一周的顺序不平衡

This table reports the alternative results for Table V using previous week order imbalances to predict the cross-section of returns over the next 12 weeks. The independent variables are the previous week's order imbalance, *Oib*(w-1). The control variables are the same as those in Table II, and are not reported for brevity. To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

这个表格使用前一周的订单不平衡来预测未来 12 周的收益截面，报告了表 v 的替代结果。自变量是前一周的订单不平衡，*Oib* (w-1)。控制变量与表 II 中的相同，并且为了简洁起见没有报告。为了考虑系数的序列相关性，采用 Newey 和 West (1987)方法对时间序列的标准误差进行调整，并按照 Newey 和 West (1994)方法选择最佳带宽。***，** 和 * 分别表示 1%，5% 和 10% 水平的显著性。

wth weeks 几个星期了		OibRT1(w-1) OibRT1(w-1)	OibRT2(w-1) OibRT2(w-1)	OibRT3(w-1) OibRT3(w-1)	OibRT4(w-1) OibRT4(w-1)	OibRT5(w-1) OibRT5(w-1)	OibINST(w-1) OibINST (w-1)
1	Estimate	-0.0449***	-0.0428***	-0.0280***	-0.0041***	0.0064***	0.0081***
	估计	-0.0449 * * *	-0.0428 * * *	-0.0280 * * *	-0.0041 * * *	0.0064 * * *	0.0081 * *
	[t-stat]	[-12.37]	[-11.26]	[-8.70]	[-3.42]	[8.05]	[12.14]
	[t-stat]	[-12.37]	[-11.26]	[-8.70]	[-3.42]	[8.05]	[12.14]
2	Estimate	-0.0176***	-0.0162***	-0.0095***		0.0034***	0.0031***
	估计	-0.0176 * * *	-0.0162 * * *	-0.0095 * * *	0.0004	0.0034 * *	0.0031 * *
	[t-stat]	[-7.64]	[-7.18]	[-5.09]	[0.39]	[5.17]	[5.15]
	[t-stat]	[-7.64]	[-7.18]	[-5.09]	[0.39]	[5.17]	[5.15]
3	Estimate	-0.0099***	-0.0082***	-0.0050**	-0.0003	0.0031***	0.0021***
	估计	-0.0099 * * *	-0.0082 * * *	-0.0050 * *	-0.0003	0.0031 * *	0.0021
	[t-stat]	[-3.83]	[-2.93]	[-2.05]	[-0.33]	[5.64]	[2.93]
	[t-stat]	[-3.83]	[-2.93]	[-2.05]	[-0.33]	[5.64]	[2.93]
4	Estimate	-0.0072***	-0.0065***	-0.0036*	-0.0002	0.0023***	
	估计	-0.0072 * * *	-0.0065 * * *	-0.0036 *	-0.0002	0.0023 * *	0.0008
	[t-stat]	[-3.48]	[-2.93]	[-1.71]	[-0.21]	[3.54]	[1.42]
	[t-stat]	[-3.48]	[-2.93]	[-1.71]	[-0.21]	[3.54]	[1.42]
5	Estimate	-0.0059***	-0.0053***	-0.0050***	-0.0025**		0.0011**
	估计	-0.0059 * * *	-0.0053 * * *	-0.0050 * * *	-0.0025 * *	0.0009	0.0011 * *
	[t-stat]	[-3.60]	[-2.93]	[-2.71]	[-2.58]	[1.57]	[2.14]
	[t-stat]	[-3.60]	[-2.93]	[-2.71]	[-2.58]	[1.57]	[2.14]
6	Estimate	-0.0098***	-0.0089***	-0.0062***	-0.0010		0.0021***
	估计	-0.0098 * * *	-0.0089 * * *	-0.0062 * * *	-0.0010	0.0003	0.0021
	[t-stat]	[-4.73]	[-4.24]	[-3.42]	[-1.04]	[0.65]	[4.14]
	[t-stat]	[-4.73]	[-4.24]	[-3.42]	[-1.04]	[0.65]	[4.14]

7	Estimate	-0.0070***	-0.0060***	-0.0035**	-0.0004		0.0018***
	估计	-0.0070 * * * *	-0.0060 * * *	-0.0035	-0.0004	0.0009	0.0018 * * *
	[t-stat]	[-4.11]	[-3.44]	[-2.21]	[-0.33]	[1.53]	[4.46]
	[t-stat]	[-4.11]	[-3.44]	[-2.21]	[-0.33]	[1.53]	[4.46]
8	Estimate	-0.0048**	-0.0033	-0.0027	-0.0002	0.0013**	0.0010*
	估计	-0.0048 * *	-0.0033	-0.0027	-0.0002	0.0013 * *	0.0010 *
	[t-stat]	[-2.49]	[-1.53]	[-1.39]	[-0.16]	[2.02]	[1.80]
	[t-stat]	[-2.49]	[-1.53]	[-1.39]	[-0.16]	[2.02]	[1.80]
9	Estimate	-0.0030	-0.0015	-0.0011		0.0010*	
	估计	-0.0030	-0.0015	-0.0011	0.0006	0.0010 *	0.0002
	[t-stat]	[-1.46]	[-0.75]	[-0.62]	[0.80]	[1.81]	[0.39]
	[t-stat]	[-1.46]	[-0.75]	[-0.62]	[0.80]	[1.81]	[0.39]
10	Estimate	-0.0035*	-0.0030	-0.0024		-0.0007	0.0010*
	估计	-0.0035 *	-0.0030	-0.0024	0.0001	—0.0007	0.0010 *
	[t-stat]	[-1.66]	[-1.38]	[-1.22]	[0.11]	[-1.25]	[1.72]
	[t-stat]	[-1.66]	[-1.38]	[-1.22]	[0.11]	[-1.25]	[1.72]
11	Estimate	-0.0011	-0.0007				
	估计	-0.0011	—0.0007	0.0009	0.0008	0.0001	0.0001
	[t-stat]	[-0.54]	[-0.29]	[0.41]	[0.77]	[0.19]	[0.10]
	[t-stat]	[-0.54]	[-0.29]	[0.41]	[0.77]	[0.19]	[0.10]
12	Estimate	-0.0027	-0.0027	-0.0004	-0.0005		
	估计	-0.0027	-0.0027	-0.0004	-0.0005	0.0000	0.0001
	[t-stat]	[-1.22]	[-1.20]	[-0.19]	[-0.63]	[0.05]	[0.20]
	[t-stat]	[-1.22]	[-1.20]	[-0.19]	[-0.63]	[0.05]	[0.20]

Appendix Table VI. Two Stage Decomposition with Additional Lags of Order Imbalances

This table presents the alternative two-stage decomposition results for Table VI with additional lags of order imbalances. Panel A reports the first-stage estimation results controlling additional two weeks order imbalance. The order imbalances are decomposed into five components as specified in the following equation:

$$, = 0 + [11 \quad , -1 + 12 \quad , -2, -6 + 13 \quad , -7, -11] + 2' \quad , -1 + 3, -1 \quad (A11)$$

$$+ 4, -1 + 10, ,$$

where the order persistence components contain additional two weeks order imbalances, and other components are same as in Table VI.

Panel B reports the second-stage decomposition of the order imbalance's predictive power, as specified in the following equation:

$$\begin{aligned} \hat{H}^h &= \hat{H}^h + \hat{H}^h + \hat{H}^h + \hat{H}^h \\ \hat{H}^h &= \hat{H}^h + \hat{H}^h + \hat{H}^h + \hat{H}^h \end{aligned} \quad (A12)$$

where the order imbalance component, $Oib(-1, Persistence^{MoreLags})$, is constructed using the previous one day and previous two weeks order imbalance in first stage, and the rest are the same as those in Table VI. To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A. First stage of projecting the order imbalance on persistence, past returns, overconfidence, and gambling proxies

		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5
		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5
Oib(-1)	Estimate	0.1517***	0.1675***	0.1496***	0.0456***	0.0917***
Oib (- 1)	估计	0.1517 * *	0.1675 * * *	0.1496 * * *	0.0456 * * *	0.0917 * * *
	[t-stat]	[24.43]	[36.07]	[37.47]	[15.01]	[28.14]
	[t-stat]	[24.43]	[36.07]	[37.47]	[15.01]	[28.14]
Oib(-6,-2)	Estimate	0.2434***	0.2246***	0.1931***	0.0950***	0.1524***
Oib (- 6,-2)	估计	0.2434 * *	0.2246 * * *	0.1931 * * *	0.0950	0.1524 * * *

	[t-stat]	[38.71]	[37.27]	[33.78]	[18.14]	[30.75]
	[t-stat]	[38.71]	[37.27]	[33.78]	[18.14]	[30.75]
Oib(-11,-7)	Estimate	0.0691***	0.0541***	0.0533***	0.0475***	0.0691***
Oib (- 11, -7)	估计	0.0691 * *	0.0541 * *	0.0533 * * *	0.0475 * * *	0.0691 * *
	[t-stat]	[12.43]	[10.33]	[11.53]	[13.47]	[22.22]
	[t-stat]	[12.43]	[10.33]	[11.53]	[13.47]	[22.22]
Ret(-1)	Estimate	0.5303***	0.6989***	0.4174***	-0.2229***	-1.3117***
Ret (- 1)	估计	0.5303 * * *	0.6989 * * *	0.4174 * * *	-0.2229 * *	-1.3117 * *
	[t-stat]	[9.16]	[15.30]	[13.53]	[-7.67]	[-30.37]
	[t-stat]	[9.16]	[15.30]	[13.53]	[-7.67]	[-30.37]
Ret(-6,-2)	Estimate	-0.2261***	-0.0271**	0.0435***	-0.0285	-0.0799***
Ret (- 6, -2)	估计	-0.2261 * * *	-0.0271	0.0435 * *	-0.0285	-0.0799 * * *
	[t-stat]	[-12.25]	[-2.54]	[3.85]	[-1.60]	[-4.95]
	[t-stat]	[-12.25]	[-2.54]	[3.85]	[-1.60]	[-4.95]
Ret(-27,-7)	Estimate	0.0077*	0.0074**	-0.0033	-0.0282***	-0.0170***
Ret (- 27,-7)	估计	0.0077 *	0.0074 * *	-0.0033	-0.0282 * * *	-0.0170 * * * *
	[t-stat]	[1.95]	[2.28]	[-1.19]	[-7.51]	[-3.99]
	[t-stat]	[1.95]	[2.28]	[-1.19]	[-7.51]	[-3.99]
Overconf(-1)	Estimate	0.0391***	0.0204**	0.0218***		-0.0533***
Overconf (- 1)	估计	0.0391 * *	0.0204 * *	0.0218 * *	0.0167	-0.0533 * * *
	[t-stat]	[3.11]	[2.17]	[2.90]	[1.63]	[-5.05]
	[t-stat]	[3.11]	[2.17]	[2.90]	[1.63]	[-5.05]
Gamble(-1)	Estimate		0.0580***	0.1430***	0.2229***	-0.0425*
Gamble (- 1)	估计	0.0172	0.0580 * *	0.1430 * * *	0.2229	-0.0425 *
	[t-stat]	[1.03]	[4.49]	[11.54]	[12.02]	[-1.94]
	[t-stat]	[1.03]	[4.49]	[11.54]	[12.02]	[-1.94]

Intercept	Estimate	-0.0131***	-0.0076***	-0.0080***	-0.0062***	0.0174***
拦截	估计	-0.0131 * * *	-0.0076 * * *	-0.0080 * * *	-0.0062 * * *	0.0174 * * *
	[t-stat]	[-4.14]	[-3.60]	[-6.05]	[-3.75]	[8.57]
	[t-stat]	[-4.14]	[-3.60]	[-6.05]	[-3.75]	[8.57]
Adj.R2		9.68%	7.64%	5.37%	1.21%	2.99%
R2		9.68%	7.64%	5.37%	1.21%	2.99%

Panel B. Second stage decomposition of order imbalance's predictive power

面板 b: 顺序不平衡预测能力的第二阶段分解

Dep.var		Ret	Ret	Ret	Ret	Ret
风险管理局		休息	休息	休息	休息	休息
Oib.var		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5
Oib.var		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5
Oib(-1,PersistenceMoreLags)						
Oib (- 1, persistence morerelags)	Estimate	-0.0284***	-0.0274***	-0.0202***	-0.0062***	0.0059***
	估计	-0.0284 * * *	-0.0274 * * *	-0.0202 * * *	-0.0062 * * *	0.0059 * * *
	[t-stat]	[-13.95]	[-13.52]	[-11.19]	[-3.95]	[8.36]
	[t-stat]	[-13.95]	[-13.52]	[-11.19]	[-3.95]	[8.36]
Oib(-1,Liquidity)	Estimate	-0.0085**	-0.0174***	-0.0255***		
离岸价格(- 1, 流动性)	估计	-0.0085 * *	-0.0174 * * *	-0.0255 * * *	0.0204	0.0037
	[t-stat]	[-2.16]	[-3.87]	[-3.03]	[1.29]	[1.50]
	[t-stat]	[-2.16]	[-3.87]	[-3.03]	[1.29]	[1.50]
Oib(-1,Overconf)	Estimate	-0.1022	-0.4609***	-0.6067***	-0.9331***	0.1815***
Oib (- 1, Overconf)	估计	-0.1022	-0.4609 * *	-0.6067 * * *	-0.9331 * * *	0.1815 * * *
	[t-stat]	[-1.63]	[-3.79]	[-4.29]	[-5.06]	[5.14]
	[t-stat]	[-1.63]	[-3.79]	[-4.29]	[-5.06]	[5.14]
Oib(-1,Gamble)	Estimate	-0.5580***	-0.1310***	-0.0538***	-0.0453***	0.3008***
Oib (- 1, Gamble)	估计	-0.5580 * * *	-0.1310 * * *	-0.0538 * *	-0.0453 * * *	0.3008 * *
	[t-stat]	[-4.87]	[-4.00]	[-4.05]	[-5.32]	[6.35]
	[t-stat]	[-4.87]	[-4.00]	[-4.05]	[-5.32]	[6.35]
Oib(-1,Other)	Estimate	-0.0084***	-0.0082***	-0.0058***	-0.0008***	0.0011***
Oib (- 1, 其他)	估计	-0.0084 * * *	-0.0082 * * *	-0.0058 * * *	-0.0008 * * *	0.0011 * * *
	[t-stat]	[-15.19]	[-15.17]	[-13.90]	[-6.06]	[9.40]
	[t-stat]	[-15.19]	[-15.17]	[-13.90]	[-6.06]	[9.40]
Adj.R2		10.54%	10.37%	10.06%	9.61%	9.48%
R2		10.54%	10.37%	10.06%	9.61%	9.48%

Interquartile return

四分位间回归

Oib(- 1,PersistenceMoreLags)					
Oib (- 1, persistencemorerelags)	-0.1387%	-0.1103%	-0.0656%	-0.0123%	0.0335%
Oib(-1,Liquidity)	-0.1387%	-0.1103%	-0.0656%	-0.0123%	0.0335%
离岸价格(- 1, 流动性)	-0.0124%	-0.0223%	-0.0195%	0.0102%	0.0089%
Oib(-1,Overconf)	-0.0124%	-0.0223%	-0.0195%	0.0102%	0.0089%
Oib (- 1, Overconf)	-0.0150%	-0.0338%	-0.0404%	-0.0428%	0.0252%
Oib(-1,Gamble)	-0.0150%	-0.0338%	-0.0404%	-0.0428%	0.0252%
Oib (- 1, Gamble)	-0.0317%	-0.0251%	-0.0254%	-0.0334%	0.0422%
Oib(-1,Other)	-0.0317%	-0.0251%	-0.0254%	-0.0334%	0.0422%
Oib (- 1, 其他)	-0.1694%	-0.1433%	-0.0980%	-0.0227%	0.0475%

附录表七。根据流动资金供应和流动资金需求跟踪零售交易组合的年度表现

本表采用与表八类似的框架，从流动资金供应和需求角度报告不同零售投资集团的年度业绩。年度总体表现可分解为流动性供应、流动性需求和交易成本，如下表所示：

$$, = , + , - , . \quad (A13)$$

我们假设如果投资者的交易方向与之前的股价变动方向相反，那么投资者提供流动性；如果投资者的交易方向与之前的股价变动方向相同，那么投资者需要流动性。流动性供给和流动性需求的总体表现定义如下：

[illegible]

[illegible]

Panel A presents the total annual trading performance by different investor groups, from liquidity provision and demand perspective. Panel B further decompose the annual trading performance into stock selection and market timing components. Panel C reports the annual trading performance around earnings announcements. The earnings announcement events include the stock on announcement day d and the previous day $d-1$. To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

小组 a 从流动资金供应和需求的角度介绍不同投资者群体的年度总交易表现。面板 b 进一步将年度交易表现分解为股票选择和市场时机组成部分。面板 c 报告每年的交易业绩围绕收益公告。收益公告事件包括公告日 d 和前一天 $d-1$ 的股票。为了考虑系数的序列相关性，采用 Newey 和 West (1987)方法对时间序列的标准误差进行调整，并按照 Newey 和 West (1994)方法选择最佳带宽。***，** 和 * 分别表示 1%，5% 和 10% 水平的显著性。

Panel A. Annual trading performance using tracking portfolios
使用跟踪投资组合的年度交易业绩

		Total 总计	Liquidity Provision 流动资金供应	Liquidity Demand 流动资金需求	Trading Cost 交易成本
RT1	Performance	-5.61%***	-0.34%	-3.89%***	-1.38%***
RT1	表演	-5.61% * * *	-0.34%	-3.89% * *	-1.38% * *
	[t-stat]	[-10.29]	[-0.82]	[-7.65]	[-23.47]
	[t-stat]	[-10.29]	[-0.82]	[-7.65]	[-23.47]
RT2	Performance	-4.61%***	-0.04%	-2.78%***	-1.80%***
RT2	表演	-4.61% * * *	-0.04%	-2.78% * * *	-1.80% * * *
	[t-stat]	[-8.96]	[-0.09]	[-6.45]	[23.07]
	[t-stat]	[-8.96]	[-0.09]	[-6.45]	[23.07]
RT3	Performance	-3.60%***	-0.25%	-1.63%***	-1.72%***
RT3	表演	-3.60% * *	-0.25%	-1.63% * * *	-1.72%
	[t-stat]	[-9.49]	[-0.90]	[-6.06]	[-20.89]
	[t-stat]	[-9.49]	[-0.90]	[-6.06]	[-20.89]
RT4	Performance	-2.80%***	-0.11%	-1.11%***	-1.58%***
RT4	表演	-2.80% * * *	-0.11%	-1.11% * * *	-1.58% * * *
	[t-stat]	[-8.51]	[-0.49]	[-4.63]	[-19.53]
	[t-stat]	[-8.51]	[-0.49]	[-4.63]	[-19.53]
RT5	Performance	-0.29%	1.89%***	-1.14%***	-1.03%***
RT5	表演	-0.29%	1.89% * * *	-1.14% * *	-1.03%
	[t-stat]	[-0.92]	[5.12]	[-3.05]	[-18.84]
	[t-stat]	[-0.92]	[5.12]	[-3.05]	[-18.84]
INST 国家科 学研究 所	Performance	1.15%***	0.84%***	0.73%***	-0.41%***
	表演	1.15% * *	0.84% * *	0.73%	-0.41% * *
	[t-stat]	[3.57]	[5.70]	[2.91]	[-20.49]
	[t-stat]	[3.57]	[5.70]	[2.91]	[-20.49]

Panel B. Annual trading performance using tracking portfolios: separate stock selection and market timing
面板 b 使用跟踪投资组合的年度交易表现: 单独的股票选择和市场时机

		Total 总计	Liquidity Provision 流动资金供应		Liquidity Demand 流动资金需求		TradingCost 交易成本
			Stock Selection 股票选择	Market Timing 市场时机	Stock Selection 股票选择	Market Timing 市场时机	
RT1	Performance	-5.61%***	0.07%	-0.41%	-4.10%***	0.22%	-1.38%***

RT1	表演	-5.61% * * *	0.07%	-0.41%	-4.10%	0.22%	-1.38% * *
	[t-stat]	[-10.29]	[0.25]	[-1.39]	[-8.51]	[1.03]	[-23.47]
	[t-stat]	[-10.29]	[0.25]	[-1.39]	[-8.51]	[1.03]	[-23.47]
RT2	Performance	-4.61%***	0.43%	-0.46%*	-2.97%***	0.19%	-1.80%***
RT2	表演	-4.61% * * *	0.43%	-0.46%	-2.97% * *	0.19%	-1.80% * * *
	[t-stat]	[-8.96]	[1.46]	[-1.79]	[-7.63]	[1.12]	[23.07]
	[t-stat]	[-8.96]	[1.46]	[-1.79]	[-7.63]	[1.12]	[23.07]
RT3	Performance	-3.60%***	0.17%	-0.42%**	-1.77%***	0.14%	-1.72%***
RT3	表演	-3.60% * *	0.17%	-0.42% * *	-1.77% * *	0.14%	-1.72%
	[t-stat]	[-9.49]	[0.85]	[-2.47]	[-7.48]	[1.02]	[-20.89]
	[t-stat]	[-9.49]	[0.85]	[-2.47]	[-7.48]	[1.02]	[-20.89]
RT4	Performance	-2.80%***	0.33%**	-0.44%***	-1.23%***	0.12%	-1.58%***
RT4	表演	-2.80% * * *	0.33% * *	-0.44% * *	-1.23% * * *	0.12%	-1.58% * * *
	[t-stat]	[-8.51]	[2.01]	[-3.32]	[-6.66]	[0.65]	[-19.53]
	[t-stat]	[-8.51]	[2.01]	[-3.32]	[-6.66]	[0.65]	[-19.53]
RT5	Performance	-0.29%	2.45%***	-0.56%***	-1.40%***	0.26%	-1.03%***
RT5	表演	-0.29%	2.45%	-0.56% * *	-1.40% * *	0.26%	-1.03%
	[t-stat]	[-0.92]	[7.96]	[-3.03]	[-5.32]	[1.18]	[-18.84]
	[t-stat]	[-0.92]	[7.96]	[-3.03]	[-5.32]	[1.18]	[-18.84]
INST							
国家科							
学研究	Performance	1.15%***	0.89%***	-0.05%	0.49%***	0.23%	-0.41%***
所	表演	1.15% * *	0.89% * *	-0.05%	0.49% * *	0.23%	-0.41% * *
	[t-stat]	[3.57]	[6.71]	[-0.62]	[3.04]	[1.32]	[-20.49]
	[t-stat]	[3.57]	[6.71]	[-0.62]	[3.04]	[1.32]	[-20.49]

Panel C. Annual trading performance using tracking portfolios: around earnings announcements

面板 c 使用跟踪投资组合的年度交易业绩: 围绕盈利公告

		Total 总计		Liquidity Provision 流动资金供应		Liquidity Demand 流动资金需求		Trading Cost 交易成本	
		EA 电子 艺界	EA/all days 电子艺界/全 天候	EA 电子艺 界	EA/all days 电子艺界/全 天候	EA 电子艺 界	EA/all days 电子艺界/全 天候	EA 电子艺 界	EA/all days 电子艺界/全 天候
RT1	Performance	-0.39%***	6.89%	-0.05%	14.43%	-0.30%***	7.60%	-0.04%***	3.06%
RT1	表演	-0.39% **	6.89%	-0.05%	14.43%	-0.30% **	7.60%	-0.04% **	3.06%
	[t-stat]	[-3.68]		[-0.86]		[-4.10]		[-5.91]	
	[t-stat]	[-3.68]		[-0.86]		[-4.10]		[-5.91]	
		-0.36%***							
RT2	Performance	-0.36% **	7.71%	-0.03%	93.81%	-0.27%***	9.54%	-0.06%***	3.12%
RT2	表演	*	7.71%	-0.03%	93.81%	-0.27% **	9.54%	-0.06% **	3.12%
	[t-stat]	[-4.08]		[-0.64]		[-3.89]		[-5.90]	
	[t-stat]	[-4.08]		[-0.64]		[-3.89]		[-5.90]	
RT3	Performance	-0.24%***	6.61%	-0.03%	11.05%	-0.16%***	9.52%	-0.05%***	3.18%
RT3	表演	-0.24% **	6.61%	-0.03%	11.05%	-0.16% **	9.52%	-0.05% **	3.18%
	[t-stat]	[-4.24]		[-0.75]		[-3.44]		[-5.86]	
	[t-stat]	[-4.24]		[-0.75]		[-3.44]		[-5.86]	
		-0.14%***							
RT4	Performance	-0.14% **	5.02%	0.00%	0.24%	-0.09%**	8.01%	-0.05%***	3.26%
RT4	表演	*	5.02%	0.00%	0.24%	-0.09%	8.01%	-0.05% **	3.26%
	[t-stat]	[-2.70]		[-0.01]		[-2.06]		[-5.86]	
	[t-stat]	[-2.70]		- 0.01		[-2.06]		[-5.86]	
RT5	Performance	0.10%	-33.28%	0.19%***	9.90%	-0.06%	4.96%	-0.03%***	3.37%
RT5	表演	0.10%	-33.28%	0.19% **	9.90%	-0.06%	4.96%	-0.03% **	3.37%
	[t-stat]	[0.91]		[3.13]		[-1.18]		[-5.87]	
	[t-stat]	[0.91]		[3.13]		[-1.18]		[-5.87]	
INST									
国家科									
学研究	Performance	0.11%*	9.80%	0.10%***	11.57%	0.03%	4.23%	-0.01%***	3.56%
所	表演	0.11% *	9.80%	0.10%	11.57%	0.03%	4.23%	-0.01% **	3.56%
	[t-stat]	[1.81]		[3.48]		[0.67]		[-5.86]	
	[t-stat]	[1.81]		[3.48]		[0.67]		[-5.86]	

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Appendix Table VIII. Gender and Age

附录表八性别和年龄

This table examines retail investors by gender and age. The sample period covers January 2019 to March 2019. The sample firms are A-share stocks with at least 15 days of trade during the previous month. Panel A reports the summary statistics of trading volumes across age and gender groups. Panel B reports the return predictions across genders and age groups. To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

这个表格按性别和年龄分析了散户投资者。样本期涵盖二零一九年一月至二零一九年三月。样本公司是 a 股股票，在上个月至少有 15 天的交易。面板 a 报告了不同年龄和性别群体的交易量统计摘要。小组 b 报告跨性别和年龄组的回报预测。为了考虑系数的序列相关性，采用 Newey 和 West (1987)方法对时间序列的标准误差进行调整，并按照 Newey 和 West (1994)方法选择最佳带宽。***，** 和 * 分别表示 1%，5% 和 10% 水平的显著性。

Panel A. Summary Statistics of gender and age groups

小组 a。性别和年龄组的总结统计

Gender 性别	Trading Volume (% of total) 交易量(占总量的百分比)	
Age 年龄	<45 < 45	>=45 > = 45
Male 男	29%	38%
Female 女	13%	20%

Panel B. Cross-sectional return predictions, by different gender and age groups

面板 b 按不同性别和年龄组的横截面回报预测

Dep.var 风险管理局		Ret 休息	Ret 休息	Ret 休息	Ret 休息
Gender 性别		Male 男	Male 男	Female 女	Female 女
Age 年龄		<45 < 45	>45 > 45	<45 < 45	>45 > 45
Oib(-1)	Estimate	-0.0026***	-0.0060***		-0.0002
Oib (- 1)	估计	-0.0026 * * *	-0.0060 * * *	0.0007	-0.0002
	[t-stat]	[-5.45]	[-3.48]	[1.46]	[-0.32]
	[t-stat]	[-5.45]	[-3.48]	[1.46]	[-0.32]
	Interquartile return	-0.05%	-0.11%	0.02%	0.00%
	四分位间回归	-0.05%	-0.11%	0.02%	0.00%

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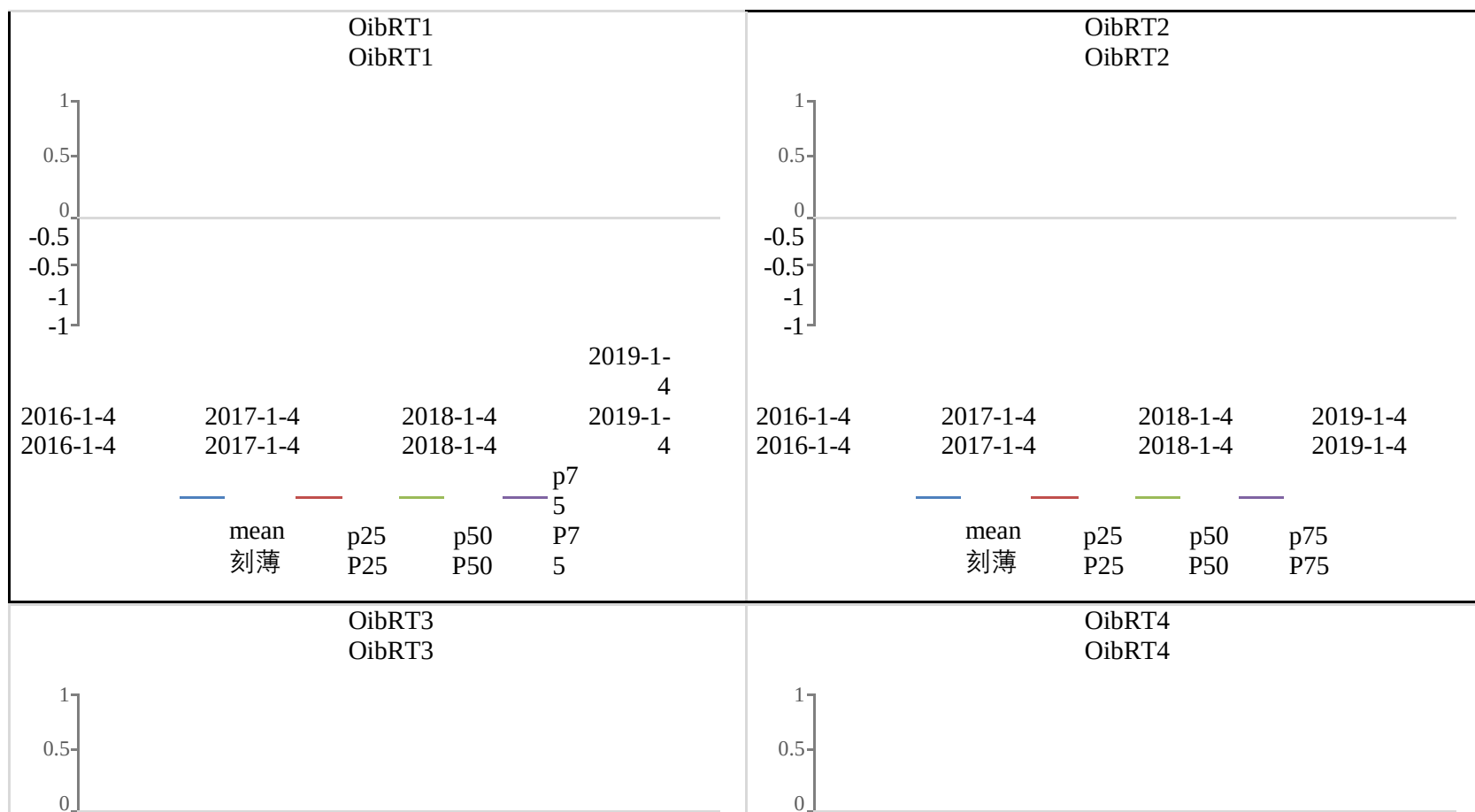
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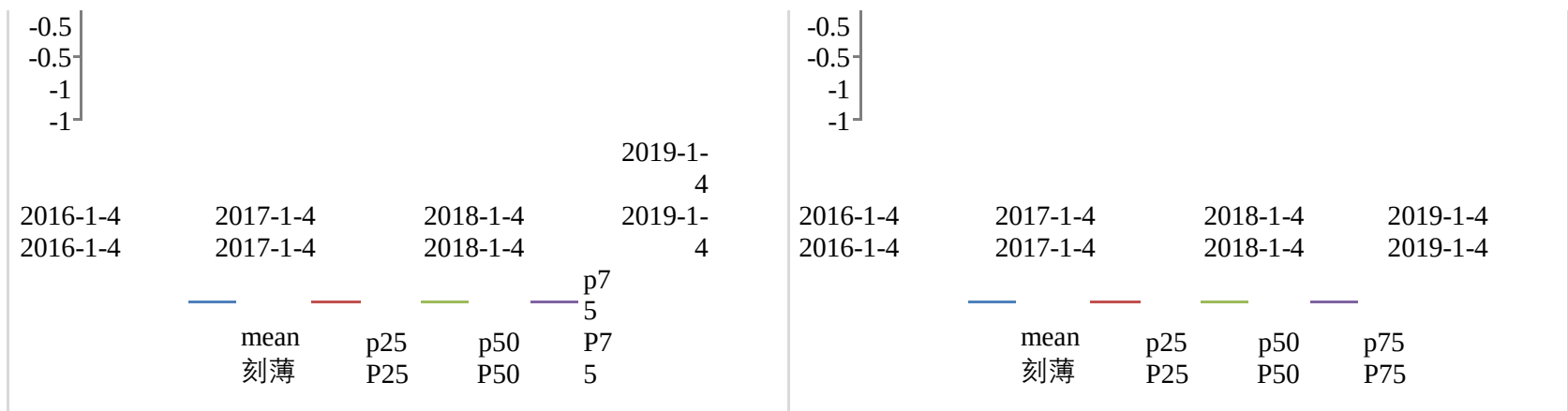
Appendix Figure I. Time Series of Different Types of Investor Order Imbalance

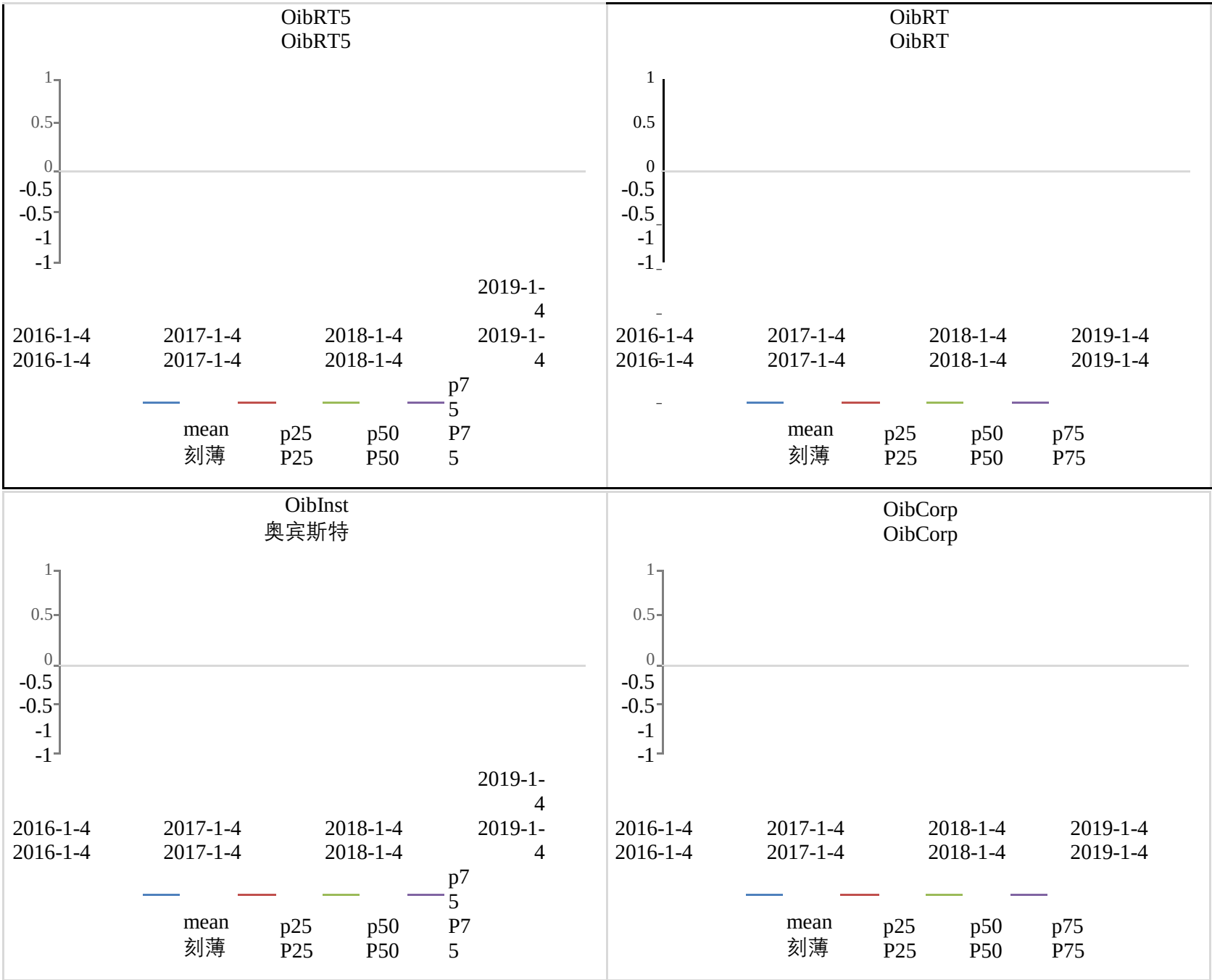
附录图一不同类型投资者订单失衡的时间序列

These figures report the time series of the different types of investor trading activities. The sample period covers January 2019 to March 2019. The sample firms are A-share stocks with at least 15 days of trade during the previous month. We present the cross-sectional mean, median, 25th percentiles, and 75th percentiles of scaled daily order imbalances for each investor group each day. Order imbalance is computed as the buy share volume minus the sell share volume divided by the buy share volume plus sell share volume for each investor group, as specified in Equation (1).

这些数字反映了不同类型投资者交易活动的时间序列。样本期包括二零一九年一月至二零一九年三月。样本公司是 a 股股票，在上个月至少有 15 天的交易。我们给出了每个投资者群体每天的截面平均值、中位数、第 25 百分位数和第 75 百分位数。订单不平衡的计算方法是买入股票数量减去卖出股票数量除以买入股票数量加上卖出股票数量，每个投资者群体，如方程(1)所规定。







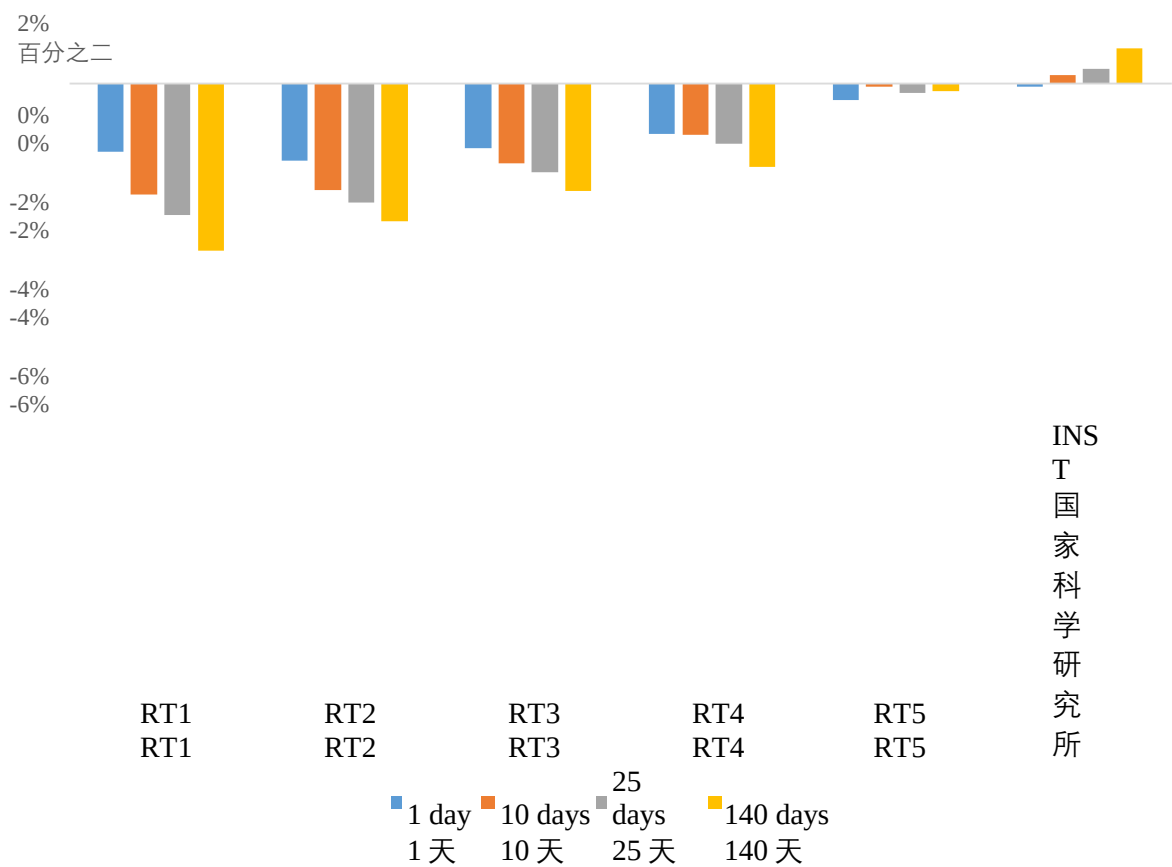
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Appendix Figure II. Annual Performances over Different Portfolio Holding Horizon
附录图二。不同投资组合持有水平的年度业绩

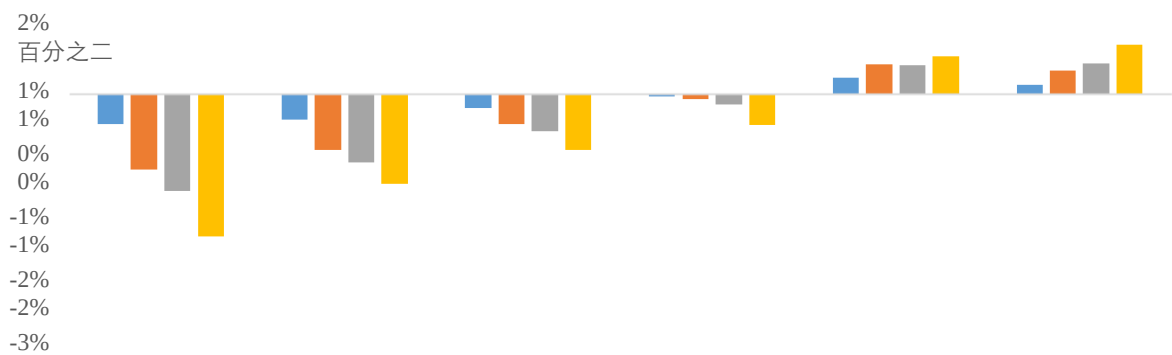
This table reports the alternative performance for different retail investor groups over different portfolio holding days. Following Barber et al. (2009) method, as specified in equation (A4) to (A10), we construct the portfolios tracking aggregate retail trading and present the annual performance for holding n days ($n = 1$ day, 10 days, 25 days, and 140 days). Panel A presents the annual total performance and Panel B and C present stock selection and market timing.

这个表格报告了不同的散户投资者群体在不同的投资组合持有日的不同表现。根据 Barber 等人(2009)的方法，按照方程(A4)至(A10)的规定，我们构造了追踪零售交易总量的投资组合，并给出了持有 n 天($n = 1$ 天，10 天，25 天和 140 天)的年度业绩。面板 a 表示年度总体表现，面板 b 和 c 表示股票选择和市场时机。

Panel A. Total trading performances over different portfolio holding days
小组 a: 不同投资组合持有日的总交易表现



Panel B. Stock selection over different portfolio holding days
面板 b: 不同投资组合持有日的股票选择



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-4%
-4%
-5%
-5%

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RT1

RT2
RT2

RT3
RT3

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RT5
RT5

1 day 10 days 25 days 140 days
1 天 10 天 25 天 140 天

Panel C. Market timing over different portfolio holding days
不同投资组合持有日的市场时机选择

0.30%
0.30%
0.20%
0.20%
0.10%
0.10%
0.00%
0.00%
-0.10%
-0.10%
-0.20%
-0.20%
-0.30%
-0.30%
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1 day 10 days 25 days 140 days
1 天 10 天 25 天 140 天

25
25 天